A Regime-Aware Trajectory Prediction Framework for 1000+ Systems Biology Models

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Abstract

Predicting long-horizon trajectories of biological dynamical systems remains challenging due to substantial system heterogeneity. Most existing machine learning approaches are system-specific, requiring retraining for each new system and exhibiting limited generalization across distinct biological regimes. To address this limitation, we create a large-scale benchmark of over 1,000 ODE-based systems biology models spanning diverse organisms, biological processes, and dynamical behaviors. Building on this benchmark, we propose a regime-aware trajectory prediction framework that enables cross-system generalization and uncertainty quantification for unseen systems. Our approach introduces structured initial states derived from biological regime priors, such as growth trends and oscillatory rhythms, into conditional flow matching, replacing the standard Gaussian source distribution. We provide theoretical justification for this initialization and empirically demonstrate state-of-the-art accuracy (31% MAE reduction), well-calibrated uncertainty (17% CRPS improvement), and efficient long-horizon inference across the benchmark.

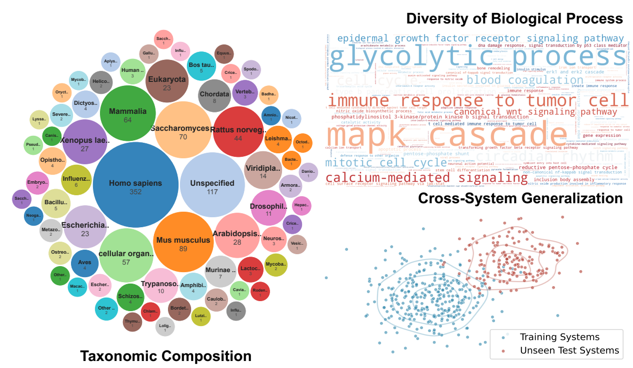

Figure 1. Diversity of SysBio-Traj across 1,050 biological models. **Left:** Taxonomic composition shows broad organism coverage (circle size indicates model count). **Top Right:** Gene Ontology biological process coverage (text size indicates model count). **Bottom Right:** Illustration of cross-system generalization task.

1. Introduction

Predicting trajectories of biological dynamical systems is fundamental for understanding complex biological processes (Ideker et al., 2001; Kitano, 2002), and these systems are traditionally modeled using ordinary differential equations (ODEs). While classical numerical solvers provide rigorous trajectory integration (Hairer & Wanner, 1996), they are hindered by high simulation costs and require explicit biological equations and ODE parameters that are usually incomplete, hypothetical, or difficult to specify in real-world biological discovery (Course & Nair, 2023). The advent of scientific machine learning has offered a promising alternative, with architectures such as Neural ODEs (Chen et al., 2018) and Neural Operators (Li et al., 2020) capable of approximating complex dynamics directly from observational data. However, these data-driven approaches are predominantly system-specific, trained to fit a single dynamical system and unable to generalize across heterogeneous biological systems (Vasiliauskaite & Antulov-Fantulin, 2024; Klötergens et al., 2025). This limitation becomes particularly prohibitive in large-scale settings involving thousands of biological systems (See our benchmark diversity in Figure 1), where per-system retraining is impractical. Consequently, a unified model is needed to handle trajectory prediction across heterogeneous biological systems.

To address the lack of scalability and transferability in system-specific models, general purpose time-series fore-

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casting architectures, such as Transformers (Liu et al., 2023; Zeng et al., 2023) and Mamba (Gu & Dao, 2024), have emerged as powerful system-agnostic tools to learn across heterogeneous datasets. While these approaches improve scalability, they are typically deterministic and produce point forecasts. Recent generative modeling paradigms, including diffusion models and flow matching approaches (Kollovieh et al., 2023; 2024), have been introduced to enable uncertainty quantification of trajectory prediction. Despite these advances, both deterministic and generative methods remain limited in long-horizon forecasting, as different future trajectories may share similar observed history, making the mapping from past to future ambiguous. Moreover, they may generate trajectories against known macroscopic behaviors, referred to as biological regimes. For example, irreversible disease progression exhibits monotonic dynamics; cell-cycle processes require sustained oscillations. This reflects a lack of biological inductive biases that enforce plausible dynamical regimes.

Motivated by these challenges, we propose a regime-aware trajectory prediction framework (RegimeFlow, see Figure 2), a unified architecture designed to master the diverse landscape of biological dynamics. Our key insight is that, despite massive heterogeneity, biological systems are not arbitrary but exhibit characteristic macroscopic behaviors (e.g., stable, monotonic, or oscillatory patterns) (Andrews et al., 2024; Tyson et al., 2003). Leveraging this, regime-level information is encoded directly in the generative prior, reducing reliance on heuristic regularization. Specifically, we construct regime-aware source distributions using Bayesian linear regression, initializing generation from biologically plausible states rather than isotropic Gaussian noise. This design significantly reduces the optimal transport distance between the source and target distributions, simplifying the learning process. To handle long-horizon dependencies, we employ a linear-complexity state-space backbone (Gu & Dao, 2024), modulated by adaptive conditioning to dynamically align the vector field with regime characteristics while preserving cross-system generalization. We evaluate RegimeFlow on a large collection of real-world biological systems (Figure 1), assessing its ability to generalize in long-horizon trajectory prediction with uncertainty quantification.

Our contributions are summarized as follows:

- We construct SysBio-Traj, a benchmark of 1,050 ODE-based biological systems spanning diverse organisms and pathways, and standardize it in a unified Python framework to support scalable evaluation of AI methods for systems biology.
- We propose RegimeFlow, a regime-aware trajectory prediction framework that encodes biological regime information through structural priors and adaptive conditioning, enabling robust cross-system generalization with long-horizon uncertainty quantification.
- We provide theoretical justification and empirical evidence that regime-aware prior initialization yields a favorable starting distribution for conditional flow matching, leading to state-of-the-art accuracy (31% MAE reduction), well-calibrated uncertainty (17% CRPS improvement), and efficient long-horizon inference.

The resulting SysBio-Traj benchmark is publicly available at `https://huggingface.co/datasets/HengRao/SysBio-Traj`, and the RegimeFlow codebase and configurations are available at `https://github.com/hengrao02/RegimeFlow`.

2. Related Work

2.1. Point Forecasting in Time Series

Advancements in time-series forecasting have produced a diverse family of architectures, each navigating the fundamental trade-off between model expressivity and computational efficiency. High-capacity Transformer-based models (Nie, 2022; Liu et al., 2022; 2023) excel at capturing long-range dependencies through attention mechanisms, but this flexibility often incurs substantial computational and memory overhead, particularly for long sequences (Kim et al., 2025). In contrast, MLP-based models (Zeng et al., 2023; Ekambaram et al., 2023; Wang et al., 2024a) emphasize architectural simplicity and efficiency, though often at the expense of representational power. More recently, State Space Models (SSMs) (Liang et al., 2024; Ahamed & Cheng, 2024; Wang et al., 2025) have emerged as an effective middle ground. By leveraging linear recurrence, they achieve linear-time complexity while retaining the ability to model long-context dependencies (Liu et al., 2025a; Mei et al., 2025). Despite their success in general domains, these approaches face two challenges in biological contexts. First, they are inherently deterministic, producing single-point forecasts without predictive uncertainty that is essential for biological decision-making. Second, they struggle in long-horizon extrapolation under distribution shifts and sparse observations, which further degrade performance when transferring across heterogeneous biological systems.

2.2. Probabilistic Forecasting with Generative Models

Generative modeling has evolved significantly, progressing from the foundational architectures of GANs and VAEs (Kingma & Welling, 2013; Karras et al., 2019; Goodfellow et al., 2020) to the high-fidelity capabilities of Diffusion Probabilistic Models (Sohl-Dickstein et al., 2015). Diffusion-based frameworks (Rasul et al., 2021; Tashiro et al., 2021; Kollovieh et al., 2023) have recently advanced conditional time-series forecasting, enabling uncertainty quantification through trajectory sampling. However, their iterative denoising process incurs prohibitive computational costs in real-world deployment (Lipman et al., 2022).

Flow Matching (FM) (Lipman et al., 2022; Tong et al., 2023b) offers a more efficient alternative by learning continuous probability paths grounded in optimal transport theory (McCann, 1997). By inducing nearly straight vector fields, FM reduces sampling cost and scales well to probabilistic forecasting. Yet, FM-based time-series models (Kollovieh et al., 2024; Hu et al., 2024; Zhang et al., 2024) typically rely on uninformative priors, neglecting critical temporal dependencies. Several recent works have attempted to introduce structural bias into generative forecasting, but limitations persist. TSFlow (Kollovieh et al., 2024) employs dataset-specific Gaussian Process priors that cannot adapt to heterogeneous regimes, while CGFM (Xu et al., 2025) constructs its source distribution using external auxiliary forecasters, increasing computational overhead. In contrast, our framework internalizes prior and conditioning adaptation within a unified architecture, eliminating the need for external predictors and providing robust, self-contained modeling across diverse biological dynamics.

3. Method

Figure 2 illustrates RegimeFlow, a regime-aware trajectory prediction framework for biological models. The key challenge is to generalize long-horizon predictions from short observation windows across heterogeneous systems. We address this by constructing a regime-aware prior and mapping it to target trajectory distributions via conditional flow matching, with adaptive conditioning to align the vector field with regime characteristics.

3.1. Preliminary: Conditional Flow Matching

Our approach builds on the conditional flow matching (CFM) framework (Lipman et al., 2022; Tong et al., 2023a), which constructs a time-dependent vector field to transport a source distribution q_0 to a target distribution q_1. Formally, CFM defines a vector field $u_t : [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$ that governs the trajectory of samples through $d\mathbf{x}_t = u_t(\mathbf{x}_t)\,dt$. This ODE induces a probability path $\{p_t\}_{t\in[0,1]}$ satisfying the boundary conditions $p_0 = q_0$ and $p_1 = q_1$. To circumvent the intractability of the marginal vector field generating p_t, CFM learns a neural approximation v_θ by regressing onto tractable conditional vector fields.

Following the method proposed in (Lipman et al., 2022), we define the conditional probability path via linear interpolation between a source sample $\mathbf{x}_0 \sim q_0$ and a target sample $\mathbf{x}_1 \sim q_1$. The resulting trajectory is expressed as $\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\mathbf{x}_1$ with $t \in [0,1]$.

This choice yields straight conditional paths with constant conditional vector field $u_t(\mathbf{x}_t \mid \mathbf{x}_0, \mathbf{x}_1) = \mathbf{x}_1 - \mathbf{x}_0$. Consequently, learning reduces to a regression problem that matches $v_\theta(t, \mathbf{x}_t)$ to this computable target field:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{\substack{t\sim\mathcal{U}[0,1] \\ \mathbf{x}_0\sim q_0,\mathbf{x}_1\sim q_1}} \left[\|v_\theta(t,\mathbf{x}_t) - (\mathbf{x}_1 - \mathbf{x}_0)\|_2^2\right].$$

3.2. Problem Formulation

We consider a universe of heterogeneous biological ODE systems $\mathcal{S}$. To rigorously evaluate the model's capability to generalize across distinct systems, we partition $\mathcal{S}$ into two disjoint sets: a source training set $\mathcal{S}_{\text{train}}$ and a target set $\mathcal{S}_{\text{test}}$ reserved for evaluation, ensuring that $\mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{test}} = \emptyset$.

In this framework, **biological regime** (denoted by $\mathcal{C}$) refers to a qualitative class of dynamical behavior exhibited by a system over time. Unlike prescribing specific trajectories or microscopic parameters (e.g., reaction rates), regimes characterize the macroscopic pattern that a biological system is constrained to follow. These include sustained oscillations (e.g., circadian rhythms or cell cycles), monotonic trends (e.g., irreversible disease progression or tumor growth), and convergence to stable equilibria (e.g., homeostatic regulation or metabolic processes). Systems within the same regime may differ in transient dynamics, but share similar long-term qualitative behavior.

For any system $S^{(i)} \in \mathcal{S}$, let $\mathbf{y} \in \mathbb{R}^{L+T}$ represent a univariate trajectory composed of a historical observation $\mathbf{y}_p \in \mathbb{R}^L$ and a future trajectory $\mathbf{y}_f \in \mathbb{R}^T$. Our goal is to learn a parameterized model on $\mathcal{S}_{\text{train}}$ that approximates the conditional target distribution $q_1(\mathbf{y}|\mathbf{y}_p, \mathcal{C})$ for unseen systems in $\mathcal{S}_{\text{test}}$, conditioned on the observation and biological regime. The conditional generative process is formalized as:

$$p_\theta(\mathbf{y}|\mathbf{y}_p,\mathcal{C}) = \int p_\theta(\mathbf{y}|\mathbf{x}_0,\mathbf{y}_p,\mathcal{C})q_0(\mathbf{x}_0|\mathbf{y}_p,\mathcal{C})\,d\mathbf{x}_0. \quad (1)$$

We introduce a regime-aware prior $q_0(\mathbf{x}_0|\mathbf{y}_p, \mathcal{C})$ that conditions the source distribution on the biological regime and the observed history via basis functions, thereby simplifying the learning of vector field v_θ. Incorporating this prior into flow matching yields a specialized training objective:

$$\mathcal{L}(\theta) = \mathbb{E}_{t,\mathbf{x}_0,\mathbf{x}_1}\left[\|v_\theta(t,\mathbf{x}_t,\mathbf{y}_p,\mathcal{C}) - (\mathbf{x}_1 - \mathbf{x}_0)\|_2^2\right], \quad (2)$$

where $\mathbf{x}_0 \sim q_0(\cdot|\mathbf{y}_p, \mathcal{C})$ and $\mathbf{x}_1 \sim q_1(\cdot|\mathbf{y}_p, \mathcal{C})$.

3.3. Regime-Aware Prior Construction

In contrast to standard generative frameworks that rely on uninformative Gaussian source distributions, biological trajectories exhibit characteristic macroscopic behaviors, including stable, monotonic, and oscillatory regimes. We encode this information into a regime-aware prior, replacing

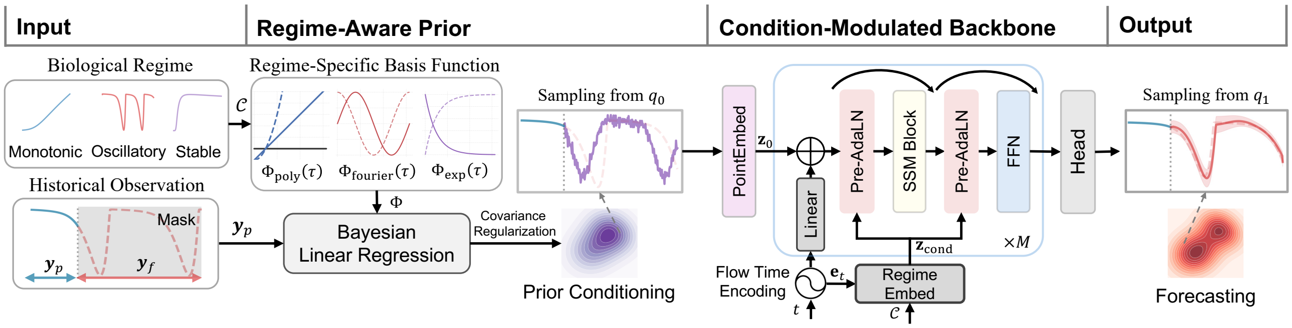

Figure 2. Overview of regime-aware trajectory prediction framework. Given historical observations $\mathbf{y}_p$, RegimeFlow encodes biological regime information via structured basis functions and Bayesian linear regression to infer a regime-aware prior q_0. Sampling from the prior initializes conditional flow matching. The backbone, adaptively conditioned on regime characteristics, learns a vector field v_θ to generate future long-horizon trajectories $\mathbf{y}_p$ via ODE integration.

generic stochastic initialization with structured initial states that are consistent with the system's dynamical patterns. The resulting transport process is biased toward biologically plausible trajectories from the outset.

3.3.1. Design of Regime-Specific Basis Functions

We ground our trajectory modeling in classical function approximation theory. The Stone-Weierstrass Theorem (Stone, 1948) establishes that continuous functions on compact intervals can be uniformly approximated to arbitrary precision by sufficiently rich families of functions, such as polynomial or trigonometric bases. While our construction employs finite, structured basis sets, this result provides theoretical intuition for representing trajectories through basis superpositions.

Accordingly, we model an observed trajectory segment $\mathbf{y}_p$ as a weighted superposition of explicit basis functions, augmented with structural inductive biases to reflect regime priors. We formalize this basis projection as a Bayesian linear regression (BLR) (Box & Tiao, 2011) problem. Let $\mathbf{\Phi}(\tau) = [\phi_1(\tau), \ldots, \phi_K(\tau)]^\top \in \mathbb{R}^K$ denote the vector of basis functions evaluated at the relative time index τ of the trajectory. We specify the **likelihood function** as:

$$\mathbf{y}_p = \mathbf{\Phi}\mathbf{w} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \beta^{-1}\mathbf{I}), \tag{3}$$

where $\mathbf{w} \in \mathbb{R}^K$ is the weight vector, β denotes the noise precision, and $\boldsymbol{\epsilon}$ accounts for both aleatoric measurement noise and approximation errors due to basis truncation.

We construct the basis $\mathbf{\Phi}(\tau)$ as a composite library of functional primitives, explicitly tailored to span canonical biological dynamical regimes. **(i)** For stable trajectories characterized by asymptotic relaxation toward steady states, we employ a dictionary of exponential decay terms $\{1 - e^{-\gamma_j \tau}\}_j$ over a discretized set of rates $\{\gamma_j\}$, with scaling absorbed by the regression weights. **(ii)** To represent oscillatory patterns, we adopt a Fourier basis $\{\sin(k\omega\tau), \cos(k\omega\tau)\}_k$, where the fundamental frequency ω is informed by domain knowledge of the biological system when available (e.g., circadian rhythms exhibit a 24-hour period). **(iii)** Monotonic trends are captured using polynomial basis $\{1, \tau, \tau^2, ...\}$. In complex scenarios where the dynamic pattern does not fall into any of the predefined regime categories, we revert to the degenerate basis $\{1\}$, yielding a time-invariant isotropic Gaussian prior. This fallback avoids imposing spurious structure when regime information is ambiguous. Regime labels are assigned per coordinate using an automated detector followed by manual validation; details are provided in Appendix C.4.

3.3.2. Bayesian Inference

Given the Gaussian likelihood and the conjugate isotropic prior $p(\mathbf{w}) = \mathcal{N}(\mathbf{0}, \alpha^{-1}\mathbf{I})$, the posterior distribution $p(\mathbf{w} \mid \mathbf{y}_p)$ remains Gaussian and admits a closed-form solution:

$$p(\mathbf{w} \mid \mathbf{y}_p) = \mathcal{N}(\mathbf{w}; \boldsymbol{\mu}_{\mathbf{w}}, \mathbf{\Sigma}_{\mathbf{w}}). \tag{4}$$

The posterior covariance and mean are given by

$$\mathbf{\Sigma}_{\mathbf{w}} = \left(\alpha\mathbf{I} + \beta\mathbf{\Phi}_p^\top\mathbf{\Phi}_p\right)^{-1}, \quad \boldsymbol{\mu}_{\mathbf{w}} = \beta\mathbf{\Sigma}_{\mathbf{w}}\mathbf{\Phi}_p^\top\mathbf{y}_p, \tag{5}$$

where $\mathbf{\Phi}_p \in \mathbb{R}^{L\times K}$ denotes the design matrix formed by evaluating K basis functions at the L observed time points. α and β denote the prior precision and the observation noise, respectively. As a basis for our prior, we first derive the standard Bayesian predictive distribution by marginalizing over the weight uncertainty. Define $\mathbf{\Phi}_f \in \mathbb{R}^{T\times K}$ as the basis matrix evaluated over the future T time points. The resulting distribution is:

$$\mathcal{N}(\mathbf{x}_0; \mathbf{\Phi}_f\boldsymbol{\mu}_{\mathbf{w}},\ \beta^{-1}\mathbf{I} + \mathbf{\Phi}_f\mathbf{\Sigma}_{\mathbf{w}}\mathbf{\Phi}_f^\top). \tag{6}$$

The predictive trajectory mean $\boldsymbol{\mu}_{\text{prior}} = \mathbf{\Phi}_f\boldsymbol{\mu}_{\mathbf{w}}$ provides a regime-informed extrapolation of the observed trajectory that reflects the macroscopic biological behavior implied by the chosen basis functions.

3.3.3. Covariance Regularization

While the Bayesian linear regression provides a full covariance matrix, directly using this dense structure for long-horizon prediction ($T > L$) can be problematic. In particular, finite basis extrapolation over extended horizons may induce numerical instability and inflated variance estimates (Bishop & Nasrabadi, 2006), potentially misleading the generative process through poorly scaled priors. To address this instability, we retain the predictive mean $\boldsymbol{\mu}_{\text{prior}}$ and regularize the covariance used to initialize the flow. We replace the anisotropic predictive covariance with an isotropic approximation to improve numerical stability.

This simplification is supported by the Wasserstein-2 decomposition in Appendix A (Eq. 19), which separates the effect of mean alignment from the centered distribution term. Thus, fixing an isotropic covariance preserves the BLR mean guidance while stabilizing the source distribution; the empirical comparison with the full BLR covariance is provided in Appendix Table 3.

Formally, we define the regime-aware prior as

$$q_0(\mathbf{x}_0|\mathbf{y}_p, \mathcal{C}) = \mathcal{N}(\mathbf{x}_0; \boldsymbol{\mu}_{\text{prior}}, \sigma^2_{\text{init}}\mathbf{I}), \tag{7}$$

where σ^2_{init} is a scalar variance hyperparameter.

This covariance regularization preserves the extrapolated trend of the trajectory mean while avoiding numerical artifacts arising from long-range uncertainty propagation. Empirically, the residuals $(\mathbf{y}_f - \boldsymbol{\mu}_{\text{prior}})$ are approximately Gaussian, as shown in Appendix Figure 9.

3.3.4. Theoretical Justification

We provide a geometric characterization of the proposed prior, demonstrating that aligning the prior with the regime subspace yields provable advantages. Specifically, we establish that under an explicit Signal-to-Noise Ratio (SNR) condition, our approach strictly reduces the expected Optimal Transport cost relative to the standard Gaussian.

First, we analyze the estimator's variance. Let $\boldsymbol{\Sigma}_{\text{design}} := \boldsymbol{\Phi}_f(\boldsymbol{\Phi}_p^\top\boldsymbol{\Phi}_p)^{-1}\boldsymbol{\Phi}_f^\top$ denote the predictive covariance matrix determined solely by the basis geometry.

Proposition 3.1 (Error Reduction under Regime-Aware Initialization)**.** *In the limit of vanishing regularization ($\alpha \to 0$), the regime-aware prior yields a strictly lower mean squared error for trajectory initialization than the uninformative Gaussian prior, provided the following SNR condition holds:*

$$\frac{\|\boldsymbol{\mu}_Q\|^2}{\sigma^2_\epsilon} > \mathrm{Tr}(\boldsymbol{\Sigma}_{\text{design}}), \tag{8}$$

where $\|\boldsymbol{\mu}_Q\|^2$ denotes the energy of the target mean trajectory and $\sigma^2_\epsilon = \beta^{-1}$ is the observation noise variance.

Proposition 3.1 establishes that when the projected trajectory energy dominates the extrapolation variance induced by the basis geometry, regime-aware initialization improves estimation accuracy. Geometrically, this corresponds to initializing the flow closer to the target manifold, thereby shortening the transport path required for generation.

Proposition 3.2 (Reduction of Expected Wasserstein-2 Cost)**.** *Under the condition in Proposition 3.1, for any target distribution Q with finite variance, the regime-aware prior P_{BLR} yields a strictly smaller expected Wasserstein-2 distance to Q than an uninformative Gaussian prior P_{GA}:*

$$\mathbb{E}_{\mathbf{y}_p}\left[W_2^2(P_{\mathrm{BLR}}(\mathbf{y}_p), Q)\right] < W_2^2(P_{\mathrm{GA}}, Q)\,. \tag{9}$$

Detailed proofs are provided in Appendix A, complemented by empirical verification in Appendix B.1. Together, these results formalize how regime-aware initialization reduces the geometric complexity of the transport problem by providing an initial trajectory distribution closer to the target dynamics. By effectively contracting the transport distance, this approach alleviates the model burden to learn highly complex mappings, thereby improving optimization stability and mitigating the ill-posedness commonly encountered in long-horizon forecasting.

A Computable SNR Criterion. The SNR condition in Proposition 3.1 depends on the unknown target mean $\boldsymbol{\mu}_Q$, and therefore cannot be checked directly at test time. We introduce a computable surrogate based on the observed history and the selected basis family:

$$\hat{\Delta}(\mathbf{y}_p) := \frac{\|\boldsymbol{\mu}_{\text{prior}}(\mathbf{y}_p)\|^2}{\hat{\sigma}^2_\epsilon} - \mathrm{Tr}(\boldsymbol{\Sigma}_{\text{design}}), \tag{10}$$

where $\boldsymbol{\mu}_{\text{prior}}(\mathbf{y}_p)$ is the predicted trajectory mean, used as a plug-in estimate of the target mean, and $\hat{\sigma}^2_\epsilon$ is estimated from the residuals obtained when fitting the observed window. This diagnostic operationalizes the SNR condition using quantities available from the observed history and the geometry of the selected basis. A positive value suggests that the regime-specific basis can provide an informative source initialization, whereas a non-positive value indicates potential regime mismatch or insufficient confidence, in which case regime-agnostic inference serves as a natural fallback or a more refined basis family may be warranted.

3.4. Condition-Modulated Backbone

To parameterize the neural vector field v_θ, we propose the Condition-Modulated backbone. This backbone harnesses the linear time complexity of Selective State Space Model (SSM) (Gu & Dao, 2024) to capture long-range temporal dependencies efficiently.

3.4.1. Input Embedding and Condition Encoding

We employ a channel-independent architecture to handle biological models of varying dimensions. Each dimension is processed as a univariate sequence derived from the intermediate flow state $\mathbf{x}_t$, while all dimensions share the same backbone parameters. This flow state $\mathbf{x}_t$, along with the flow time t and biological regime $\mathcal{C}$, is projected into a shared latent space $\mathbb{R}^D$. The univariate input is encoded into the latent space $\mathbf{z}_0$ via an MLP-based point embedding: $\mathbf{z}_0 = \text{MLP}_{\text{emd}}(\mathbf{x}_t)$. Concurrently, the flow time $t \in [0, 1]$ is encoded via sinusoidal positional embeddings to yield $\mathbf{e}_t \in \mathbb{R}^D$ (Kollovieh et al., 2024). The initial hidden state of the backbone ($\mathbf{h}_0$) is formed by injecting a layer-specific time bias: $\mathbf{h}_0 = \mathbf{z}_0 + \text{Linear}_{\text{time}}(\mathbf{e}_t)$, where $\text{Linear}_{\text{time}}$ denotes the time linear projection.

To encode the biological regime $\mathcal{C} = (c_{\text{pat}}, c_{\text{freq}})$, we retrieve pattern embeddings $\mathbf{e}_{\text{pat}}$ for the discrete pattern variable $c_{\text{pat}} \in \{1, \ldots, N_{\text{pat}}\}$ from a learnable matrix, and project the continuous frequency $c_{\text{freq}} \in \mathbb{R}_{\geq 0}$ into embeddings $\mathbf{e}_{\text{freq}}$ via a Fourier feature mapping with learnable frequencies (Tancik et al., 2020).

To facilitate classifier-free guidance (Ho & Salimans, 2022), regime embeddings (pattern and oscillatory frequency) are randomly masked during training and replaced with learnable null tokens, enabling the model to learn an explicit representation for missing regime information and capture both regime-conditioned and regime-agnostic dynamics. The processed regime features $(\widetilde{\mathbf{e}}_{\text{pat}}, \widetilde{\mathbf{e}}_{\text{freq}})$ are concatenated with the flow time embedding $\mathbf{e}_t$ and projected into a global conditioning vector: $\mathbf{z}_{\text{cond}} = \text{MLP}_{\text{cond}}([\widetilde{\mathbf{e}}_{\text{pat}}, \widetilde{\mathbf{e}}_{\text{freq}}, \mathbf{e}_t])$. $\mathbf{z}_{\text{cond}}$ serves to regulate the vector field dynamics.

3.4.2. Condition-Modulated Layer

The backbone consists of M stacked condition-modulated layers designed to integrate regime prior directly into the latent dynamics. Each layer utilizes a pre-normalized residual structure, comprising an SSM block followed by a Feed-Forward Network (FFN). Conditioning is achieved using Adaptive Layer Normalization (AdaLN) (Perez et al., 2018), which modulates intermediate representations based on the global conditioning vector $\mathbf{z}_{\text{cond}}$. For the l-th layer, AdaLN computes dimension-wise scale ($\mathbf{s}_l$) and shift ($\mathbf{d}_l$) from $\mathbf{z}_{\text{cond}}$ and applies them to normalized hidden states:

$$\begin{aligned} (\mathbf{s}_l, \mathbf{d}_l) &= \text{Linear}_{\text{ada}}^{(l)}(\mathbf{z}_{\text{cond}}), \\ \text{AdaLN}(\mathbf{h}, \mathbf{z}_{\text{cond}}) &= \text{RMSNorm}(\mathbf{h}) \odot (1 + \mathbf{s}_l) + \mathbf{d}_l. \end{aligned} \tag{11}$$

Following (Peebles & Xie, 2023), we initialize the projection $\text{Linear}_{\text{ada}}^{(l)}$ with zeros, starting the modulation as an identity transformation ($\mathbf{s}_l = \mathbf{0}, \mathbf{d}_l = \mathbf{0}$), allowing the backbone to initially train in a regime-agnostic manner and improving convergence stability.

With this modulation, the state updates for the l-th layer are:

$$\begin{aligned} \mathbf{h}'_l &= \mathbf{h}_{l-1} + \text{SSM}\left(\text{AdaLN}(\mathbf{h}_{l-1}, \mathbf{z}_{\text{cond}})\right), \\ \mathbf{h}_l &= \mathbf{h}'_l + \text{FFN}\left(\text{AdaLN}(\mathbf{h}'_l, \mathbf{z}_{\text{cond}})\right). \end{aligned} \tag{12}$$

Finally, to preserve multi-scale representations across network depths, the target vector field is predicted by decoding the aggregated sum of all layer outputs, rather than relying solely on the final hidden state.

4. Experiments

We evaluate RegimeFlow using our benchmark designed to rigorously assess long-horizon trajectory inference across diverse biological systems. All reported metrics are presented as the mean and standard deviation ($\mu \pm \sigma$) over three independent seeds, providing a quantitative measure of predictive stability and robustness.

4.1. Benchmark Construction and Task Formulation

To evaluate performance across the broad spectrum of biological complexity, we assemble SysBio-Traj, a large-scale benchmark of 1,050 ODE systems spanning diverse taxonomic domains and dynamical regimes sourced from (Malik-Sheriff et al., 2020) (see Appendix D for the full list of biological systems). As illustrated in Appendix Figure 15, this benchmark exhibits substantial heterogeneity, with system dimensions ranging from 1 to 1,000 and reaction counts spanning three orders of magnitude. For each system, ground-truth trajectories are generated by numerically solving the ODEs and sampling the solution at 512 uniformly spaced time points.

To emulate real-world scenarios where observational data is limited, we formulate a challenging long-horizon forecasting task: predicting trajectories over $T = 256$ time points from a given $L = 96$ observed points. Specifically for oscillatory systems, we forecast one full cycle from observations covering only part of a cycle, testing the model's ability to extrapolate full oscillatory dynamics when periodicity is not explicitly apparent. We adopt a randomized system-wise train/validation/test split (70%/10%/20%), such that models are trained and evaluated on disjoint sets of biological systems. This setup enforces a rigorous out-of-distribution (OOD) generalization. To accommodate the wide variation in system dimensionality, we employ a channel-independent modeling strategy with instance-wise normalization.

4.2. Baselines and Implementation Details

We compare our framework against two prominent paradigms: point forecasting and probabilistic generative models. Point forecasting baselines include MLP-based models such as DLinear (Zeng et al., 2023) and TimeMixer (Wang et al., 2024a), Transformer-based models

Table 1. Cross-system generalization performance on MSE and MAE across regime-specific subsets, complex subset outside the three regimes, and overall. Each entry reports the mean with standard deviation. Best and second-best performances are highlighted in bold and <u>underline</u>, respectively. Models marked with † additionally condition on biological regime, including pattern and frequency.

	Overall		Complex		Stable		Oscillatory		Monotonic	
Model	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓
Point Forecasting										
DLinear	$0.061_{\pm0.002}$	$0.174_{\pm0.004}$	$0.079_{\pm0.003}$	$0.188_{\pm0.010}$	$0.046_{\pm0.001}$	$0.154_{\pm0.005}$	$0.156_{\pm0.013}$	$0.290_{\pm0.011}$	$0.051_{\pm0.005}$	$0.176_{\pm0.013}$
iTransformer	$0.045_{\pm0.002}$	$0.100_{\pm0.005}$	$0.070_{\pm0.010}$	$0.138_{\pm0.020}$	$0.032_{\pm0.002}$	$0.072_{\pm0.004}$	$0.127_{\pm0.028}$	$0.233_{\pm0.018}$	$0.030_{\pm0.008}$	$0.100_{\pm0.018}$
PatchTST	$0.041_{\pm0.001}$	$0.095_{\pm0.005}$	$0.062_{\pm0.006}$	$0.133_{\pm0.021}$	$0.028_{\pm0.001}$	$0.070_{\pm0.004}$	$0.128_{\pm0.024}$	$0.233_{\pm0.016}$	$0.021_{\pm0.006}$	$0.085_{\pm0.016}$
TimeMixer	$0.051_{\pm0.004}$	$0.124_{\pm0.008}$	$0.070_{\pm0.008}$	$0.145_{\pm0.017}$	$0.038_{\pm0.004}$	$0.096_{\pm0.009}$	$0.112_{\pm0.013}$	$0.234_{\pm0.014}$	$0.053_{\pm0.010}$	$0.163_{\pm0.020}$
TimeXer	$0.043_{\pm0.001}$	$0.100_{\pm0.007}$	$0.069_{\pm0.009}$	$0.139_{\pm0.020}$	$0.030_{\pm0.003}$	$0.074_{\pm0.008}$	$0.123_{\pm0.028}$	$0.231_{\pm0.018}$	$0.025_{\pm0.006}$	$0.098_{\pm0.015}$
SMamba	$0.045_{\pm0.001}$	$0.105_{\pm0.004}$	$0.068_{\pm0.010}$	$0.141_{\pm0.023}$	$0.033_{\pm0.002}$	$0.078_{\pm0.003}$	$0.114_{\pm0.016}$	$0.228_{\pm0.013}$	$0.035_{\pm0.005}$	$0.115_{\pm0.010}$
BiMamba4TS	$0.045_{\pm0.001}$	$0.099_{\pm0.004}$	$0.070_{\pm0.010}$	$0.140_{\pm0.021}$	$0.031_{\pm0.002}$	$0.070_{\pm0.004}$	$0.125_{\pm0.022}$	$0.237_{\pm0.015}$	$0.030_{\pm0.005}$	$0.102_{\pm0.013}$
NSformer	$0.028_{\pm0.002}$	$0.079_{\pm0.005}$	$0.048_{\pm0.002}$	$0.116_{\pm0.007}$	$0.017_{\pm0.001}$	$0.055_{\pm0.005}$	$0.100_{\pm0.008}$	$0.221_{\pm0.006}$	$0.011_{\pm0.001}$	$0.053_{\pm0.002}$
Time-Series Foundation Models (Zero-Shot)										
TimeMoE	$0.120_{\pm0.004}$	$0.231_{\pm0.010}$	$0.099_{\pm0.013}$	$0.191_{\pm0.032}$	$0.118_{\pm0.003}$	$0.221_{\pm0.010}$	$0.138_{\pm0.005}$	$0.268_{\pm0.005}$	$0.138_{\pm0.012}$	$0.289_{\pm0.023}$
Sundial	$0.111_{\pm0.003}$	$0.217_{\pm0.007}$	$0.095_{\pm0.012}$	$0.185_{\pm0.031}$	$0.106_{\pm0.002}$	$0.203_{\pm0.008}$	$0.141_{\pm0.009}$	$0.270_{\pm0.009}$	$0.133_{\pm0.011}$	$0.281_{\pm0.022}$
TimesFM	$0.099_{\pm0.005}$	$0.152_{\pm0.004}$	$0.109_{\pm0.009}$	$0.145_{\pm0.021}$	$0.086_{\pm0.002}$	$0.138_{\pm0.006}$	$0.249_{\pm0.032}$	$0.299_{\pm0.014}$	$0.045_{\pm0.004}$	$0.112_{\pm0.007}$
Chronos	$0.097_{\pm0.004}$	$0.167_{\pm0.000}$	$0.144_{\pm0.021}$	$0.181_{\pm0.012}$	$0.078_{\pm0.003}$	$0.147_{\pm0.002}$	$0.234_{\pm0.013}$	$0.309_{\pm0.007}$	$0.054_{\pm0.004}$	$0.153_{\pm0.005}$
Probabilistic Forecasting w/o Regime Condition (Best Performing: Ours)										
CSDI	$0.099_{\pm0.020}$	$0.254_{\pm0.034}$	$0.100_{\pm0.030}$	$0.250_{\pm0.048}$	$0.103_{\pm0.022}$	$0.264_{\pm0.038}$	$0.137_{\pm0.012}$	$0.304_{\pm0.014}$	$0.053_{\pm0.013}$	$0.172_{\pm0.030}$
TSDiff	$0.042_{\pm0.003}$	$0.118_{\pm0.004}$	$0.058_{\pm0.007}$	$0.146_{\pm0.014}$	$0.030_{\pm0.003}$	$0.096_{\pm0.004}$	$0.124_{\pm0.010}$	$0.263_{\pm0.012}$	$0.023_{\pm0.001}$	$0.092_{\pm0.004}$
TSFlow	$0.026_{\pm0.003}$	$0.075_{\pm0.011}$	$0.044_{\pm0.001}$	$0.111_{\pm0.009}$	$0.017_{\pm0.004}$	$0.055_{\pm0.012}$	$0.088_{\pm0.007}$	$0.208_{\pm0.007}$	$0.008_{\pm0.002}$	$0.044_{\pm0.008}$
Ours	$0.023_{\pm0.002}$	$0.067_{\pm0.004}$	<u>0.039</u>$_{\pm0.001}$	<u>0.102</u>$_{\pm0.008}$	$0.015_{\pm0.002}$	<u>0.046</u>$_{\pm0.003}$	$0.081_{\pm0.008}$	$0.196_{\pm0.009}$	$0.008_{\pm0.002}$	<u>0.041</u>$_{\pm0.005}$
Probabilistic Forecasting w/ Regime Condition (Best Performing: Ours)										
CSDI†	$0.076_{\pm0.010}$	$0.204_{\pm0.012}$	$0.100_{\pm0.009}$	$0.236_{\pm0.011}$	$0.072_{\pm0.015}$	$0.196_{\pm0.019}$	$0.105_{\pm0.011}$	$0.257_{\pm0.024}$	$0.060_{\pm0.024}$	$0.183_{\pm0.046}$
TSDiff†	$0.034_{\pm0.008}$	$0.109_{\pm0.017}$	$0.088_{\pm0.033}$	$0.201_{\pm0.054}$	$0.014_{\pm0.003}$	$0.068_{\pm0.011}$	$0.118_{\pm0.002}$	$0.292_{\pm0.001}$	$0.014_{\pm0.001}$	$0.078_{\pm0.009}$
TSFlow†	<u>0.016</u>$_{\pm0.004}$	<u>0.064</u>$_{\pm0.013}$	$0.039_{\pm0.005}$	$0.111_{\pm0.022}$	<u>0.008</u>$_{\pm0.002}$	$0.046_{\pm0.010}$	<u>0.044</u>$_{\pm0.005}$	<u>0.139</u>$_{\pm0.017}$	<u>0.008</u>$_{\pm0.001}$	$0.050_{\pm0.005}$
Ours†	$\mathbf{0.012}_{\pm0.002}$	$\mathbf{0.044}_{\pm0.006}$	$\mathbf{0.034}_{\pm0.002}$	$\mathbf{0.096}_{\pm0.011}$	$\mathbf{0.006}_{\pm0.001}$	$\mathbf{0.027}_{\pm0.003}$	$\mathbf{0.035}_{\pm0.002}$	$\mathbf{0.111}_{\pm0.008}$	$\mathbf{0.006}_{\pm0.000}$	$\mathbf{0.033}_{\pm0.002}$

Table 2. Uncertainty quantification measured by CRPS (↓) across regime-specific subsets, complex subset, and overall. Best results are in **bold**. (†: condition on biological regime.)

Model	Overall	Complex	Stable	Oscillatory	Monotonic
CSDI†	0.138 (±0.013)	0.166 (±0.018)	0.127 (±0.019)	0.186 (±0.015)	0.135 (±0.040)
TSDiff†	0.077 (±0.015)	0.151 (±0.049)	0.048 (±0.009)	0.198 (±0.001)	0.057 (±0.006)
TSFlow†	0.047 (±0.010)	0.086 (±0.016)	0.034 (±0.008)	0.100 (±0.011)	0.037 (±0.003)
Ours†	**0.039** (±0.006)	**0.086** (±0.010)	**0.023** (±0.003)	**0.098** (±0.007)	**0.028** (±0.002)

including iTransformer (Liu et al., 2023), PatchTST (Nie, 2022), TimeXer (Wang et al., 2024b), and NSformer (Liu et al., 2022), as well as state-space models such as SMamba (Wang et al., 2025) and BiMamba4TS (Liang et al., 2024). For probabilistic baselines, we consider diffusion models, TSDiff (Kollovieh et al., 2023) and CSDI (Tashiro et al., 2021), and flow matching methods, TSFlow (Kollovieh et al., 2024). We additionally evaluate four zero-shot time-series foundation models (TSFMs), including TimeMoE (Shi et al., 2025), Sundial (Liu et al., 2025b), TimesFM (Das et al., 2024), and Chronos (Ansari et al., 2024); detailed settings and analysis are provided in Appendix B.7. Full baseline descriptions and implementation settings are provided in Appendix C.

We assess point prediction accuracy using mean squared error (MSE) and mean absolute error (MAE). To evaluate distributional fidelity, we employ the continuous ranked probability score (CRPS) (Gneiting & Raftery, 2007), which is approximated using 100 Monte Carlo samples per test instance. All experiments were implemented in PyTorch and executed on a single NVIDIA A6000 GPU.

4.3. Main Results

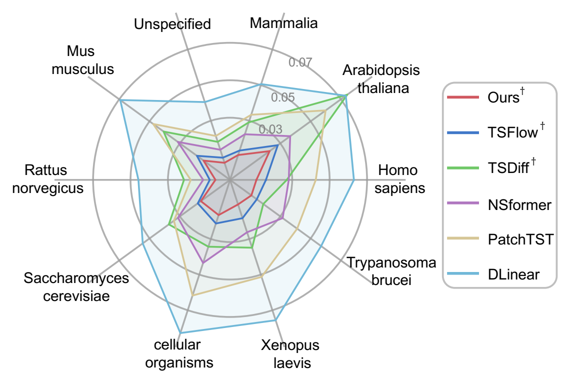

Figure 3. MSE comparison between Ours† and five competitive baselines across ten most frequent biological system taxonomies.

Table 1 summarizes the forecasting performance on unseen biological systems. When incorporating regime information (denoted by †), our method consistently outperforms all

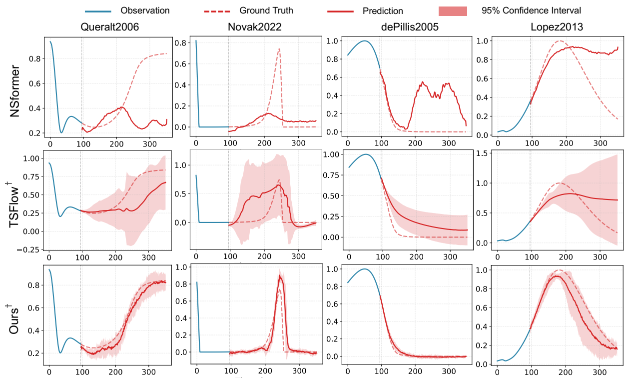

Figure 4. Visualization of trajectory forecasts by Ours†, TSFlow†, and NSformer on four representative biological systems.

SOTA baselines across individual regimes, complex systems outside the three canonical regimes, and overall. It achieves overall reductions of 25% in MSE and 31% in MAE relative to the best-performing baseline, TSFlow†. These improvements persist across diverse regimes (MSE: 14%$\sim$32%), underscoring our superior capability for cross-system generalization. Figure 3 further shows our method attains the lowest MSE across all ten most frequent biological system taxonomies compared to baselines.

To demonstrate robustness and uncertainty across distinct dynamical behaviors, we visualize predictions on four representative biological systems: bistable mitotic exit switch (Queralt et al., 2006), oscillatory endoreplication (Novak & Tyson, 2022), tumor-immune interactions (de Pillis et al., 2005), and a high-dimensional apoptosis model (Lopez et al., 2013). As shown in Figure 4, our framework produces the most accurate predictions with well-calibrated uncertainty. In contrast, TSFlow† exhibits degraded predictive performance with substantially wider confidence intervals, likely due to its reliance on fixed, system-agnostic priors, while NSformer struggles to accommodate diverse regimes. Consistent with these qualitative observations, our approach achieves an average 17% reduction in CRPS relative to the strongest baseline, TSFlow† (Table 2), demonstrating superior uncertainty calibration and closer alignment with the target distribution.

Notably, even without regime information, our framework remains competitive and consistently outperforms all baselines across individual subsets and overall. RegimeFlow reduces MSE and MAE by 12% and 11% relative to the best-performing baseline, TSFlow. This advantage likely stems from the input-dependent dynamics of the SSM-based backbone (Gu et al., 2021; Gu & Dao, 2024). Point forecasting architectures, while effective on single-domain benchmarks (Lai et al., 2018; Makridakis et al., 2020), exhibit fundamental limitations in this cross-system setting and often collapse to trivial mean trends across diverse regimes.

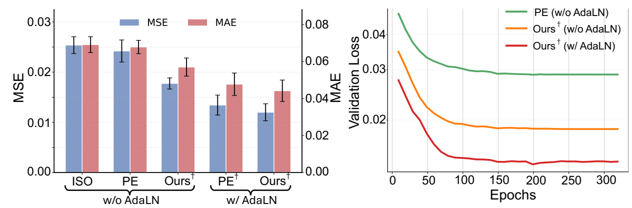

Figure 5. **Left:** Ablation analysis of prior designs (ISO, PE, Our regime-aware) and adaptive conditioning (AdaLN). **Right:** Validation loss under different prior initializations and conditioning.

Among point forecasting baselines, NSformer achieves the strongest performance, likely due to its series stationarization and de-stationary attention mechanisms, which enhance robustness to scale shifts. Conversely, methods that rely heavily on inter-variable correlations, such as iTransformer (Liu et al., 2023), are disadvantaged as biological coupling relationships often vary drastically across systems.

Table 1 also reports zero-shot results from four representative TSFMs. Despite their larger parameter scales (128–200 M), these models remain substantially less accurate on SysBio-Traj; the strongest TSFM, Chronos, obtains an overall MSE of 0.097, compared with 0.012 for Ours†. Detailed analyses are provided in Appendix B.7.

4.4. Effectiveness of Regime-Aware Conditioning

To assess the contribution of each component, we independently ablate the regime-aware prior and adaptive conditioning design. In Figure 5 (left), replacing the Isotropic (ISO) kernel with the Periodic (PE) kernel yields only a marginal improvement in accuracy. In contrast, substituting the above kernels with our regime-aware prior leads to a substantially larger reduction in error (MSE: 0.025$\rightarrow$0.018), indicating that our prior provides more informative guidance for modeling complex dynamical behaviors than dataset-specific priors. Under our prior, introducing adaptive conditioning (AdaLN) further reduces the error (MSE: 0.018$\rightarrow$0.012), highlighting its importance in leveraging prior information. Figure 5 (right) shows that our initialization converges to a consistently lower validation loss than the PE kernel under identical model settings, highlighting the optimization benefits of the regime-aware prior.

4.5. Sensitivity, Efficiency, and Extrapolation Analysis

We first assess the model's sensitivity to the regime information availability. During training, regime information is randomly masked as a form of regularization, discouraging over-reliance on explicit prior metadata and encouraging the backbone to infer regime-relevant features directly from raw time-series observations. When regime information is masked, system dynamics are treated as unknown and the source distribution is initialized from the observed history as

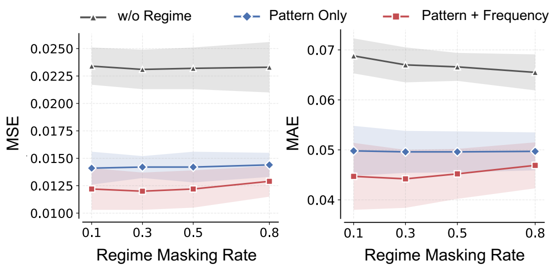

Figure 6. Sensitivity to regime information availability. Our model is trained with different regime masking rates and tested under varying regime conditions (no regime, pattern, pattern & frequency).

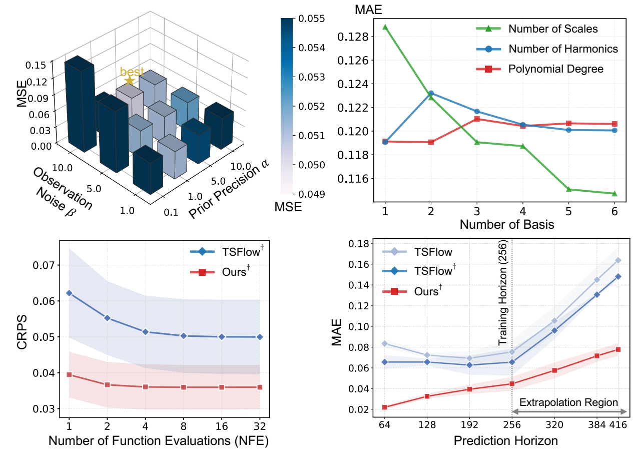

Figure 7. **Top left:** Sensitivity to BLR hyperparameters. **Top right:** Effect of basis complexity (numbers of harmonics, scales, and polynomial degrees). **Bottom left:** Inference efficiency (CRPS vs. Number of Function Evaluation). **Bottom right:** Extrapolation performance under extended forecasting horizons.

$\mathcal{N}(\mathbf{y}_L, \sigma^2_{\text{init}}\mathbf{I})$. Figure 6 reveals a trade-off: higher masking improves regime-agnostic accuracy but slightly degrades regime-aware performance. A masking rate of 0.3 offers a good balance, maintaining low regime-aware error while preserving competitive regime-agnostic performance.

We further investigate the sensitivity of Bayesian linear regression (BLR) hyperparameters. Figure 7 (top left) identifies prior precision $\alpha = 5.0$ and observation noise $\beta = 10.0$ as the optimal choices. The top-right panel shows that a quadratic polynomial coupled with a single harmonic component offers sufficient performance, while adding additional scales further improves accuracy. In terms of inference efficiency (bottom left), our method achieves near-optimal CRPS with a single function evaluation step, significantly outperforming TSFlow†, which requires ≥ 16 steps. This efficiency suggests that the proposed prior initializes the flow closer to the target distribution, thereby reducing the complexity of ODE integration. A full computational cost comparison across all models is provided in Appendix B.6, where RegimeFlow is 9.3× faster than TSFlow† during inference with a single function evaluation. Finally, the extrapolation analysis (bottom right) demonstrates that our method maintains low error when extending the prediction horizon from 256 to 416 time steps. In contrast, TSFlow exhibits a much steeper error increase, highlighting the improved robustness of regime-aware initialization under horizon shifts.

4.6. Additional Robustness Studies

Beyond the main benchmark results and sensitivity analyses discussed above, Appendix B provides supplementary stress tests and component-level ablations. Under measurement noise and irregular sampling, RegimeFlow maintains consistent advantages over TSFlow† (Appendix B.10); on representative transient and hybrid trajectories, it accommodates dynamics beyond canonical regimes, supporting that the regime-aware prior acts as soft guidance rather than a rigid constraint (Appendix B.8).

The appendix also provides component-level ablations on covariance regularization, backbone choice, and TSFlow kernel configurations (Appendix Sections B.2, B.3, and B.4). For the NFE, extrapolation, and theory-bound analyses discussed above, Appendix Sections B.5, B.9, and B.1 provide detailed protocols and additional quantitative results. Together, these studies show that RegimeFlow remains effective beyond the clean benchmark setting, while the ablations confirm the complementary benefits of the regime-aware prior and sequence backbone.

5. Conclusion

This work introduced RegimeFlow, a unified framework for cross-system biological trajectory prediction. By replacing an uninformative source distribution with a regime-aware prior, RegimeFlow improves long-horizon forecasting accuracy, uncertainty calibration, and inference efficiency across SysBio-Traj, a large curated benchmark of biological systems. Beyond model performance, SysBio-Traj provides a standardized testbed for evaluating cross-system generalization in systems biology. Several limitations motivate future work. RegimeFlow currently adopts a channel-independent formulation to support transfer across systems with heterogeneous dimensions and system-specific interaction topologies, leaving topology-aware relational modeling as an important extension. Although representative transient and hybrid behaviors can be handled through soft regime guidance (Appendix B.8), parameter-dependent regime transitions and highly irregular or chaotic dynamics may require more adaptive, data-driven regime priors beyond the current basis library. Future work will also extend SysBio-Traj and RegimeFlow toward stochastic, spatio-temporal, and multi-modal biological settings.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning and its application to Systems Biology. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Theoretical Analysis and Proofs

In this appendix, we formally derive the theoretical conditions under which the regime-aware prior yields a strictly lower estimation risk than an uninformative baseline. We focus our analysis on the limit of vanishing regularization ($\alpha \to 0$). In this regime, the prediction is governed solely by the projection of the history onto the mechanism subspace, allowing us to isolate and quantify the geometric advantage of the proposed framework.

A.1. Proof of Proposition 3.1

Proposition 3.1 (Error Reduction under Regime-Aware Initialization). *Consider the observation regime where the history length L is sufficient to identify the K basis coefficients ($L \geq K$ and $\mathbf{\Phi}_p$ has full column rank). Let $\mathbf{\Sigma}_{\text{design}} \coloneqq \mathbf{\Phi}_f(\mathbf{\Phi}_p^\top \mathbf{\Phi}_p)^{-1}\mathbf{\Phi}_f^\top$ denote the predictive covariance matrix determined solely by the basis geometry. In the limit of vanishing regularization ($\alpha \to 0$), the Regime-Aware Prior strictly reduces the Mean Squared Error (MSE) compared to the uninformed zero-mean prior if and only if the Signal-to-Noise Ratio (SNR) satisfies:*

$$\frac{\|\boldsymbol{\mu}_Q\|^2}{\sigma_\epsilon^2} > \mathrm{Tr}(\mathbf{\Sigma}_{\text{design}}), \tag{13}$$

where $\|\boldsymbol{\mu}_Q\|^2$ denotes the energy of the ground truth trajectory, and σ_ϵ^2 represents the variance of the observation noise.

Proof. We assess the efficacy of the regime-aware prior by comparing the MSE of the proposed estimator against that of the zero-mean baseline. For the uninformative estimator $\hat{\boldsymbol{\mu}}_{\text{GA}} = \mathbf{0}$, the risk corresponds strictly to the energy of the ground truth trajectory $\boldsymbol{\mu}_Q = \mathbf{\Phi}_f \mathbf{w}^*$:

$$R_{\text{GA}} \coloneqq \mathbb{E}_{\mathbf{y}_p}[\|\mathbf{0} - \boldsymbol{\mu}_Q\|^2] = \|\boldsymbol{\mu}_Q\|^2. \tag{14}$$

Next, consider the regime-aware prior $\boldsymbol{\mu}_{\text{prior}} = \mathbf{\Phi}_f \hat{\mathbf{w}}$. In the limit of vanishing regularization ($\alpha \to 0$), then:

$$\hat{\mathbf{w}} = (\mathbf{\Phi}_p^\top \mathbf{\Phi}_p)^{-1}\mathbf{\Phi}_p^\top \mathbf{y}_p. \tag{15}$$

Under the linear observation model $\mathbf{y}_p = \mathbf{\Phi}_p \mathbf{w}^* + \boldsymbol{\epsilon}$, this estimator is unbiased, satisfying $\mathbb{E}[\hat{\mathbf{w}}] = \mathbf{w}^*$. Consequently, the bias term in the risk decomposition vanishes, and the total risk is governed solely by the predictive variance:

$$\begin{aligned} R_{\text{BLR}} &\coloneqq \mathbb{E}_{\mathbf{y}_p}[\|\mathbf{\Phi}_f(\hat{\mathbf{w}} - \mathbf{w}^*)\|^2] \\ &= \mathbb{E}_{\boldsymbol{\epsilon}}[\|\mathbf{\Phi}_f(\mathbf{\Phi}_p^\top \mathbf{\Phi}_p)^{-1}\mathbf{\Phi}_p^\top \boldsymbol{\epsilon}\|^2]. \end{aligned} \tag{16}$$

Using the property that $\mathbf{x}^\top \mathbf{x} = \mathrm{Tr}(\mathbf{x}\mathbf{x}^\top)$ and the noise assumption $\mathbb{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^\top] = \sigma_\epsilon^2 \mathbf{I}$, we expand the expectation:

$$\begin{aligned} R_{\text{BLR}} &= \mathbb{E}[\mathrm{Tr}\left(\mathbf{\Phi}_f(\mathbf{\Phi}_p^\top \mathbf{\Phi}_p)^{-1}\mathbf{\Phi}_p^\top \boldsymbol{\epsilon}\boldsymbol{\epsilon}^\top \mathbf{\Phi}_p(\mathbf{\Phi}_p^\top \mathbf{\Phi}_p)^{-1}\mathbf{\Phi}_f^\top\right)] \\ &= \mathrm{Tr}\left(\mathbf{\Phi}_f(\mathbf{\Phi}_p^\top \mathbf{\Phi}_p)^{-1}\mathbf{\Phi}_p^\top \mathbb{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^\top]\mathbf{\Phi}_p(\mathbf{\Phi}_p^\top \mathbf{\Phi}_p)^{-1}\mathbf{\Phi}_f^\top\right) \\ &= \sigma_\epsilon^2 \,\mathrm{Tr}\left(\mathbf{\Phi}_f(\mathbf{\Phi}_p^\top \mathbf{\Phi}_p)^{-1}(\mathbf{\Phi}_p^\top \mathbf{\Phi}_p)(\mathbf{\Phi}_p^\top \mathbf{\Phi}_p)^{-1}\mathbf{\Phi}_f^\top\right) \\ &= \sigma_\epsilon^2 \,\mathrm{Tr}\left(\mathbf{\Phi}_f(\mathbf{\Phi}_p^\top \mathbf{\Phi}_p)^{-1}\mathbf{\Phi}_f^\top\right) \\ &= \sigma_\epsilon^2 \,\mathrm{Tr}(\mathbf{\Sigma}_{\text{design}}). \end{aligned} \tag{17}$$

Therefore, the condition for strict superiority, $R_{\text{BLR}} < R_{\text{GA}}$, holds if and only if $\sigma_\epsilon^2 \,\mathrm{Tr}(\mathbf{\Sigma}_{\text{design}}) < \|\boldsymbol{\mu}_Q\|^2$. Rearranging terms yields the stated inequality. □

A.2. Proof of Proposition 3.2

Proposition 3.2 (Reduction of Expected Wasserstein-2 Cost). *Under the condition in Proposition 3.1, for target measure Q with finite second moments (not necessarily Gaussian), the regime-aware prior P_{BLR} strictly reduces the expected Wasserstein-2 distance to Q compared to the standard Gaussian prior P_{GA}:*

$$\mathbb{E}_{\mathbf{y}_p}\left[W_2^2(P_{BLR}(\mathbf{y}_p), Q)\right] < W_2^2(P_{GA}, Q). \tag{18}$$

Proof. We compare the transport costs associated with the uninformed prior $P_{\text{GA}} = \mathcal{N}(\mathbf{0}, \sigma^2_{\text{init}}\mathbf{I})$ and the regime-aware prior $P_{\text{BLR}}(\mathbf{y}_p) = \mathcal{N}(\boldsymbol{\mu}_{\text{prior}}, \sigma^2_{\text{init}}\mathbf{I})$. Observe that both distributions share the same isotropic covariance $\sigma^2_{\text{init}}\mathbf{I}$, differing only in their first moments.

We invoke the translation invariance property of the Wasserstein-2 distance. For any probability measure μ with finite second moments and any target measure Q on a Euclidean space, the squared distance decomposes orthogonally into the distance between their means and the distance between their centered distributions (Remark 2.19 in (Peyré et al., 2019)):

$$W_2^2(\mu, Q) = \|\mathbb{E}[\mu] - \mathbb{E}[Q]\|_2^2 + W_2^2(\bar{\mu}, \bar{Q}), \tag{19}$$

where $\bar{\mu}$ and $\bar{Q}$ denote the centered distributions. Since the covariance of P_{BLR} is fixed to $\sigma^2_{\text{init}}\mathbf{I}$ independent of the observation $\mathbf{y}_p$, its centered distribution is deterministic and identical to that of P_{GA}. Specifically, $\bar{P}_{\text{BLR}} = \bar{P}_{\text{GA}} = \mathcal{N}(\mathbf{0}, \sigma^2_{\text{init}}\mathbf{I})$. Consequently, the shape component of the transport cost is a constant, which we denote by $C_{\text{shape}} := W_2^2(\mathcal{N}(\mathbf{0}, \sigma^2_{\text{init}}\mathbf{I}), \bar{Q})$.

For the deterministic uninformed prior, the transport cost is thus:

$$W_2^2(P_{\text{GA}}, Q) = \|\mathbf{0} - \boldsymbol{\mu}_Q\|^2 + C_{\text{shape}} = \|\boldsymbol{\mu}_Q\|^2 + C_{\text{shape}}. \tag{20}$$

For the regime-aware prior, the cost is a random variable dependent on the historical observation $\mathbf{y}_p$. Taking the expectation over the observation noise yields:

$$\begin{aligned}\mathbb{E}_{\mathbf{y}_p}\left[W_2^2(P_{\text{BLR}}(\mathbf{y}_p), Q)\right] &= \mathbb{E}_{\mathbf{y}_p}\left[\|\boldsymbol{\mu}_{\text{prior}} - \boldsymbol{\mu}_Q\|^2 + C_{\text{shape}}\right] \\ &= R_{\text{BLR}} + C_{\text{shape}}.\end{aligned} \tag{21}$$

Defining the reduction in expected transport cost as $\Delta\mathcal{E} := W_2^2(P_{\text{GA}}, Q) - \mathbb{E}_{\mathbf{y}_p}[W_2^2(P_{\text{BLR}}, Q)]$, we find:

$$\Delta\mathcal{E} = (\|\boldsymbol{\mu}_Q\|^2 + C_{\text{shape}}) - (R_{\text{BLR}} + C_{\text{shape}}) = \|\boldsymbol{\mu}_Q\|^2 - R_{\text{BLR}}. \tag{22}$$

Under the condition established in Proposition 3.1, we have $R_{\text{BLR}} < \|\boldsymbol{\mu}_Q\|^2$, which implies $\Delta\mathcal{E} > 0$.

Thus, the regime-aware prior strictly reduces the expected Wasserstein-2 transport cost under the conditions of Proposition 3.1. $\square$

B. Additional Experiments

B.1. Empirical Validation of Theoretical Bounds

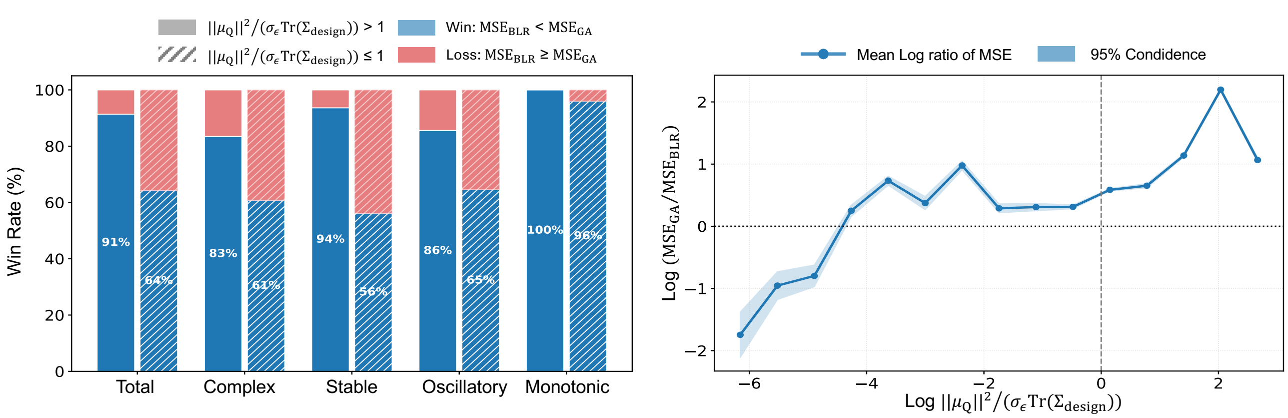

Figure 8. **Left:** Empirical win rate of the our BLR-induced regime-aware prior over an isotropic Gaussian initialization (ISO, $\mathcal{N}(\mathbf{0}, \sigma^2_{\text{init}}\mathbf{I})$) across regimes. **Right:** Empirical relationship between the theoretical bound term in Proposition 3.1 and the observed error ratio.

To assess the practical utility of Proposition 3.1 in the finite-sample regime, we analyze the relationship between the theoretical bounds and the empirical performance gap of the BLR-induced and isotropic Gaussian (ISO, $\mathcal{N}(\mathbf{0}, \sigma^2_{\text{init}}\mathbf{I})$)

Table 3. Comparing a fixed-variance prior to the BLR-induced covariance Σ_{BLR} on our benchmark. Each entry is reported as mean and ±std. Best mean results are in **bold** and second-best are underlined.

	Overall			Complex			Stable			Oscillatory			Monotonic		
Model	MSE↓	MAE↓	CRPS↓	MSE↓	MAE↓	CRPS↓	MSE↓	MAE↓	CRPS↓	MSE↓	MAE↓	CRPS↓	MSE↓	MAE↓	CRPS↓
Ours† (σ^2_{init})	**0.012** ±0.002	**0.044** ±0.006	**0.039** ±0.006	**0.034** ±0.002	**0.096** ±0.011	**0.086** ±0.010	**0.006** ±0.001	**0.027** ±0.003	**0.023** ±0.003	**0.035** ±0.002	**0.111** ±0.008	**0.098** ±0.007	**0.006** ±0.000	**0.033** ±0.002	**0.028** ±0.002
Ours† (Σ_{BLR})	0.014 ±0.003	0.050 ±0.007	0.046 ±0.007	0.042 ±0.007	0.110 ±0.018	0.103 ±0.018	0.007 ±0.001	0.032 ±0.002	0.029 ±0.002	0.036 ±0.003	0.113 ±0.007	0.107 ±0.008	0.006 ±0.001	0.036 ±0.001	0.033 ±0.001

Table 4. Backbone ablation comparing SSM-based and self-attention models on our benchmark. Each entry is reported as mean and ±std. Best mean results are in **bold** and second-best are underlined.

	Overall			Complex			Stable			Oscillatory			Monotonic		
Model	MSE↓	MAE↓	CRPS↓	MSE↓	MAE↓	CRPS↓	MSE↓	MAE↓	CRPS↓	MSE↓	MAE↓	CRPS↓	MSE↓	MAE↓	CRPS↓
Ours† (SSM)	**0.012** ±0.002	**0.044** ±0.006	**0.039** ±0.006	0.034 ±0.002	**0.096** ±0.011	**0.086** ±0.010	**0.006** ±0.001	**0.027** ±0.003	**0.023** ±0.003	**0.035** ±0.002	**0.111** ±0.008	**0.098** ±0.007	**0.006** ±0.000	**0.033** ±0.002	**0.028** ±0.002
Ours† (Attention)	0.012 ±0.001	0.046 ±0.006	0.041 ±0.005	**0.033** ±0.000	0.097 ±0.007	0.087 ±0.006	0.006 ±0.001	0.028 ±0.003	0.025 ±0.003	0.036 ±0.001	0.114 ±0.008	0.100 ±0.007	0.006 ±0.001	0.034 ±0.002	0.030 ±0.002

initializations. As illustrated in Figure 8 (Left), the BLR prior outperforms the ISO baseline across the majority of dynamic regimes. Crucially, this empirical advantage exhibits a strong alignment with our theoretical derivation. Figure 8 (Right) reveals a positive correlation between the magnitude of the theoretical bound and the observed log-error ratio. This result indicates that although Proposition 3.1 is derived under an asymptotic limit ($\alpha \to 0$), it serves as a robust and informative proxy for $\alpha > 0$ settings. Thus, this bound effectively identifies scenarios where the mechanistic prior yields the most significant reduction in the Wasserstein transport cost, further validating the BLR initialization's superiority over the uninformed isotropic baseline.

B.2. Empirical Benefits of a Covariance-Regularized Prior

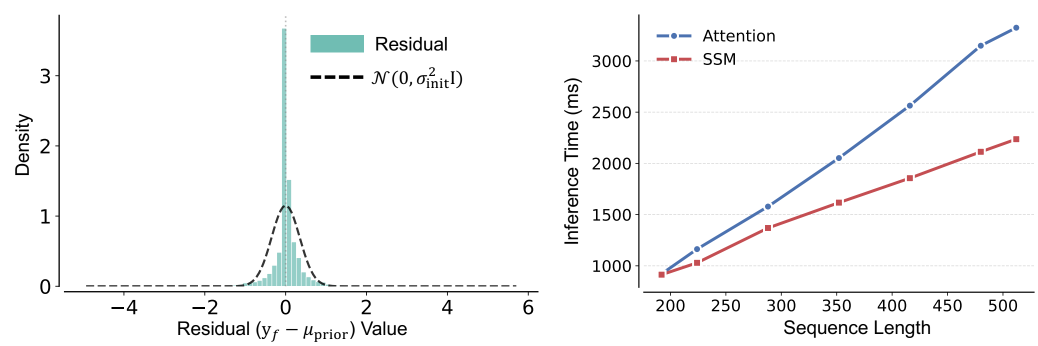

Figure 9. **Left:** Residual distribution of $\mathbf{y} - \boldsymbol{\mu}_{\mathrm{prior}}$ under the regime-aware prior. **Right:** Inference-time scaling of SSM and self-attention backbones with increasing prediction horizon, highlighting the superior long-horizon efficiency of SSM-based models.

In practice, we instantiate the prior with a fixed variance σ^2_{init} rather than the BLR-induced Σ_{BLR}. As shown in Table 3, using the full BLR predictive covariance already yields strong accuracy (overall MSE 0.014), confirming that the Bayesian predictive distribution provides useful source information. Regularizing this covariance to an isotropic form further improves overall MSE/MAE/CRPS from $0.014/0.050/0.046$ to $0.012/0.044/0.039$, consistent with the mean-alignment argument in Section 3.3.4. Furthermore, Figure 9(left) shows that the residual $\mathbf{y} - \boldsymbol{\mu}_{\mathrm{prior}}$ is well-approximated by a Gaussian, supporting the constant-variance approximation in this regime.

B.3. Architecture Analysis: Attention and SSM

We ablate the sequence module by replacing the Selective State Space Model (SSM) with a standard self-attention block. As shown in Table 4, the accuracy is largely preserved under this substitution (MSE/MAE/CRPS: $0.012/0.044/0.039$ for SSM

Table 5. Performance comparison of TSFlow with different kernels at NFE=1 and NFE=16. Each entry is reported as mean and ±std. Best mean results are in **bold** and second-best are underlined.

	Overall			Complex			Stable			Oscillatory			Monotonic		
Model	MSE↓	MAE↓	CRPS↓	MSE↓	MAE↓	CRPS↓	MSE↓	MAE↓	CRPS↓	MSE↓	MAE↓	CRPS↓	MSE↓	MAE↓	CRPS↓
NFE = 1															
TSFlow-ISO	0.026 ±0.002	0.081 ±0.017	0.074 ±0.015	0.042 ±0.001	**0.112** ±0.013	0.104 ±0.011	0.017 ±0.003	0.062 ±0.021	0.057 ±0.019	0.085 ±0.013	0.208 ±0.007	0.197 ±0.008	0.009 ±0.004	0.051 ±0.020	0.045 ±0.018
TSFlow-OU	**0.025** ±0.003	0.081 ±0.012	0.074 ±0.010	0.041 ±0.002	0.115 ±0.011	0.107 ±0.010	0.017 ±0.005	0.063 ±0.016	0.057 ±0.013	0.079 ±0.008	0.200 ±0.005	0.190 ±0.006	0.009 ±0.002	0.050 ±0.007	0.044 ±0.005
TSFlow-SE	0.026 ±0.001	0.082 ±0.014	0.074 ±0.011	**0.040** ±0.001	0.112 ±0.013	**0.103** ±0.011	0.018 ±0.002	0.065 ±0.017	0.058 ±0.014	**0.079** ±0.013	**0.198** ±0.012	**0.187** ±0.013	0.009 ±0.003	0.053 ±0.017	0.046 ±0.014
TSFlow-PE	0.025 ±0.003	**0.078** ±0.011	**0.072** ±0.010	0.042 ±0.001	0.112 ±0.009	0.104 ±0.009	**0.017** ±0.003	**0.059** ±0.012	**0.054** ±0.011	0.084 ±0.006	0.208 ±0.006	0.195 ±0.005	**0.008** ±0.002	**0.047** ±0.010	**0.041** ±0.009
NFE = 16															
TSFlow-ISO	0.027 ±0.002	0.080 ±0.017	0.059 ±0.011	0.043 ±0.002	0.112 ±0.012	0.086 ±0.008	0.018 ±0.004	0.060 ±0.020	0.043 ±0.014	0.088 ±0.013	0.208 ±0.008	0.159 ±0.012	0.010 ±0.004	0.051 ±0.019	0.040 ±0.014
TSFlow-OU	**0.026** ±0.003	0.077 ±0.012	0.056 ±0.008	0.043 ±0.002	0.113 ±0.010	0.086 ±0.013	0.018 ±0.005	0.057 ±0.015	0.039 ±0.010	0.083 ±0.007	0.199 ±0.004	0.152 ±0.007	0.010 ±0.002	0.049 ±0.010	0.037 ±0.009
TSFlow-SE	0.027 ±0.002	0.079 ±0.010	0.057 ±0.004	**0.043** ±0.001	0.112 ±0.012	**0.085** ±0.007	0.019 ±0.002	0.061 ±0.012	0.042 ±0.005	**0.081** ±0.011	**0.198** ±0.011	**0.150** ±0.013	0.010 ±0.004	0.053 ±0.016	0.040 ±0.010
TSFlow-PE	0.026 ±0.003	**0.075** ±0.011	**0.055** ±0.007	0.044 ±0.001	**0.111** ±0.009	0.086 ±0.008	**0.017** ±0.004	**0.055** ±0.012	**0.038** ±0.008	0.088 ±0.007	0.208 ±0.007	0.158 ±0.006	**0.008** ±0.002	**0.044** ±0.008	**0.033** ±0.006

vs. $0.012/0.046/0.041$ for Attention), indicating that the principal error reductions are not driven solely by the choice of sequence model. However, Figure 9 (right) indicates that SSM achieves lower inference time as the prediction length grows, making it preferable for long-term trajectory generation when computational scaling is a dominant constraint.

B.4. TSFlow Baseline Results under Different Prior Kernels

To establish a rigorous baseline, we evaluated the TSFlow architecture against the full spectrum of kernel configurations proposed in the original study. As detailed in Table 5, the variant equipped with the periodic kernel (TSFlow-PE) demonstrates superior pointwise accuracy compared to alternative instantiations on overall results. Crucially, this performance advantage extends to distributional metrics. Table 5 also confirms that TSFlow-PE achieves the lowest overall CRPS, validating its selection as the primary comparator for both deterministic and probabilistic assessments.

B.5. Sensitivity to Number of Function Evaluations

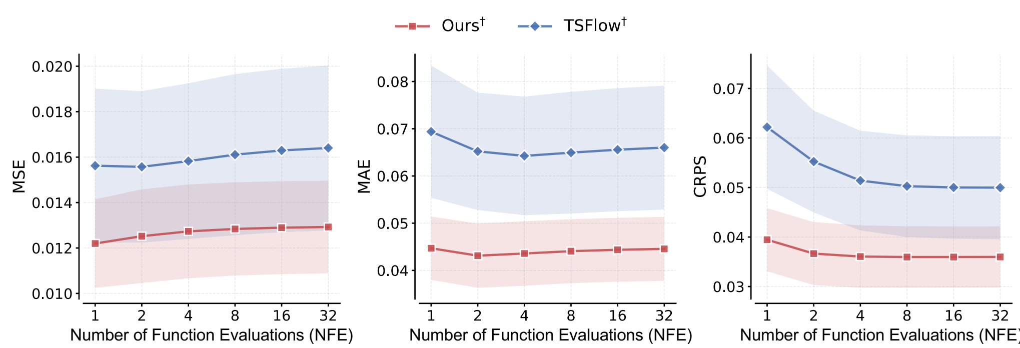

Figure 10. Impact of the Number of Function Evaluations (NFE) on forecasting performance. Our method achieves near-optimal overall results with a single step (NFE=1). In contrast, the baseline TSFlow requires significantly more steps (NFE $\geq$ 16) to converge to the target distribution.

We analyze the impact of the Number of Function Evaluations (NFE) in Figure 10. Our method achieves near-optimal deterministic and probabilistic performance at NFE $= 1$. This rapid convergence stems from the regime-aware prior, which structurally approximates the target to minimize the source data distributional mismatch, maximizing inference speed while maintaining competitive accuracy. In contrast, TSFlow initiates with uninformative prior, necessitating significantly more

Table 6. Computational cost comparison on a single NVIDIA RTX A6000 GPU. Training time is reported in hours, and inference time is reported per sample.

Model	Family	Params	Train (h)	Infer. (ms)	NFE
DLinear	Point	49.7 K	0.3	0.092	–
iTransformer	Point	4.9 M	0.5	0.966	–
PatchTST	Point	4.7 M	0.6	0.800	–
TimeMixer	Point	133 K	0.5	3.135	–
TimeXer	Point	11.5 M	0.5	2.104	–
SMamba	Point	3.0 M	0.5	2.300	–
BiMamba4TS	Point	3.1 M	1.5	2.496	–
NSformer	Point	2.0 M	5	3.000	–
CSDI†	Diffusion	413 K	8	90.8	50
TSDiff†	Diffusion	946 K	10	552.6	100
TSFlow†	Flow	1.9 M	6	71.9	16
Ours†	**Flow**	**1.8 M**	**7**	**7.7**	**1**

Table 7. Comparison with zero-shot time-series foundation models. Each entry reports the mean with standard deviation. Best and second-best performances are highlighted in **bold** and underline, respectively.

	Overall		Complex		Stable		Oscillatory		Monotonic	
Model	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓
TimeMoE	0.120$\pm$0.004	0.231$\pm$0.010	0.099$\pm$0.013	0.191$\pm$0.032	0.118$\pm$0.003	0.221$\pm$0.010	<u>0.138</u>$\pm$0.005	<u>0.268</u>$\pm$0.005	0.138$\pm$0.012	0.289$\pm$0.023
Sundial	0.111$\pm$0.003	0.217$\pm$0.007	<u>0.095</u>$\pm$0.012	0.185$\pm$0.031	0.106$\pm$0.002	0.203$\pm$0.008	0.141$\pm$0.009	0.270$\pm$0.009	0.133$\pm$0.011	0.281$\pm$0.022
TimesFM	0.099$\pm$0.005	<u>0.152</u>$\pm$0.004	0.109$\pm$0.009	<u>0.145</u>$\pm$0.021	0.086$\pm$0.002	<u>0.138</u>$\pm$0.006	0.249$\pm$0.032	0.299$\pm$0.014	<u>0.045</u>$\pm$0.004	<u>0.112</u>$\pm$0.007
Chronos	<u>0.097</u>$\pm$0.004	0.167$\pm$0.000	0.144$\pm$0.021	0.181$\pm$0.012	<u>0.078</u>$\pm$0.003	0.147$\pm$0.002	0.234$\pm$0.013	0.309$\pm$0.007	0.054$\pm$0.004	0.153$\pm$0.005
Ours†	**0.012**$\pm$0.002	**0.044**$\pm$0.006	**0.034**$\pm$0.002	**0.096**$\pm$0.011	**0.006**$\pm$0.001	**0.027**$\pm$0.003	**0.035**$\pm$0.002	**0.111**$\pm$0.008	**0.006**$\pm$0.000	**0.033**$\pm$0.002

steps (NFE ≥ 16) to adequately resolve distributional features. Consequently, to ensure a fair comparison at peak capacity, we adopt NFE $= 1$ for our framework and NFE $= 16$ for TSFlow (similarly to (Kollovieh et al., 2024)) in all experiments.

B.6. Computational Cost Comparison

We provide a comprehensive computational cost comparison across all evaluated models in Table 6. All measurements were conducted on a single NVIDIA RTX A6000 GPU.

When considered together with the predictive results in Table 1, these measurements show a clear trade-off between predictive performance and computational cost. Point forecasting methods provide fast training and inference (e.g., DLinear: 0.3 h training and 0.092 ms per sample), but have higher predictive errors on the main benchmark. Existing probabilistic baselines provide uncertainty-aware forecasts but often incur substantially higher inference cost, with diffusion-based CSDI† and TSDiff† requiring 50–100 function evaluations and over 90 ms per sample. TSFlow† reduces the NFE to 16 but still requires 71.9 ms per sample. In contrast, RegimeFlow (Ours†) performs inference in 7.7 ms with a single function evaluation (NFE = 1), making it 9.3$\times$ faster than TSFlow†. With 1.8 M parameters and approximately 7 h training time, RegimeFlow keeps training cost comparable to standard probabilistic baselines while substantially reducing inference overhead.

B.7. Comparison with Time-Series Foundation Models

We further compare RegimeFlow against four representative zero-shot time-series foundation models (TSFMs): TimeMoE (Shi et al., 2025), Sundial (Liu et al., 2025b), TimesFM (Das et al., 2024), and Chronos (Ansari et al., 2024). These models are pretrained on large-scale general-purpose time-series corpora and support zero-shot inference, making them natural baselines for evaluating the advantage of domain-specific biological priors in cross-system generalization. All TSFMs are evaluated in the zero-shot setting on the same test trajectories and prediction horizons used in the main benchmark, without task-specific fine-tuning.

Table 7 reports the results. Despite their massive pretraining scale (128–200 M parameters), all four TSFMs yield substantially higher errors than RegimeFlow. The strongest foundation model in the overall comparison, Chronos, achieves an overall MSE of 0.097, compared to 0.012 for Ours† with only 1.8 M parameters. The gap is also pronounced on the Oscillatory subset:

Table 8. Performance on transient and hybrid dynamical regimes under the $L = 96$, $T = 416$ extrapolation setting. Each entry reports the mean.

		Ours†		TSFlow†	
Case	**System**	MSE↓	MAE↓	MSE↓	MAE↓
Complex Oscil.	Golomb2006	0.024	0.092	0.135	0.275
Oscil.→Stable	Sharp2013	0.035	0.116	0.046	0.159
Damped Oscil.	Radulescu2008	0.051	0.151	0.086	0.216

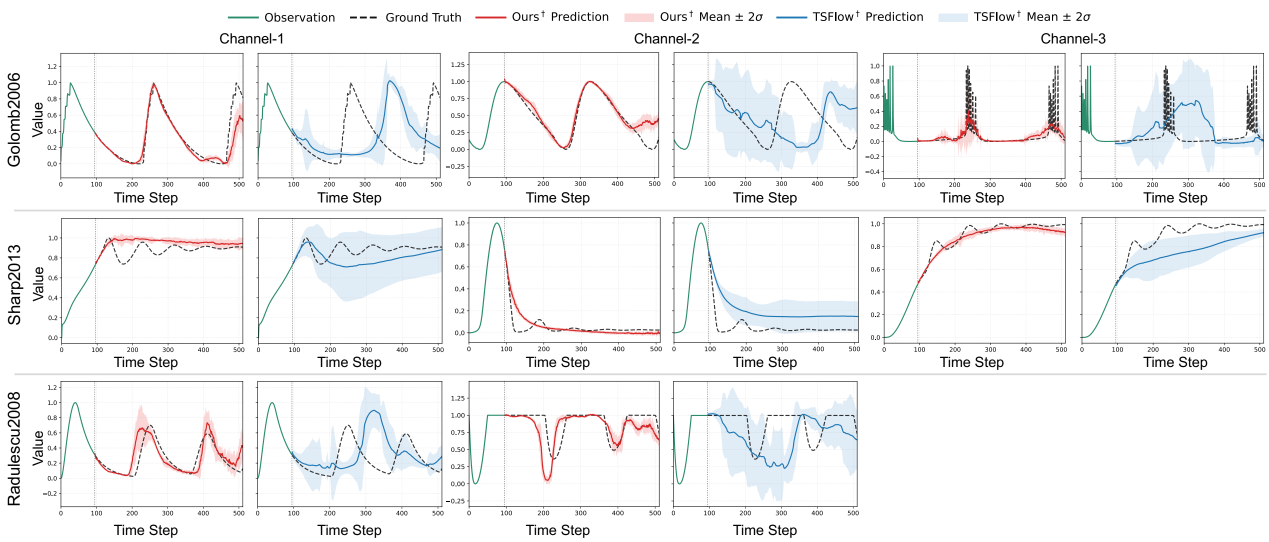

Figure 11. Representative predictions on transient and hybrid regimes under the $L = 96$, $T = 416$ extrapolation setting. The selected cases include complex oscillations (Golomb2006), oscillation-to-stable transition (Sharp2013), and damped oscillations (Radulescu2008).

the best TSFM reaches an MSE of 0.138, whereas Ours† achieves 0.035. Chronos and TimesFM show even larger errors on this regime, suggesting that generic pretraining alone does not reliably capture sustained biological oscillations. This pattern indicates that general-purpose TSFMs, while strong zero-shot forecasting baselines, do not encode the biological regime structure required by this benchmark. In contrast, RegimeFlow combines a regime-aware prior with a condition-modulated sequence backbone, providing structural guidance tailored to long-horizon biological dynamics.

B.8. Validation on Transient and Hybrid Dynamical Regimes

To assess whether RegimeFlow remains effective beyond the three canonical regime categories, we evaluate it on three representative biological systems that exhibit transient or hybrid dynamical behaviors under a longer extrapolation setting with $L = 96$ observed points and a $T = 416$ prediction horizon: **Golomb2006** (complex oscillations with no dominant single-regime pattern) (Golomb et al., 2006), **Sharp2013** (oscillatory dynamics that transition to a stable steady state) (Sharp et al., 2013), and **Radulescu2008** (decreasing-amplitude oscillations that dampen over time) (Radulescu et al., 2008). These cases are deliberately chosen because they do not fit neatly into a single macroscopic regime, testing whether the regime-aware prior provides flexible guidance rather than a rigid constraint. As shown in Table 8 and Figure 11, RegimeFlow obtains lower errors than TSFlow† across all three representative transient cases. On Golomb2006 (complex oscillation), RegimeFlow reduces MSE from 0.135 to 0.024, an 82% improvement, suggesting that the structured prior remains useful even when the dynamics do not follow a single canonical pattern. On Sharp2013 (oscillation-to-stable transition), both methods show higher error, as expected for a regime-switching system; RegimeFlow's MSE of 0.035 versus 0.046 for TSFlow† suggests that the prior provides partial guidance without over-constraining the flow. On Radulescu2008 (decreasing oscillation), RegimeFlow achieves an MSE of 0.051 compared to 0.086 for TSFlow†, a 41% improvement. These results indicate that RegimeFlow's regime-aware prior can still capture the underlying structure of the dynamics, even when the system exhibits transient or hybrid behaviors that do not fit neatly into a single macroscopic regime category.

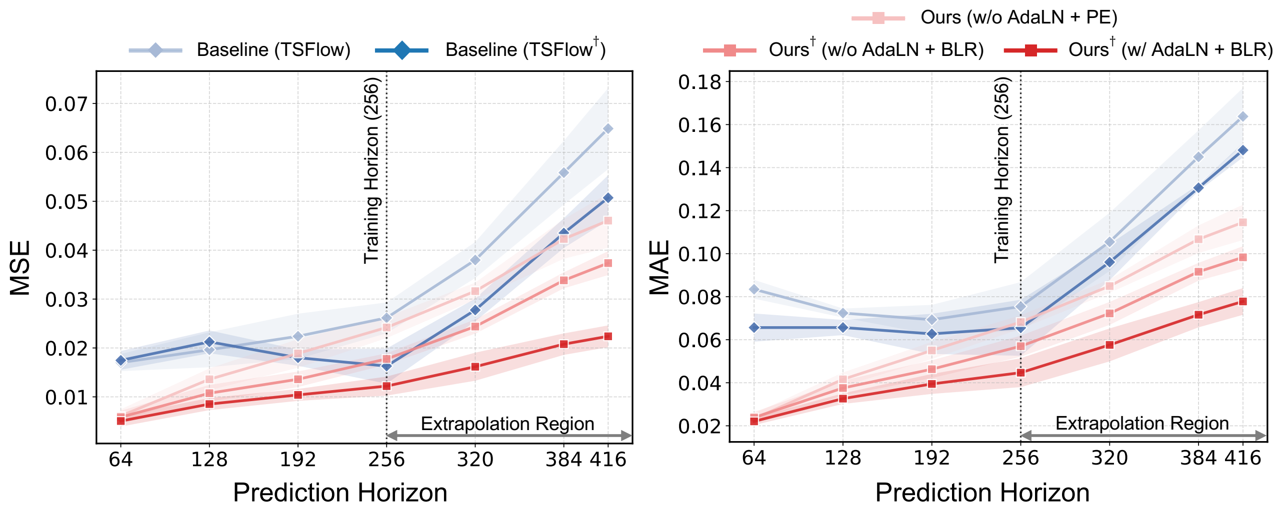

Figure 12. Extrapolation performance across extended prediction lengths. PE denotes the baseline initialized with a Periodic Kernel prior, while BLR represents our proposed regime-aware prior induced by Bayesian Linear Regression. Models trained on a fixed length ($T = 256$) are evaluated on longer trajectories, demonstrating the superior generalization of our proposed framework.

B.9. Robustness to Extrapolation Length

B.9.1. Quantitative Evaluation of Long-Horizon Extrapolation

To evaluate the generalization capability of our framework, we investigate its performance when extrapolating to prediction lengths significantly beyond the training setup ($T_{\text{train}} = 256$). As illustrated in Figure 12, our full model (w/ AdaLN + BLR) maintains high predictive fidelity across extended lengths, effectively mitigating the degradation typically observed in autoregressive or iterative generation.

Disentangling the drivers of this robustness yields two primary insights. First, the architectural superiority of our framework is demonstrated by the performance gap between our Periodic (PE) variant (w/o AdaLN + PE) and the TSFlow baseline. Unlike the standard S4-based architecture employed in TSFlow, our backbone leverages the enhanced receptivity of modern Selective State Space Models to capture long-range dependencies. Second, a comparison of the conditioned variants underscores the essential role of the prior. The model initialized with the BLR-induced distribution (w/o AdaLN + BLR) consistently outperforms both the TSFlow† model and our own model initialized with the PE kernel (w/o AdaLN + PE). This result highlights the superior efficacy of the BLR prior in capturing underlying dynamic patterns compared to the PE kernel. This indicates that the mechanistic prior encodes invariant dynamic properties that persist beyond the training window, whereas the standard PE kernel struggles to generalize to unseen temporal coordinates.

B.9.2. Qualitative Visualization of Extended Trajectory Predictions

As shown in Figure 13, we select five dynamical systems with qualitatively different regime patterns, each of which can be described by mechanistic ordinary differential equation models commonly used in prior studies (Tomasz et al., 2004; Fribourg et al., 2014; Pritchard & Birch, 2014; Yildirim & Mackey, 2003; Bungay et al., 2006).

Overall, our method demonstrates strong adaptability: even with a long extrapolation horizon of $T = 416$, it can still capture the global trajectory trend with high fidelity. When the underlying series exhibits regime changes, Ours†(w/o AdaLN) already tracks most regime-wise trend variations well, highlighting the effectiveness of the regime-aware prior during extrapolation. When the series is relatively regime-agnostic, the comparison between Ours-PE(w/o AdaLN) and TSFlow shows that our backbone yields more accurate predictions, supporting the advantage of our backbone design.

As illustrated in Figure 14, we selected five representative oscillatory systems exhibiting distinct temporal patterns, including cell division cycles and genetic network dynamics (Tyson, 1991; Locke et al., 2005; Li et al., 2009; Adams et al., 2012; Conradie et al., 2010).

The comparative analysis reveals that our proposed method consistently achieves superior predictive fidelity across these

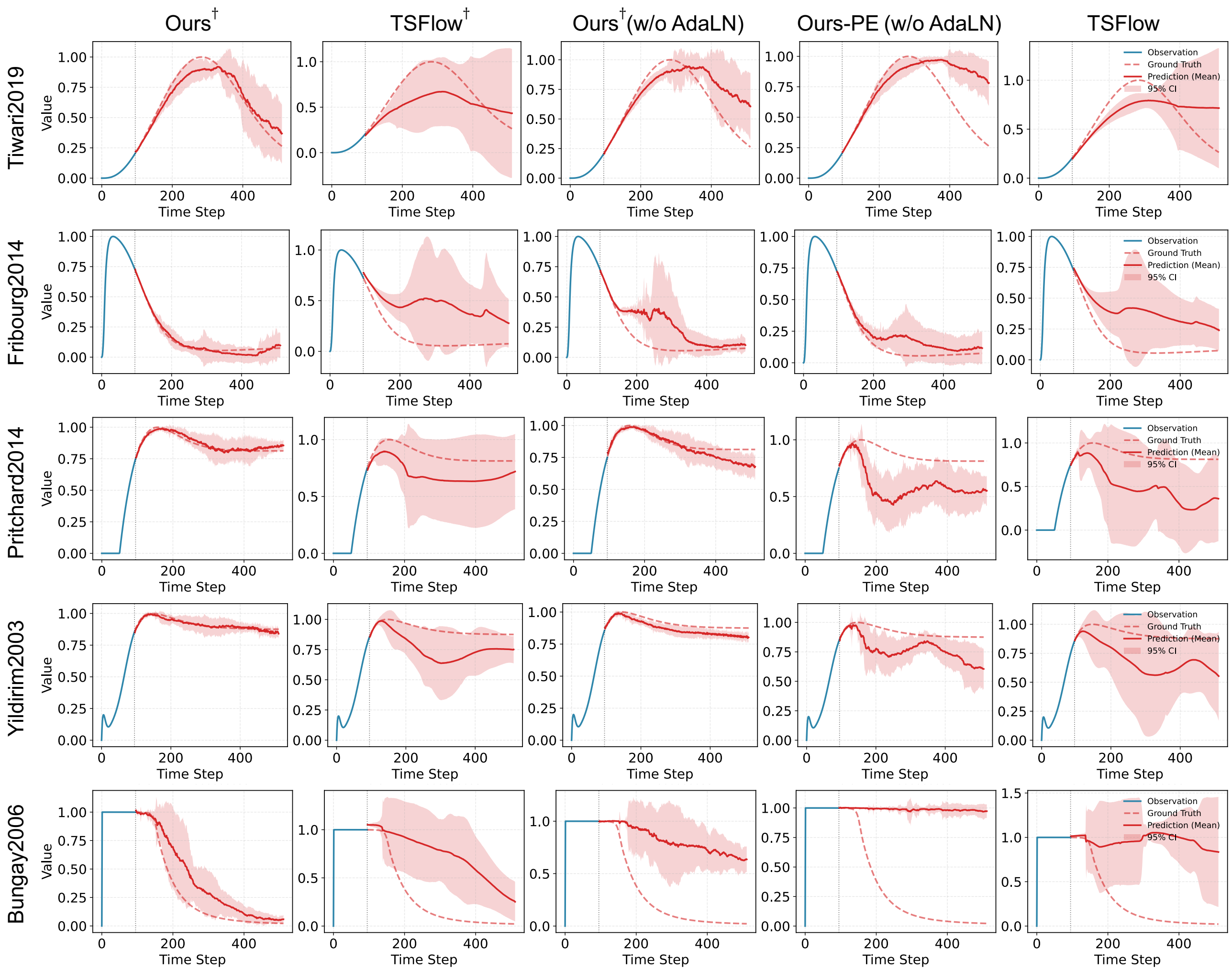

Figure 13. Qualitative comparison of extrapolation trajectories ($T = 416$) across five systems exhibiting distinct regimes. We compare Ours† against the state-of-the-art baseline TSFlow and two ablations: Ours† (w/o AdaLN) and Ours-PE (w/o AdaLN). Ours-PE denotes the variant initialized with a standard Periodic Kernel prior, while Ours† employs our proposed regime-aware prior. The visualization highlights the model's capability to capture long-term trends under diverse dynamic patterns.

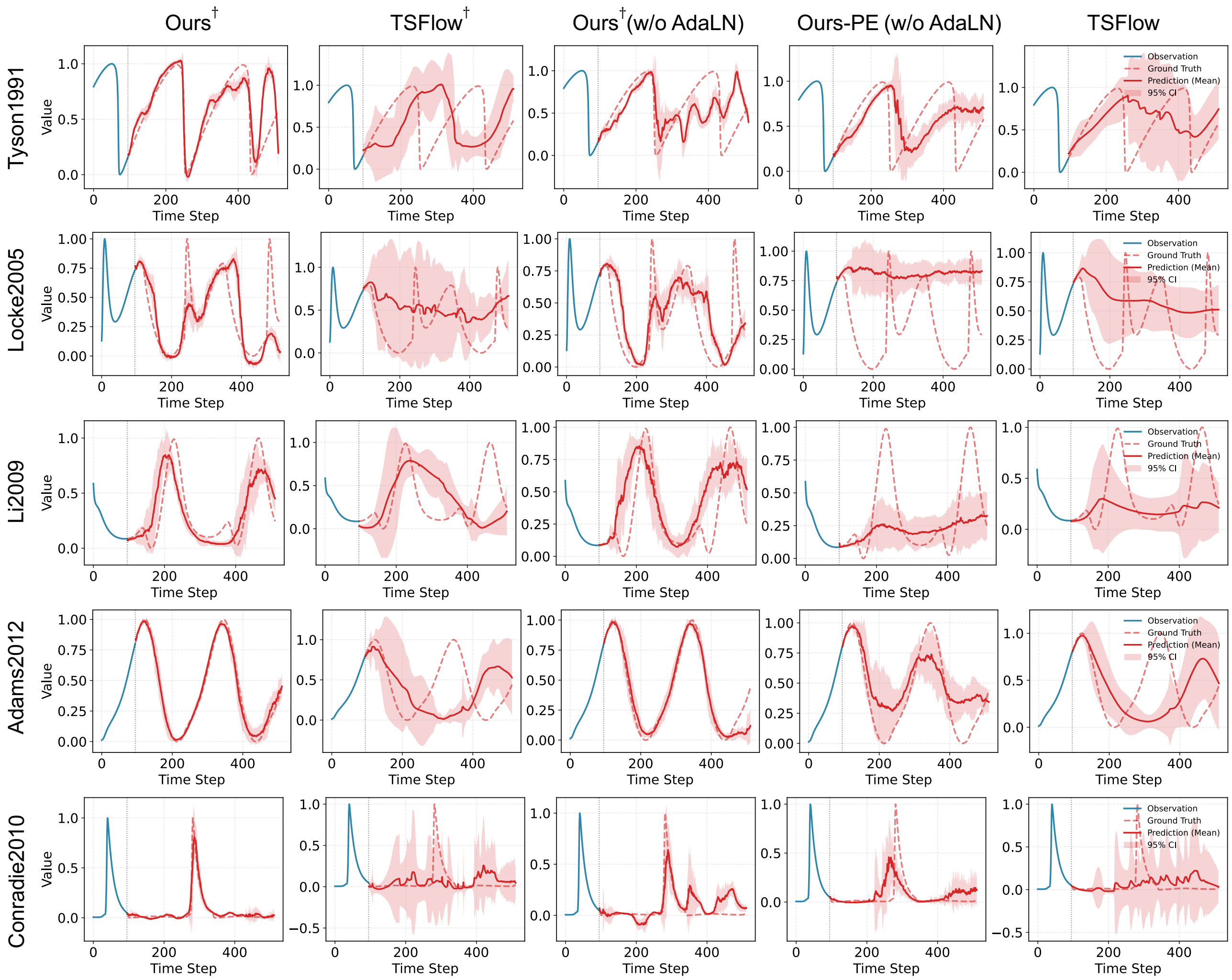

Figure 14. Qualitative visualization of predicted trajectories across five distinct oscillatory biological systems. We compare the extrapolation performance ($T = 416$) of Ours† against the baseline TSFlow† and ablation variants (Ours† w/o AdaLN, Ours-PE w/o AdaLN). The results demonstrate that our regime-aware prior initialization significantly enhances the capture of diverse oscillatory patterns compared to standard periodic kernel methods.

Table 9. Robustness evaluation under measurement noise and irregular sampling. Each entry reports the mean with standard deviation. ΔMSE is computed on the overall MSE as $(1 - \text{Ours}^\dagger/\text{TSFlow}^\dagger) \times 100\%$. The Clean setting from the main benchmark (Table 1) is included as reference.

		Overall			Complex		Stable		Oscillatory		Monotonic	
Setting	Model	MSE↓	MAE↓	ΔMSE	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓	MSE↓	MAE↓
Clean	Ours†	0.012$_{\pm 0.002}$	0.044$_{\pm 0.006}$	25.0%	0.034$_{\pm 0.002}$	0.096$_{\pm 0.011}$	0.006$_{\pm 0.001}$	0.027$_{\pm 0.003}$	0.035$_{\pm 0.002}$	0.111$_{\pm 0.008}$	0.006$_{\pm 0.000}$	0.033$_{\pm 0.002}$
	TSFlow†	0.016$_{\pm 0.004}$	0.064$_{\pm 0.013}$		0.039$_{\pm 0.005}$	0.111$_{\pm 0.022}$	0.008$_{\pm 0.002}$	0.046$_{\pm 0.010}$	0.044$_{\pm 0.005}$	0.139$_{\pm 0.017}$	0.008$_{\pm 0.001}$	0.050$_{\pm 0.005}$
Noisy-0.1	Ours†	0.041$_{\pm 0.004}$	0.134$_{\pm 0.009}$	8.9%	0.089$_{\pm 0.019}$	0.200$_{\pm 0.024}$	0.031$_{\pm 0.005}$	0.117$_{\pm 0.013}$	0.074$_{\pm 0.014}$	0.188$_{\pm 0.017}$	0.027$_{\pm 0.003}$	0.121$_{\pm 0.004}$
	TSFlow†	0.045$_{\pm 0.003}$	0.140$_{\pm 0.006}$		0.089$_{\pm 0.011}$	0.199$_{\pm 0.018}$	0.031$_{\pm 0.005}$	0.117$_{\pm 0.009}$	0.081$_{\pm 0.007}$	0.197$_{\pm 0.015}$	0.046$_{\pm 0.010}$	0.154$_{\pm 0.019}$
Noisy-0.3	Ours†	0.059$_{\pm 0.002}$	0.178$_{\pm 0.003}$	11.9%	0.098$_{\pm 0.014}$	0.233$_{\pm 0.015}$	0.050$_{\pm 0.002}$	0.164$_{\pm 0.008}$	0.099$_{\pm 0.012}$	0.234$_{\pm 0.012}$	0.040$_{\pm 0.010}$	0.160$_{\pm 0.022}$
	TSFlow†	0.067$_{\pm 0.014}$	0.187$_{\pm 0.021}$		0.104$_{\pm 0.018}$	0.230$_{\pm 0.025}$	0.048$_{\pm 0.012}$	0.161$_{\pm 0.023}$	0.110$_{\pm 0.009}$	0.242$_{\pm 0.007}$	0.094$_{\pm 0.044}$	0.232$_{\pm 0.053}$
Irregular-48	Ours†	0.038$_{\pm 0.005}$	0.123$_{\pm 0.010}$	29.6%	0.074$_{\pm 0.013}$	0.181$_{\pm 0.025}$	0.027$_{\pm 0.002}$	0.103$_{\pm 0.008}$	0.102$_{\pm 0.006}$	0.241$_{\pm 0.008}$	0.014$_{\pm 0.002}$	0.075$_{\pm 0.005}$
	TSFlow†	0.054$_{\pm 0.009}$	0.163$_{\pm 0.018}$		0.097$_{\pm 0.010}$	0.214$_{\pm 0.016}$	0.039$_{\pm 0.007}$	0.144$_{\pm 0.016}$	0.124$_{\pm 0.010}$	0.263$_{\pm 0.013}$	0.042$_{\pm 0.014}$	0.139$_{\pm 0.026}$
Irregular-24	Ours†	0.080$_{\pm 0.004}$	0.202$_{\pm 0.008}$	15.8%	0.111$_{\pm 0.015}$	0.234$_{\pm 0.021}$	0.072$_{\pm 0.001}$	0.195$_{\pm 0.005}$	0.150$_{\pm 0.006}$	0.297$_{\pm 0.007}$	0.036$_{\pm 0.005}$	0.131$_{\pm 0.009}$
	TSFlow†	0.095$_{\pm 0.012}$	0.233$_{\pm 0.017}$		0.133$_{\pm 0.016}$	0.271$_{\pm 0.023}$	0.074$_{\pm 0.011}$	0.210$_{\pm 0.017}$	0.195$_{\pm 0.012}$	0.348$_{\pm 0.011}$	0.089$_{\pm 0.019}$	0.226$_{\pm 0.027}$

diverse oscillation modes. Crucially, the performance gap between Ours† (including its variant w/o AdaLN) and the regime-agnostic Ours-PE (w/o AdaLN) underscores the critical role of the regime-aware prior. This regime-aware prior provides a robust structural initialization that is essential for accurate long-term extrapolation. In contrast, although TSFlow† also incorporates regime information, its performance is constrained by a less accurate initial trajectory shape relative to our approach. This discrepancy in initialization induces optimization bias in later stages, preventing the baseline from fully recovering the true oscillatory dynamics.

B.10. Robustness to Measurement Noise and Irregular Sampling

To assess the robustness of RegimeFlow under conditions that approximate real-world experimental data, we evaluate both RegimeFlow (Ours†) and TSFlow† under two forms of data degradation: (1) **measurement noise**, where additive Gaussian perturbations $x + s\epsilon$ with $\epsilon \sim \mathcal{N}(0, 1)$ and $s \in \{0.1, 0.3\}$ are applied to the observed trajectories; and (2) **irregular sampling**, where the observation window is randomly subsampled, retaining only 48 or 24 out of the original 96 time points.

As shown in Table 9, RegimeFlow consistently outperforms TSFlow† across all overall robustness settings. Under mild noise (s=0.1), the regime-aware prior preserves an 8.9% MSE advantage; under stronger noise (s=0.3), the advantage remains meaningful at 11.9%, indicating that the regime-aware prior partially attenuates noise through its closed-form projection onto the basis subspace. Under irregular sampling, RegimeFlow exhibits particularly strong robustness: with only 24 irregular observations (Irregular-24), RegimeFlow achieves an MSE of 0.080 compared to 0.095 for TSFlow† (15.8% improvement), and with 48 observations the improvement reaches 29.6%. This resilience is attributable to the regime-aware prior's ability to recover the underlying trend from sparse observations via the selected basis family, providing a structurally informed initialization even when data density is low. Table 9 also reports regime-wise breakdowns for completeness, while ΔMSE is computed from the overall results.

C. Implementation Details

In this section, we provide a comprehensive overview of the experimental setup to facilitate reproducibility. We detail the baseline models, hyperparameter configurations, and training protocols used in our evaluation. All experiments were implemented in PyTorch and executed on a workstation equipped with a single NVIDIA A6000 GPU.

C.1. Baseline Descriptions

To evaluate the effectiveness of our proposed framework, we benchmarked against a diverse set of state-of-the-art methods, categorized into Point Time-Series Forecasting architectures and Generative Probabilistic Forecasting models.

C.1.1. Point Time-Series Forecasting.

We acknowledge the *Time-Series-Library*[1] for providing a unified codebase for time-series forecasting. All point forecasting baselines reported in this work were implemented based on this framework to ensure a fair and standardized comparison.

- **DLinear** (Zeng et al., 2023): A simple yet effective decomposition-based linear model that separates time series into trend and seasonal components.
- **iTransformer** (Liu et al., 2023): An inverted Transformer architecture that embeds the entire temporal sequence of each variate as a token to capture multivariate correlations.
- **PatchTST** (Nie, 2022): A channel-independent Transformer model that utilizes patching to segment time series into sub-series tokens, preserving local semantic information.
- **TimeMixer** (Wang et al., 2024a): An MLP-based architecture designed for multi-scale time-series mixing.
- **TimeXer** (Wang et al., 2024b): A Transformer-based framework specialized in empowering time-series forecasting by effectively incorporating exogenous variables.
- **NSformer** (Liu et al., 2022): A Transformer-based model specifically designed to address the challenges of non-stationary time series.
- **SMamba** (Wang et al., 2025): A selective state-space model (SSM) adapted for time-series forecasting, leveraging the efficiency of the Mamba architecture.
- **BiMamba4TS** (Liang et al., 2024): A bidirectional state-space model that extends Mamba to capture temporal dependencies from both forward and backward directions.

C.1.2. Generative Probabilistic Forecasting.

- **CSDI** (Tashiro et al., 2021): A conditional score-based diffusion model originally designed for time-series imputation.
- **TSDiff** (Kollovieh et al., 2023): A diffusion-based model that introduces a self-guidance mechanism to regularize the generation process.
- **TSFlow** (Kollovieh et al., 2024): A conditional flow matching model for probabilistic time series forecasting that combines Gaussian process priors.

C.1.3. Time-Series Foundation Models (Zero-Shot).

We evaluate four representative zero-shot time-series foundation models (TSFMs) to assess how large-scale general-purpose pretraining performs relative to domain-specific biological priors in cross-system trajectory prediction. All TSFMs are evaluated in the zero-shot setting on the same test trajectories.

- **TimeMoE** (Shi et al., 2025): A decoder-only sparse mixture-of-experts (MoE) Transformer, pretrained on over 300 billion time points spanning nine domains. Public checkpoint: `https://huggingface.co/Maple728/TimeMoE-200M`.
- **Sundial** (Liu et al., 2025b): A patch-based decoder-only Transformer, trained with a flow-matching objective (TimeFlow Loss) on one trillion time points. Public checkpoint: `https://huggingface.co/thuml/sundial-base-128m`.
- **TimesFM** (Das et al., 2024): A decoder-only patched Transformer, pretrained by Google Research on approximately 100 billion time points from web, retail, traffic, and energy domains. Public checkpoint: `https://huggingface.co/google/timesfm-1.0-200m`.
- **Chronos** (Ansari et al., 2024): A T5-based encoder-decoder Transformer that tokenizes time series via scaling and quantization, trained with a cross-entropy loss on 84 billion tokens from public datasets augmented with synthetic Gaussian process data. Public checkpoint: `https://huggingface.co/amazon/chronos-t5-base`.

[1] `https://github.com/thuml/Time-Series-Library`

Table 10. Hyperparameter configurations for RegimeFlow and baseline models. **Family** categorizes the modeling paradigm. **NFE** denotes the Number of Function Evaluations per inference step. D and M correspond to the backbone's hidden dimension and layer count, respectively.

Model	Family	Hidden Dim (D)	Layers (M)	LR	NFE
DLinear	Point	512	2	1×10^{-3}	-
iTransformer	Point	512	3	2×10^{-4}	-
PatchTST	Point	512	2	2×10^{-4}	-
TimeMixer	Point	16	3	2×10^{-4}	-
TimeXer	Point	512	4	1×10^{-4}	-
NSformer	Point	256	2	2×10^{-4}	-
S-Mamba	Point	256	4	1×10^{-4}	-
BiMamba4TS	Point	256	4	1×10^{-4}	-
CSDI	Diffusion	64	4	1×10^{-3}	50
TSDiff	Diffusion	128	6	1×10^{-4}	100
TSFlow	Flow	256	4	8×10^{-4}	16
Ours	**Flow**	**96**	**4**	$\mathbf{8 \times 10^{-4}}$	**1**

C.2. Hyperparameter Settings

We summarize the key hyperparameters used for Ours (RegimeFlow) and all baselines in Table 10. For Ours, we use $M = 4$ condition-modulated residual blocks and the backbone hidden size is $D = 96$. We employ a BLR-induced regime-aware prior with $(\alpha, \beta) = (5, 10)$. Specifically, we employ an exponential basis set $\{1-e^{-0.2\tau}, 1-e^{-2.0\tau}, 1-e^{-5.0\tau}\}$, augmented with polynomial terms of degree 2 and a single harmonic component. The prior scale is initialized to $\sigma_{\text{init}} = 0.247$. All models are optimized with AdamW.

C.3. Training Protocol

C.3.1. Optimization Configuration.

All experiments were conducted using 32-bit floating-point precision (FP32). The models were trained using learning rates ranging from 10^{-4} to 10^{-3} and a weight decay of 10^{-6}. To ensure training stability, we employed a maximum batch size of 2048 and applied Exponential Moving Average (EMA) to the model parameters with a decay rate of 0.995, commencing after 200 optimization steps. The learning schedule included a warmup period of 50 epochs, and validation evaluations were performed every 10 epochs.

C.3.2. Data and Training Setup.

The dataset was partitioned into training, validation, and test sets with ratios of 0.7, 0.1, and 0.2, respectively. We configured the data sequences with an input observation length of 96, a prediction horizon of 256, and a sliding window stride of 16. All models were trained for a maximum of 500 epochs, utilizing an early stopping strategy monitoring the validation MSE loss with a patience of 5 epochs.

C.4. Regime Assignment Procedure

Regime labels are assigned **per coordinate trajectory**, not per biological system. For a model with multiple state variables, each coordinate is analyzed independently, so different variables from the same system may receive different $(c_{\text{pat}}, c_{\text{freq}})$ labels. The assignment follows a two-step procedure.

Automated trajectory analysis. For each coordinate trajectory, we first apply a rule-based detector to assign the discrete pattern label c_{pat} and, when applicable, the frequency label c_{freq}. Because biological trajectories can be sinusoidal, pulse-like, square-wave, damped, or only partially observed, we combine multiple criteria rather than relying on a single test. Oscillatory trajectories are detected using peak/valley periodicity, zero-crossing regularity around the trajectory mean, and FFT-based spectral evidence for a dominant periodic component. For oscillatory coordinates, c_{freq} is estimated from the dominant periodic component; otherwise, $c_{\text{freq}} = 0$. Monotonic trajectories are identified by a persistent global trend and consistent directional changes across local segments. Stable trajectories are identified by near-zero tail slope and decreasing local variation, indicating convergence to a steady state. Coordinates that do not satisfy these criteria are assigned to the Complex category, for which we use the intercept-only fallback basis $\mathbf{\Phi} = \{\mathbf{1}\}$.

Manual validation. All automatically generated labels were manually reviewed by visually inspecting trajectory plots, with particular attention to borderline cases such as weak oscillations, damped oscillations, and slow convergence. When the automated label was clearly inconsistent with the observed trajectory shape, it was corrected.

The six fine-grained trajectory types are mapped to the four regime groups used in the paper as follows: complex $\rightarrow$ Complex; increasing-stable and decreasing-stable $\rightarrow$ Stable; oscillation $\rightarrow$ Oscillatory; monotonic increasing and monotonic decreasing $\rightarrow$ Monotonic. In the released SysBio-Traj metadata, these assignments are stored for each coordinate trajectory together with the trajectory type, bounds, and estimated period when applicable.

D. Benchmark Statistics and Specifications for 1,050 Systems Biology Models

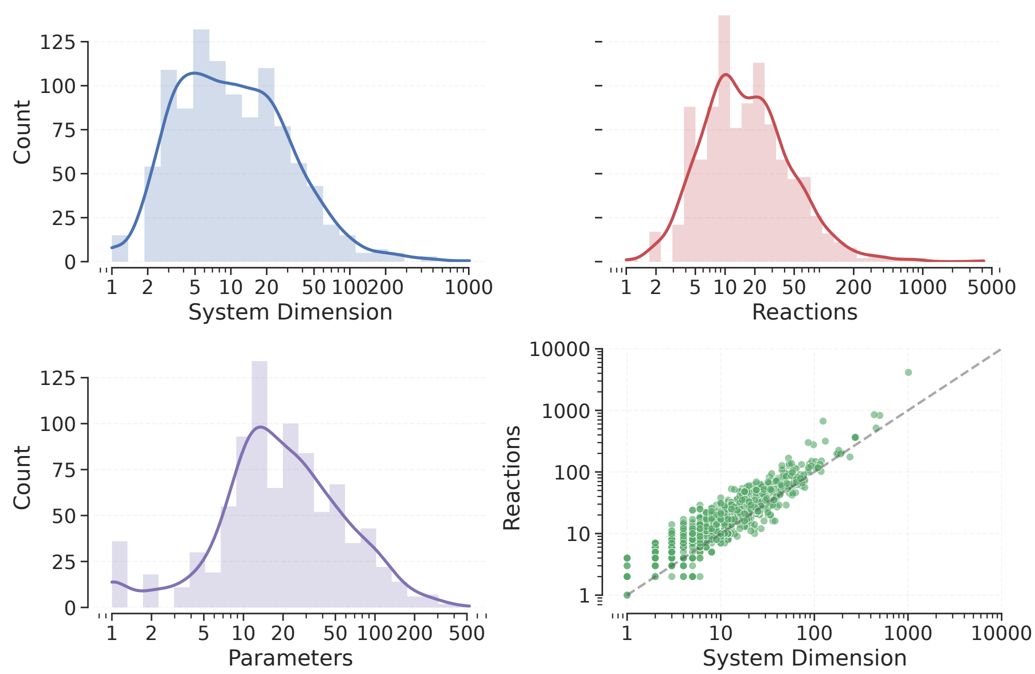

Figure 15. Statistical overview of the curated biological ODE benchmark. **Top Left:** Distribution of system dimensionality (number of state variables). **Top Right:** Distribution of reaction counts per system. **Bottom Left:** Distribution of parameter space size (number of ODE parameters). **Bottom Right:** Joint distribution (scatter plot) of system dimensions versus biological reactions.

In this section we analyze the structural and dynamical composition of our benchmark suite, which contains 1,050 distinct biological ODE systems sourced from (Malik-Sheriff et al., 2020).

D.1. Benchmark Overview

Figure 1 (left) shows that the dataset spans a broad taxonomic spectrum, with mammalian systems accounting for the majority, that particularly those modeling *Homo sapiens* and *Mus musculus*. Figure 15 illustrates the statistical profile of our benchmark, comprising 1,050 biological ODE systems. It highlights the structural heterogeneity across system dimensions, reaction counts, and parameter distributions.

To rigorously assess generalization, we adopt a strict system-wise split in which training and test sets contain disjoint biological systems. This partition forces the model to learn system invariant dynamical patterns rather than memorizing system specific behavior, ensuring that predictive performance reflects genuine extrapolation capability.

Trajectory generation. Each system is modeled in the Systems Biology Markup Language (SBML) format and simulated using the Tellurium library (Medley et al., 2018). The SBML file is parsed to recover the governing ODEs, parameters, and initial conditions; the system dynamics are then numerically integrated to produce trajectories sampled at 512 uniformly spaced time points. Regime labels are assigned per coordinate using the automated detection and manual validation procedure described in Appendix C.4, and are stored alongside the trajectory data and model source files in the released benchmark.

D.2. Systems Biology Model Specifications and Metadata

Table 11. Full list of 1,050 ODE-based models, including model ID, paper reference, biological domain, system and parameter dimensions, and Trajectory Time Span. "-" indicates unclassified taxonomy or biological process.

Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000003	Goldbeter1991	Taxonomy-Amphibia	3	15	[0.0, 60.0]
BIOMD0000000004	Goldbeter1991	Taxonomy-Amphibia	5	15	[0.0, 70.0]
BIOMD0000000005	Tyson1991	Taxonomy-Opisthokonta	6	10	[0.0, 100.0]
BIOMD0000000006	Tyson1991	Taxonomy-Opisthokonta	2	5	[0.0, 110.0]
BIOMD0000000008	Gardner1998	Taxonomy-Amphibia	5	23	[0.0, 40.0]
BIOMD0000000009	Huang1996	Taxonomy-Xenopus laevis	26	31	[0.0, 80.0]
BIOMD0000000010	Kholodenko2000	Taxonomy-Xenopus laevis	8	22	[0.0, 3500.0]
BIOMD0000000011	Levchenko2000	Taxonomy-Xenopus laevis	22	30	[0.0, 290.2]
BIOMD0000000012	Elowitz2000	Taxonomy-Escherichia coli	6	16	[0.0, 450.0]
BIOMD0000000013	Poolman2004	Taxonomy-Nicotiana tabacum	18	77	[0.0, 0.4]
BIOMD0000000014	Levchenko2000	Taxonomy-Xenopus laevis	86	300	[0.0, 241.0]
BIOMD0000000015	Curto1998	Taxonomy-Homo sapiens	16	111	[0.0, 20000.0]
BIOMD0000000016	Goldbeter1995	Taxonomy-Drosophila melanogaster	6	18	[0.0, 60.0]
BIOMD0000000017	Hoefnagel2002	Taxonomy-Lactococcus lactis	11	71	[0.0, 0.4]
BIOMD0000000018	Morrison1989	Taxonomy-Homo sapiens|KEGG Pathway-One carbon pool by folate|KEGG Pathway-Folate biosynthesis	20	107	[0.0, 500000.0]
BIOMD0000000019	Schoeberl2002	Taxonomy-Homo sapiens	93	90	[0.0, 300.0]
BIOMD0000000020	Hodgkin1952	Taxonomy-Loligo forbesii|Gene Ontology-giant axon	4	27	[0.0, 25.0]
BIOMD0000000021	Leloup1999	Taxonomy-Drosophila melanogaster	10	48	[0.0, 80.0]
BIOMD0000000022	Ueda2001	Taxonomy-Drosophila melanogaster	10	66	[0.0, 70.0]
BIOMD0000000023	Rohwer2001	Taxonomy-Saccharum officinarum	5	54	[0.0, 1949.2]
BIOMD0000000026	Markevich2004	Taxonomy-Xenopus laevis	11	16	[0.0, 1698.4]
BIOMD0000000027	Markevich2004	Taxonomy-Xenopus laevis	5	9	[0.0, 1327.2]
BIOMD0000000028	Markevich2004	Taxonomy-Xenopus laevis	16	27	[0.0, 1760.4]
BIOMD0000000029	Markevich2004	Taxonomy-Xenopus laevis	6	15	[0.0, 1843.2]
BIOMD0000000030	Markevich2004	Taxonomy-Xenopus laevis	18	32	[0.0, 1243.4]
BIOMD0000000031	Markevich2004	Taxonomy-Xenopus laevis	6	9	[0.0, 250.0]
BIOMD0000000032	Kofahl2004	Taxonomy-Saccharomyces cerevisiae|Gene Ontology-MAPK cascade	36	47	[0.0, 225.0]
BIOMD0000000033	Brown2004	Taxonomy-cellular organisms|Gene Ontology-epidermal growth factor receptor signaling pathway|Gene Ontology-neurotrophin TRK receptor signaling pathway	32	48	[0.0, 100.0]
BIOMD0000000035	Vilar2002	Taxonomy-cellular organisms	9	16	[0.0, 60.0]
BIOMD0000000036	Tyson1999	Taxonomy-Drosophila melanogaster	2	11	[0.0, 60.0]
BIOMD0000000037	Marwan2003	Taxonomy-Physarum polycephalum	11	12	[0.0, 95.6]
BIOMD0000000038	Rohwer2000	Taxonomy-Escherichia coli	13	20	[0.0, 0.004]
BIOMD0000000039	Marhl2000	Taxonomy-Xenopus laevis	5	11	[0.0, 30.0]
BIOMD0000000041	Kongas2007	Taxonomy-Oryctolagus	10	25	[0.0, 0.5]
BIOMD0000000042	Nielsen1998	Taxonomy-Saccharomyces cerevisiae	15	25	[0.0, 30.0]
BIOMD0000000043	Borghans1997	Taxonomy-Rattus	5	13	[0.0, 5.0]
BIOMD0000000044	Borghans1997	Taxonomy-Rattus	4	17	[0.0, 7.0]
BIOMD0000000045	Borghans1997	Taxonomy-Rattus	4	15	[0.0, 55.0]
BIOMD0000000046	Olsen2003	Taxonomy-Armoracia rusticana	12	15	[0.0, 1000.0]
BIOMD0000000047	Oxhamre2005	Taxonomy-Rattus	3	17	[0.0, 50.0]
BIOMD0000000048	Kholodenko1999	Taxonomy-Rattus norvegicus	23	50	[0.0, 103.2]

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Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000049	Sasagawa2005	Taxonomy-Rattus|Gene Ontology-MAPK cascade|Gene Ontology-epidermal growth factor receptor signaling pathway|Gene Ontology-Ras protein signal transduction|Gene Ontology-neurotrophin TRK receptor signaling pathway	94	234	[0.0, 1385.2]
BIOMD0000000050	Martins2003	Taxonomy-cellular organisms	14	16	[0.0, 351.8]
BIOMD0000000051	Chassagnole2002	Taxonomy-Escherichia coli|Gene Ontology-glycolytic process|Gene Ontology-pentose-phosphate shunt|Gene Ontology-phosphoenolpyruvate-dependent sugar phosphotransferase system|KEGG Pathway-Glycolysis / Gluconeogenesis|KEGG Pathway-Pentose phosphate pathway	18	142	[0.0, 800.0]
BIOMD0000000052	Brands2002	Taxonomy-cellular organisms	11	11	[0.0, 1487.2]
BIOMD0000000054	Ataullahkhanov 1996	Taxonomy-Homo sapiens|Gene Ontology-regulation of sodium ion transport|Gene Ontology-regulation of glycolytic process|Gene Ontology-AMP biosynthetic process	3	10	[0.0, 253.6]
BIOMD0000000055	Locke2005	Taxonomy-Arabidopsis	13	67	[0.0, 52.0]
BIOMD0000000056	Chen2004	Taxonomy-Saccharomyces cerevisiae	50	164	[0.0, 225.0]
BIOMD0000000057	Sneyd2002	Taxonomy-Rattus rattus	6	33	[0.0, 4.6]
BIOMD0000000058	Bindschadler 2001	Taxonomy-Rattus norvegicus|Gene Ontology-calcium-mediated signaling|Gene Ontology-inositol phosphate-mediated signaling	4	29	[0.0, 20.0]
BIOMD0000000059	Fridlyand2003	Taxonomy-Mus musculus|Gene Ontology-regulation of insulin secretion|Gene Ontology-regulation of calcium-mediated signaling	8	97	[0.0, 220000.0]
BIOMD0000000060	Keizer1996	Taxonomy-cellular organisms|Gene Ontology-ryanodine-sensitive calcium-release channel activity|Gene Ontology-calcium-mediated signaling|Gene Ontology-calcium-induced calcium release activity	4	11	[0.0, 8.0]
BIOMD0000000061	Hynne2001	Taxonomy-Saccharomyces cerevisiae	22	69	[0.0, 3.0]
BIOMD0000000062	Bhartiya2003	Taxonomy-Escherichia coli	3	18	[0.0, 168.0]
BIOMD0000000063	Galazzo1990	Taxonomy-Saccharomyces cerevisiae|Gene Ontology-glycolytic process	5	61	[0.0, 0.8]
BIOMD0000000064	Teusink2000	Taxonomy-Saccharomyces cerevisiae	19	85	[0.0, 2.4]
BIOMD0000000065	Yildirim2003	Taxonomy-Escherichia coli	8	23	[0.0, 200.4]
BIOMD0000000067	Fung2005	Taxonomy-Escherichia coli (strain K12)	7	21	[0.0, 100.0]
BIOMD0000000068	Curien2003	Taxonomy-Arabidopsis thaliana|Gene Ontology-threonine biosynthetic process|Gene Ontology-L-methionine biosynthetic process from O-phospho-L-homoserine and cystathionine|KEGG Pathway-Glycine, serine and threonine metabolism|KEGG Pathway-map00271	4	6	[0.0, 800.0]
BIOMD0000000069	Fuss2006	Taxonomy-Homo sapiens	10	20	[0.0, 100.0]

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Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000070	Holzhutter2004	Taxonomy-Homo sapiens|Gene Ontology-glycolytic process|Gene Ontology-pentose-phosphate shunt|Gene Ontology-glutathione metabolic process|KEGG Pathway-Glycolysis / Gluconeogenesis - Homo sapiens (human)|KEGG Pathway-Pentose phosphate pathway - Homo sapiens (human)	40	166	[0.0, 9.6]
BIOMD0000000071	Bakker2001	Taxonomy-Trypanosoma brucei	23	72	[0.0, 0.8]
BIOMD0000000072	Yi2003	Taxonomy-Saccharomyces cerevisiae	7	8	[0.0, 1500.0]
BIOMD0000000073	Leloup2003	Taxonomy-Mammalia	16	80	[0.0, 52.0]
BIOMD0000000074	Leloup2003	Taxonomy-Mammalia	19	95	[0.0, 70.0]
BIOMD0000000075	Xu2003	Taxonomy-Mus musculus	10	30	[0.0, 87.8]
BIOMD0000000076	Cronwright2002	Taxonomy-Saccharomyces cerevisiae	1	18	[0.0, 1.6]
BIOMD0000000077	Blum2000	Taxonomy-cellular organisms	8	10	[0.0, 12.0]
BIOMD0000000078	Leloup2003	Taxonomy-Mammalia	16	81	[0.0, 60.0]
BIOMD0000000079	Goldbeter2006	Taxonomy-Homo sapiens	3	11	[0.0, 23.0]
BIOMD0000000081	Suh2004	Taxonomy-Homo sapiens|Gene Ontology-lipid metabolic process|Gene Ontology-phospholipase C-activating G-protein coupled acetylcholine receptor signaling pathway	21	35	[0.0, 12.0]
BIOMD0000000083	Leloup2003	Taxonomy-Mammalia	19	64	[0.0, 100.0]
BIOMD0000000084	Hornberg2005	Taxonomy-Mammalia|KEGG Pathway-MAPK signaling pathway - Rattus norvegicus (rat)	8	18	[0.0, 34.0]
BIOMD0000000089	Locke2006	Taxonomy-Arabidopsis thaliana	16	78	[0.0, 55.0]
BIOMD0000000090	Wolf2001	Taxonomy-Saccharomyces cerevisiae	13	28	[0.0, 85.0]
BIOMD0000000091	Proctor2005	Taxonomy-cellular organisms	15	21	[0.0, 150.0]
BIOMD0000000093	Yamada2003	Taxonomy-Mus musculus	33	72	[0.0, 1000.0]
BIOMD0000000094	Yamada2003	Taxonomy-Mus musculus	33	70	[0.0, 4500.0]
BIOMD0000000096	Zeilinger2006	Taxonomy-Arabidopsis	19	94	[0.0, 80.0]
BIOMD0000000097	Zeilinger2006	Taxonomy-Arabidopsis	19	93	[0.0, 75.0]
BIOMD0000000098	Goldbeter1990	Taxonomy-cellular organisms	2	13	[0.0, 5.0]
BIOMD0000000099	Laub1998	Taxonomy-Dictyostelium	7	14	[0.0, 14.5]
BIOMD0000000100	Rozi2003	Taxonomy-cellular organisms|Gene Ontology-phosphorylase kinase regulator activity|KEGG Pathway-Calcium signaling pathway - Homo sapiens (human)	4	30	[0.0, 3.0]
BIOMD0000000101	Vilar2006	Taxonomy-cellular organisms|Reactome-Signaling by TGF-beta Receptor Complex|KEGG Pathway-TGF-beta signaling pathway - Homo sapiens (human)	6	9	[0.0, 1935.6]
BIOMD0000000102	Legewie2006	Taxonomy-cellular organisms	13	41	[0.0, 2000.0]
BIOMD0000000103	Legewie2006	Taxonomy-cellular organisms	17	49	[0.0, 2000.0]
BIOMD0000000104	Klipp2002	Taxonomy-cellular organisms	6	2	[0.0, 10.4]
BIOMD0000000105	Proctor2007	Taxonomy-Eukaryota	35	17	[0.0, 300000.0]
BIOMD0000000106	Yang2007	Taxonomy-Homo sapiens	25	54	[0.0, 49.6]
BIOMD0000000107	Novak1993	Taxonomy-Amphibia	9	36	[0.0, 200.0]
BIOMD0000000109	Haberichter2007	Taxonomy-Mammalia	57	61	[0.0, 1981.6]
BIOMD0000000110	Qu2003	Taxonomy-Eukaryota	15	30	[0.0, 150.0]
BIOMD0000000111	Novak2001	Taxonomy-Schizosaccharomycetaceae	10	53	[0.0, 300.0]
BIOMD0000000112	Clarke2006	Taxonomy-Neovison vison	10	17	[0.0, 600.0]
BIOMD0000000113	Dupont1992	Taxonomy-cellular organisms	4	19	[0.0, 2.5]
BIOMD0000000114	Somogyi1990	Taxonomy-Rattus	2	8	[0.0, 200.0]
BIOMD0000000115	Somogyi1990	Taxonomy-Rattus	2	8	[0.0, 15.0]

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Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000117	Dupont1991	Taxonomy-cellular organisms	2	15	[0.0, 40.0]
BIOMD0000000118	Golomb2006	Taxonomy-Rattus	5	45	[0.0, 420.0]
BIOMD0000000119	Golomb2006	Taxonomy-Rattus	9	69	[0.0, 230.0]
BIOMD0000000121	Clancy2001	Taxonomy-Cavia porcellus	6	19	[0.0, 1500.0]
BIOMD0000000122	Fisher2006	Taxonomy-Mammalia	12	23	[0.0, 300.0]
BIOMD0000000123	Fisher2006	Taxonomy-Mammalia	14	22	[0.0, 679.2]
BIOMD0000000124	Wu2006	Taxonomy-Rattus	7	57	[0.0, 800.0]
BIOMD0000000125	Komarova2005	Taxonomy-Opisthokonta	5	7	[0.0, 10.0]
BIOMD0000000126	Clancy2002	Taxonomy-Chordata	9	24	[0.0, 300.0]
BIOMD0000000128	Bertram2006	Taxonomy-Rattus	7	50	[0.0, 500.0]
BIOMD0000000131	Izhikevich2004	Taxonomy-Mammalia	2	8	[0.0, 200.0]
BIOMD0000000134	Izhikevich2004	Taxonomy-Mammalia	2	8	[0.0, 180.0]
BIOMD0000000135	Izhikevich2004	Taxonomy-Mammalia	2	8	[0.0, 200.0]
BIOMD0000000137	Sedaghat2002	Taxonomy-Homo sapiens	21	40	[0.0, 70.0]
BIOMD0000000138	Tabak2007	Taxonomy-Chordata	4	40	[0.0, 500.0]
BIOMD0000000143	Olsen2003B	Taxonomy-Mammalia	20	24	[0.0, 52.0]
BIOMD0000000144	Calzone2007	Taxonomy-Drosophila	17	46	[0.0, 300.0]
BIOMD0000000145	Wang2007	Taxonomy-cellular organisms|Gene Ontology-positive regulation of cytosolic calcium ion concentration involved in phospholipase C-activating G-protein coupled signaling pathway|KEGG Pathway-Calcium signaling pathway - Homo sapiens (human)	7	32	[0.0, 600.0]
BIOMD0000000146	Hatakeyama2003	Taxonomy-Cricetinae|Gene Ontology-regulation of MAP kinase activity|Gene Ontology-regulation of protein kinase B signaling	36	75	[0.0, 15000.0]
BIOMD0000000147	ODea2007	Taxonomy-Mus musculus	24	73	[0.0, 445.8]
BIOMD0000000148	Komarova2003	Taxonomy-Chordata	7	13	[0.0, 120.0]
BIOMD0000000149	Kim2007	Taxonomy-Homo sapiens|Gene Ontology-MAPK cascade|Gene Ontology-canonical Wnt signaling pathway|KEGG Pathway-MAPK signaling pathway - Homo sapiens (human)|KEGG Pathway-Wnt signaling pathway - Homo sapiens (human)	28	59	[0.0, 2100.0]
BIOMD0000000151	Singh2006	Taxonomy-Eukaryota|Gene Ontology-MAPK cascade|Gene Ontology-JAK-STAT cascade|KEGG Pathway-MAPK signaling pathway - Mus musculus (mouse)|KEGG Pathway-Jak-STAT signaling pathway - Mus musculus (mouse)	66	105	[0.0, 2000.0]
BIOMD0000000156	Zatorsky2006	Taxonomy-Homo sapiens	3	7	[0.0, 15.0]
BIOMD0000000157	Zatorsky2006	Taxonomy-Homo sapiens	3	8	[0.0, 17.0]
BIOMD0000000158	Zatorsky2006	Taxonomy-Homo sapiens	3	14	[0.0, 13.0]
BIOMD0000000159	Zatorsky2006	Taxonomy-Homo sapiens	3	7	[0.0, 50.0]
BIOMD0000000160	Xie2007	Taxonomy-Drosophila melanogaster	24	47	[0.0, 52.0]
BIOMD0000000161	Eungdamrong 2007	Taxonomy-cellular organisms|Gene Ontology-Ras protein signal transduction	43	131	[0.0, 250.0]
BIOMD0000000162	Hernjak2005	Taxonomy-Chordata	24	202	[0.0, 35.0]
BIOMD0000000163	Zi2007	Taxonomy-cellular organisms	16	20	[0.0, 494.0]
BIOMD0000000164	SmithAE2002	Taxonomy-Mammalia	26	64	[0.0, 20.4]
BIOMD0000000165	Saucerman2006	Taxonomy-Rattus norvegicus	36	62	[0.0, 192.2]
BIOMD0000000166	Zhu2007	Taxonomy-cellular organisms	3	22	[0.0, 1.25]

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Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000167	Mayya2005	Taxonomy-cellular organisms|KEGG Pathway-Jak-STAT signaling pathway - Homo sapiens (human)	8	14	[0.0, 5000.0]
BIOMD0000000168	Obeyesekere 1999	Taxonomy-Eukaryota	7	21	[0.0, 60.0]
BIOMD0000000169	Aguda1999	Taxonomy-Mammalia	11	32	[0.0, 1000.0]
BIOMD0000000170	Weimann2004	Taxonomy-Mammalia	7	27	[0.0, 52.0]
BIOMD0000000171	Leloup1998	Taxonomy-Drosophila melanogaster	10	41	[0.0, 50.0]
BIOMD0000000172	Pritchard2002	Taxonomy-Saccharomyces cerevisiae	17	87	[0.0, 1.2]
BIOMD0000000174	Zerial2008	Taxonomy-Homo sapiens	4	18	[0.0, 1000.0]
BIOMD0000000175	Birtwistle2007	Taxonomy-Homo sapiens	118	240	[0.0, 800.0]
BIOMD0000000176	Conant2007	Taxonomy-Saccharomyces cerevisiae| Gene Ontology-glycolytic process|KEGG Pathway-Glycolysis / Gluconeogenesis - Saccharomyces cerevisiae (budding yeast)	17	98	[0.0, 5.2]
BIOMD0000000177	Conant2007	Taxonomy-Saccharomyces cerevisiae| Gene Ontology-glycolytic process|KEGG Pathway-Glycolysis / Gluconeogenesis - Saccharomyces cerevisiae (budding yeast)	18	97	[0.0, 6.2]
BIOMD0000000178	Lebeda2008	Taxonomy-Mus musculus|Taxonomy-Rattus norvegicus|Taxonomy-Homo sapiens	5	5	[0.0, 30000.0]
BIOMD0000000179	Kim2007	Taxonomy-cellular organisms	7	18	[0.0, 30.0]
BIOMD0000000180	Kim2007	Taxonomy-cellular organisms	8	20	[0.0, 80.0]
BIOMD0000000181	Sriram2007	Taxonomy-Opisthokonta	6	25	[0.0, 225.0]
BIOMD0000000182	Neves2008	Taxonomy-Rattus norvegicus|Gene Ontology-MAPK cascade|Gene Ontology-activation of adenylate cyclase activity	37	83	[0.0, 1643.6]
BIOMD0000000184	Lavrentovich 2008	Taxonomy-Mammalia|Gene Ontology-calcium-mediated signaling	3	14	[0.0, 400.0]
BIOMD0000000185	Locke2008	Taxonomy-Mammalia	8	19	[0.0, 55.0]
BIOMD0000000188	Proctor2008	Taxonomy-Homo sapiens	18	21	[0.0, 40000.0]
BIOMD0000000189	Proctor2008	Taxonomy-Homo sapiens	16	16	[0.0, 50000.0]
BIOMD0000000190	Caso2006	Taxonomy-Mammalia|Taxonomy-Rodentia	11	56	[0.0, 4000.0]
BIOMD0000000191	ez2008	Taxonomy-Mammalia	2	18	[0.0, 120.0]
BIOMD0000000192	rlich2003	Taxonomy-Homo sapiens	11	16	[0.0, 60.0]
BIOMD0000000195	Tyson2001	Taxonomy-Saccharomyces cerevisiae	11	37	[0.0, 330.0]
BIOMD0000000197	Bartholome2007	Taxonomy-Mammalia	5	12	[0.0, 5000.0]
BIOMD0000000198	Stone1996	Taxonomy-Bos taurus	11	16	[0.0, 1.2]
BIOMD0000000199	Santolini2001	Taxonomy-Rattus norvegicus	9	10	[0.0, 8.0]
BIOMD0000000201	Goldbeter2008	Taxonomy-Amniota	22	71	[0.0, 225.0]
BIOMD0000000202	ChenXF2008	Taxonomy-cellular organisms	10	38	[0.0, 60.0]
BIOMD0000000203	Chickarmane 2006	Taxonomy-Mus musculus|Taxonomy-Homo sapiens	5	32	[0.0, 5.2]
BIOMD0000000204	Chickarmane 2006	Taxonomy-Mus musculus|Taxonomy-Homo sapiens	5	32	[0.0, 5.2]
BIOMD0000000205	Ung2008	Taxonomy-Eukaryota	194	313	[0.0, 5000.0]
BIOMD0000000206	Wolf2000	Taxonomy-Saccharomyces cerevisiae	9	18	[0.0, 0.35]
BIOMD0000000207	Romond1999	Taxonomy-Eukaryota	6	32	[0.0, 2500.0]
BIOMD0000000208	Deineko2003	Taxonomy-Mammalia	6	18	[0.0, 1000.0]
BIOMD0000000209	Chickarmane 2008	Taxonomy-Homo sapiens	8	49	[0.0, 94.0]
BIOMD0000000210	Chickarmane 2008	Taxonomy-Homo sapiens	8	46	[0.0, 1329.2]
BIOMD0000000211	Albert2005	Taxonomy-Trypanosoma brucei	24	81	[0.0, 1.2]

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Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000212	Curien2009	Taxonomy-Arabidopsis thaliana	8	64	[0.0, 1933.2]
BIOMD0000000213	Nijhout2004	Taxonomy-Mammalia	6	38	[0.0, 0.3]
BIOMD0000000214	Akman2008	Taxonomy-Neurospora crassa	16	38	[0.0, 55.0]
BIOMD0000000216	Hong2009	Taxonomy-Mammalia	5	18	[0.0, 55.0]
BIOMD0000000217	Bruggeman2005	Taxonomy-Escherichia coli	12	98	[0.0, 4.4]
BIOMD0000000218	Singh2006	Taxonomy-Mycobacterium tuberculosis	9	66	[0.0, 150.0]
BIOMD0000000219	Singh2006	Taxonomy-Mycobacterium tuberculosis	9	70	[0.0, 197.6]
BIOMD0000000220	Albeck2008	Taxonomy-Homo sapiens	58	77	[0.0, 1000000.0]
BIOMD0000000221	Singh2006	Taxonomy-Escherichia coli	8	49	[0.0, 2.5]
BIOMD0000000222	Singh2006	Taxonomy-Escherichia coli	8	49	[0.0, 48.6]
BIOMD0000000223	Borisov2009	Taxonomy-Homo sapiens	86	162	[0.0, 972.8]
BIOMD0000000224	Meyer1991	Taxonomy-Rattus rattus	4	11	[0.0, 90.0]
BIOMD0000000225	Westermark2003	Taxonomy-Mammalia	2	19	[0.0, 500.0]
BIOMD0000000226	Radulescu2008	Taxonomy-Mus musculus	14	31	[0.0, 18000.0]
BIOMD0000000227	Radulescu2008	Taxonomy-Mus musculus	45	88	[0.0, 200000.0]
BIOMD0000000228	Swat2004	Taxonomy-Mammalia	9	40	[0.0, 1970.0]
BIOMD0000000229	Ma2002	Taxonomy-Dictyostelium discoideum|KEGG Pathway-Bacterial chemotaxis - Xanthomonas campestris pv. campestris ATCC 33913	7	14	[0.0, 17.0]
BIOMD0000000230	Ihekwaba2004	Taxonomy-Mus musculus	24	64	[0.0, 16000.0]
BIOMD0000000231	Valero2006	Taxonomy-Bos taurus	5	9	[0.0, 200.4]
BIOMD0000000233	Wilhelm2009	Taxonomy-cellular organisms	2	4	[0.0, 10.0]
BIOMD0000000236	Westermark2003	Taxonomy-Mammalia	3	26	[0.0, 420.0]
BIOMD0000000237	Schaber2006	Taxonomy-Saccharomyces cerevisiae	24	46	[0.0, 600.0]
BIOMD0000000238	Overgaard2007	Taxonomy-Macaca fascicularis	3	56	[0.0, 350.0]
BIOMD0000000239	Jiang2007	Taxonomy-Homo sapiens	59	306	[0.0, 464.4]
BIOMD0000000240	Veening2008	Taxonomy-Octodon degus|Taxonomy-Bacillus subtilis	6	23	[0.0, 50000.0]
BIOMD0000000241	Shi1993	Taxonomy-Homo sapiens	5	19	[0.0, 30.0]
BIOMD0000000242	Bai2003	Taxonomy-Murinae	6	26	[0.0, 75.0]
BIOMD0000000243	Neumann2010	Taxonomy-Homo sapiens	23	17	[0.0, 800.0]
BIOMD0000000244	Kotte2010	Taxonomy-Escherichia coli	47	213	[0.0, 16000.0]
BIOMD0000000247	Ralser2007	Taxonomy-Saccharomyces cerevisiae	24	135	[0.0, 15.8]
BIOMD0000000249	Restif2006	Taxonomy-Bordetella pertussis|Taxonomy-Homo sapiens	9	21	[0.0, 20.0]
BIOMD0000000250	Nakakuki2010	Taxonomy-Homo sapiens	47	141	[0.0, 1250.0]
BIOMD0000000253	Teusink1998	Taxonomy-Saccharomyces cerevisiae	3	26	[0.0, 7.0]
BIOMD0000000254	Bier2000	Taxonomy-Saccharomyces cerevisiae	4	7	[0.0, 140.0]
BIOMD0000000255	Chen2009	Taxonomy-Homo sapiens	500	239	[0.0, 10000.0]
BIOMD0000000256	Rehm2006	Taxonomy-Homo sapiens|Brenda Tissue Ontology-HeLa cell	27	99	[0.0, 28.8]
BIOMD0000000257	Piedrafita2010	Taxonomy-cellular organisms	8	19	[0.0, 400.0]
BIOMD0000000258	Ortega2006	Taxonomy-cellular organisms	3	9	[0.0, 21.6]
BIOMD0000000259	Tiago2010	Taxonomy-Mus musculus	17	29	[0.0, 400.0]
BIOMD0000000260	Tiago2010	Taxonomy-Mus musculus	17	29	[0.0, 600.0]
BIOMD0000000261	Tiago2010	Taxonomy-Mus musculus	17	29	[0.0, 600.0]
BIOMD0000000262	Fujita2010	Taxonomy-Rattus norvegicus	9	24	[0.0, 1300.0]
BIOMD0000000263	Fujita2010	Taxonomy-Rattus norvegicus	9	24	[0.0, 1962.0]
BIOMD0000000264	Fujita2010	Taxonomy-Rattus norvegicus	11	26	[0.0, 738.4]
BIOMD0000000265	Conradie2010	Taxonomy-Mammalia	23	88	[0.0, 30.0]
BIOMD0000000267	Lebeda2008	Taxonomy-Mus musculus|Taxonomy-Rattus norvegicus|Taxonomy-Homo sapiens	4	4	[0.0, 398.8]

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Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000268	Reed2008	Taxonomy-Homo sapiens|Brenda Tissue Ontology-hepatocyte	34	159	[0.0, 60.0]
BIOMD0000000269	Liu2010	Taxonomy-Arabidopsis thaliana	9	32	[0.0, 100.0]
BIOMD0000000270	Schilling2009	Taxonomy-Mus musculus	33	39	[0.0, 80.0]
BIOMD0000000271	Becker2010	Taxonomy-Murinae|Brenda Tissue Ontology-hematopoietic cell line	6	10	[0.0, 600.0]
BIOMD0000000272	Becker2010	Taxonomy-Murinae|Brenda Tissue Ontology-hematopoietic cell line	6	9	[0.0, 50000.0]
BIOMD0000000273	Pokhilko2010	Taxonomy-Arabidopsis thaliana	19	99	[0.0, 55.0]
BIOMD0000000274	Rattanakul2003	Taxonomy-Chordata	3	13	[0.0, 8000.0]
BIOMD0000000275	Goldbeter2007	Taxonomy-Vertebrata	5	24	[0.0, 43.2]
BIOMD0000000276	Shrestha2010	Taxonomy-Homo sapiens	3	21	[0.0, 700.0]
BIOMD0000000277	Shrestha2010	Taxonomy-Homo sapiens	3	21	[0.0, 800.0]
BIOMD0000000281	Chance1960	Taxonomy-Homo sapiens|Brenda Tissue Ontology-ascites tumor cell	32	54	[0.0, 65.6]
BIOMD0000000282	Chance1952	Taxonomy-Equus caballus	5	4	[0.0, 0.4]
BIOMD0000000283	Chance1943	Taxonomy-Armoracia rusticana	4	2	[0.0, 32.6]
BIOMD0000000284	Hofmeyer1986	Taxonomy-Bacteria	6	0	[0.0, 48.2]
BIOMD0000000286	Proctor2010	Taxonomy-Mammalia	54	58	[0.0, 75000.0]
BIOMD0000000287	Passos2010	Taxonomy-Homo sapiens	21	36	[0.0, 300000.0]
BIOMD0000000288	Wang2009	Taxonomy-Homo sapiens	19	45	[0.0, 136.4]
BIOMD0000000289	Alexander2010	Taxonomy-Metazoa	4	15	[0.0, 150.0]
BIOMD0000000290	Alexander2010	Taxonomy-Metazoa	4	16	[0.0, 200.0]
BIOMD0000000291	Nikolaev2005	Taxonomy-Eukaryota	10	13	[0.0, 245.0]
BIOMD0000000292	Rovers1995	Taxonomy-Viridiplantae	3	6	[0.0, 500.0]
BIOMD0000000293	Proctor2010	Taxonomy-Homo sapiens	131	92	[0.0, 877.6]
BIOMD0000000294	Restif2007	Taxonomy-Bordetella pertussis|Taxonomy-Homo sapiens	11	22	[0.0, 100.0]
BIOMD0000000295	Akman2008	Taxonomy-Neurospora crassa	5	17	[0.0, 50.0]
BIOMD0000000296	Balagaddé2008	Taxonomy-Escherichia coli	5	12	[0.0, 350.0]
BIOMD0000000297	Ciliberto2003	Taxonomy-Schizosaccharomyces pombe	19	81	[0.0, 290.0]
BIOMD0000000298	Leloup1999	Taxonomy-Drosophila melanogaster	10	39	[0.0, 55.0]
BIOMD0000000299	Leloup1999	Taxonomy-Neurospora crassa	3	11	[0.0, 50.0]
BIOMD0000000300	Schmierer2010	Taxonomy-Homo sapiens	3	16	[0.0, 7.6]
BIOMD0000000301	Friedland2009	Taxonomy-Escherichia coli (strain K12)	8	36	[0.0, 980.0]
BIOMD0000000303	Liu2011	Taxonomy-Homo sapiens	42	90	[0.0, 50000.0]
BIOMD0000000304	Plant1981	Taxonomy-Aplysia|Brenda Tissue Ontology-abdominal ganglion|Brenda Tissue Ontology-neuron	5	34	[0.0, 23000.0]
BIOMD0000000305	Kolomeisky2003	Taxonomy-Eukaryota	6	15	[0.0, 0.3]
BIOMD0000000309	Tyson2003	Taxonomy-cellular organisms	1	9	[0.0, 2000.0]
BIOMD0000000312	Tyson2003	Taxonomy-cellular organisms	2	5	[0.0, 40.0]
BIOMD0000000313	Raia2010	Taxonomy-Homo sapiens|Reactome-Signaling by Interleukins|KEGG Pathway-Jak-STAT signaling pathway - Homo sapiens (human)	15	18	[0.0, 50000.0]
BIOMD0000000314	Raia2011	Taxonomy-Homo sapiens|Reactome-Signaling by Interleukins|KEGG Pathway-Jak-STAT signaling pathway - Homo sapiens (human)	11	12	[0.0, 50000.0]
BIOMD0000000315	Montagne2011	Taxonomy-cellular organisms	19	47	[0.0, 250.0]
BIOMD0000000318	Yao2008	Taxonomy-Mammalia	7	31	[0.0, 20.0]
BIOMD0000000319	Decroly1982	Taxonomy-cellular organisms	3	7	[0.0, 220.0]
BIOMD0000000320	Grange2001	Taxonomy-Rattus norvegicus	9	31	[0.0, 18.4]
BIOMD0000000321	Grange2001	Taxonomy-Rattus norvegicus	3	18	[0.0, 12.0]

Continued on next page

Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000322	Kim2011	Taxonomy-cellular organisms	4	14	[0.0, 160.0]
BIOMD0000000323	Kim2011	Taxonomy-cellular organisms	3	6	[0.0, 25.0]
BIOMD0000000325	Palini2011	Taxonomy-Saccharomyces cerevisiae	5	15	[0.0, 3000.0]
BIOMD0000000326	DellOrco2009	Taxonomy-Mus musculus	71	96	[0.0, 0.4]
BIOMD0000000327	Whitcomb2004	Taxonomy-Homo sapiens|Brenda Tissue Ontology-pancreatic duct	4	44	[0.0, 998.4]
BIOMD0000000328	Bucher2011	Taxonomy-Homo sapiens	18	30	[0.0, 2000.0]
BIOMD0000000329	Kummer2000	Taxonomy-Rattus norvegicus	3	12	[0.0, 35.0]
BIOMD0000000330	Larsen2004	Taxonomy-Rattus norvegicus	5	21	[0.0, 50.0]
BIOMD0000000331	Larsen2004	Taxonomy-Rattus norvegicus	7	27	[0.0, 31.0]
BIOMD0000000332	Bungay2006	Taxonomy-Homo sapiens	78	113	[0.0, 423.8]
BIOMD0000000333	Bungay2006	Taxonomy-Homo sapiens	54	75	[0.0, 747.2]
BIOMD0000000334	Bungay2003	Taxonomy-Homo sapiens	74	110	[0.0, 268.8]
BIOMD0000000337	Pfeiffer2001	Taxonomy-cellular organisms	3	2	[0.0, 36.4]
BIOMD0000000342	Zi2011	Taxonomy-Homo sapiens	19	34	[0.0, 8.0]
BIOMD0000000344	Proctor2011	Taxonomy-Chordata	50	65	[0.0, 759.8]
BIOMD0000000346	FitzHugh1961	Taxonomy-cellular organisms|Brenda Tissue Ontology-nerve	2	4	[0.0, 25.0]
BIOMD0000000347	Bachmann2011	Taxonomy-Mus musculus|Brenda Tissue Ontology-erythroid progenitor cell	26	32	[0.0, 519.0]
BIOMD0000000348	Fridlyand2010	Taxonomy-Homo sapiens	7	97	[0.0, 1939.6]
BIOMD0000000350	Troein2011	Taxonomy-Ostreococcus tauri	14	29	[0.0, 55.0]
BIOMD0000000351	Vernoux2011	Taxonomy-Arabidopsis	6	23	[0.0, 1448.6]
BIOMD0000000352	Vernoux2011	Taxonomy-Arabidopsis	6	23	[0.0, 1750.0]
BIOMD0000000354	Abell2011	Taxonomy-Drosophila	6	20	[0.0, 88.0]
BIOMD0000000355	Abell2011	Taxonomy-Drosophila	9	22	[0.0, 90.0]
BIOMD0000000357	Lee2010	Taxonomy-Eukaryota	9	12	[0.0, 582.0]
BIOMD0000000358	Stortelder1997	Taxonomy-Eukaryota	12	14	[0.0, 10.4]
BIOMD0000000359	Panteleev2002	Taxonomy-Eukaryota	9	15	[0.0, 1455.2]
BIOMD0000000360	Panteleev2002B	Taxonomy-Eukaryota	9	15	[0.0, 1497.6]
BIOMD0000000361	Panteleev2002	Taxonomy-Eukaryota	8	9	[0.0, 20.0]
BIOMD0000000363	Lee2010	Taxonomy-Eukaryota	4	4	[0.0, 1749.4]
BIOMD0000000364	Lee2010	Taxonomy-Homo sapiens	14	22	[0.0, 5000.0]
BIOMD0000000366	Orfao2008	Taxonomy-Eukaryota	12	14	[0.0, 23.8]
BIOMD0000000368	Beltrami1995	Taxonomy-Eukaryota	8	10	[0.0, 29.4]
BIOMD0000000369	Beltrami1995	Taxonomy-Eukaryota	7	10	[0.0, 50.0]
BIOMD0000000370	Vinod2011	Taxonomy-Saccharomyces cerevisiae	32	105	[0.0, 150.0]
BIOMD0000000371	DeVries2000	Taxonomy-Homo sapiens	3	24	[0.0, 4000.0]
BIOMD0000000372	Tolic2000	Taxonomy-Homo sapiens	6	29	[0.0, 250.0]
BIOMD0000000373	Bertram2004	Taxonomy-Homo sapiens	7	113	[0.0, 50000.0]
BIOMD0000000374	Bertram1995	Taxonomy-Homo sapiens	5	50	[0.0, 30000.0]
BIOMD0000000375	Mears1997	Taxonomy-Homo sapiens	5	61	[0.0, 31000.0]
BIOMD0000000376	Bertram2007	Taxonomy-Homo sapiens	11	151	[0.0, 5000.0]
BIOMD0000000377	Bertram2000	Taxonomy-Homo sapiens	4	32	[0.0, 5500.0]
BIOMD0000000378	Chay1997	Taxonomy-Homo sapiens	6	49	[0.0, 40.0]
BIOMD0000000379	DallaMan2007	Taxonomy-Homo sapiens	12	57	[0.0, 944.2]
BIOMD0000000382	Sturis1991	Taxonomy-Homo sapiens	6	13	[0.0, 250.0]
BIOMD0000000383	Arnold2011	Taxonomy-Viridiplantae	2	11	[0.0, 10.2]
BIOMD0000000384	Arnold2011	Taxonomy-Viridiplantae	2	14	[0.0, 30.0]
BIOMD0000000385	Arnold2011	Taxonomy-Viridiplantae	2	20	[0.0, 2.0]
BIOMD0000000386	Arnold2011	Taxonomy-Viridiplantae	2	15	[0.0, 1176.8]
BIOMD0000000387	Arnold2011	Taxonomy-Viridiplantae	2	17	[0.0, 24.6]
BIOMD0000000388	Arnold2011	Taxonomy-Viridiplantae	5	16	[0.0, 1226.4]
BIOMD0000000389	Arnold2011	Taxonomy-Viridiplantae	19	19	[0.0, 800.0]

Continued on next page

Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000390	Arnold2011	Taxonomy-Viridiplantae	7	19	[0.0, 34.4]
BIOMD0000000391	Arnold2011	Taxonomy-Viridiplantae	16	75	[0.0, 0.3]
BIOMD0000000392	Arnold2011	Taxonomy-Viridiplantae|Taxonomy-Mammalia	23	135	[0.0, 31.2]
BIOMD0000000393	Arnold2011	Taxonomy-Viridiplantae	28	181	[0.0, 350.0]
BIOMD0000000394	Sivakumar2011	Taxonomy-Mammalia	23	38	[0.0, 16.0]
BIOMD0000000395	Sivakumar2011	Taxonomy-Mammalia	23	25	[0.0, 81.6]
BIOMD0000000396	Sivakumar2011	Taxonomy-Mammalia	36	56	[0.0, 1668.6]
BIOMD0000000397	Sivakumar2011	Taxonomy-Mammalia	50	56	[0.0, 208.0]
BIOMD0000000398	Sivakumar2011	Taxonomy-Mammalia	27	37	[0.0, 6.8]
BIOMD0000000399	Jenkinson2011	Taxonomy-Homo sapiens	93	90	[0.0, 437.2]
BIOMD0000000400	Cooling2007	Taxonomy-Mus musculus	13	55	[0.0, 10000.0]
BIOMD0000000401	Ayati2010	Taxonomy-Chordata	3	15	[0.0, 153.2]
BIOMD0000000402	Ayati2010	Taxonomy-Chordata	4	21	[0.0, 5000.0]
BIOMD0000000405	Cookson2011	Taxonomy-cellular organisms	6	5	[0.0, 47.2]
BIOMD0000000406	Moriya2011	Taxonomy-Schizosaccharomyces pombe	24	118	[0.0, 190.0]
BIOMD0000000407	Schliemann2011	Taxonomy-Homo sapiens|Brenda Tissue Ontology-rhabdomyosarcoma cell line	47	106	[0.0, 6000.0]
BIOMD0000000409	Queralt2006	Taxonomy-Saccharomyces cerevisiae	13	54	[0.0, 521.6]
BIOMD0000000410	Wegner2012	Taxonomy-Mus musculus|Brenda Tissue Ontology-hepatocyte	53	151	[0.0, 300.0]
BIOMD0000000411	Heiland2012	Taxonomy-Chlamydomonas reinhardtii	7	35	[0.0, 4.8]
BIOMD0000000412	Pokhilko2012	Taxonomy-Arabidopsis thaliana	79	120	[0.0, 60.0]
BIOMD0000000413	Band2012	Taxonomy-Arabidopsis	5	10	[0.0, 175.0]
BIOMD0000000414	Band2012	Taxonomy-Arabidopsis	1	4	[0.0, 200.0]
BIOMD0000000416	Muraro2011	Taxonomy-Arabidopsis thaliana|Gene Ontology-auxin-activated signaling pathway|Gene Ontology-regulation of cytokinin-activated signaling pathway	32	50	[0.0, 250.0]
BIOMD0000000417	Ratushny2012	Taxonomy-Saccharomyces cerevisiae	1	12	[0.0, 67.6]
BIOMD0000000418	Ratushny2012	Taxonomy-Saccharomyces cerevisiae	1	11	[0.0, 134.8]
BIOMD0000000419	Ratushny2012	Taxonomy-Saccharomyces cerevisiae	3	13	[0.0, 134.8]
BIOMD0000000420	Ratushny2012	Taxonomy-Saccharomyces cerevisiae	2	13	[0.0, 90.0]
BIOMD0000000421	Ratushny2012	Taxonomy-Saccharomyces cerevisiae	2	13	[0.0, 90.0]
BIOMD0000000422	Middleton2012	Taxonomy-Arabidopsis	22	47	[0.0, 141.0]
BIOMD0000000423	Nyman2012	Taxonomy-Murinae|Brenda Tissue Ontology-adipocyte	9	19	[0.0, 26.0]
BIOMD0000000424	Faratian2009	Taxonomy-Homo sapiens	55	114	[0.0, 1537.6]
BIOMD0000000425	Tan2012	Taxonomy-Escherichia coli	1	6	[0.0, 21.2]
BIOMD0000000426	Mosca2012	Taxonomy-Homo sapiens|Brenda Tissue Ontology-HeLa cell	21	186	[0.0, 0.0005]
BIOMD0000000427	Bianconi2012	Taxonomy-Homo sapiens	21	54	[0.0, 1500.0]
BIOMD0000000428	Achcar2012	Taxonomy-Trypanosoma brucei	26	84	[0.0, 12.2]
BIOMD0000000429	Schaber2012	Taxonomy-Saccharomyces cerevisiae	15	88	[0.0, 5000.0]
BIOMD0000000430	Sarma2012	Taxonomy-cellular organisms	27	45	[0.0, 1380.4]
BIOMD0000000431	Sarma2012	Taxonomy-cellular organisms	27	45	[0.0, 1275.2]
BIOMD0000000432	Sarma2012	Taxonomy-cellular organisms	11	26	[0.0, 916.0]
BIOMD0000000433	Sarma2012	Taxonomy-cellular organisms	11	26	[0.0, 535.2]
BIOMD0000000434	McAuley2012	Taxonomy-Homo sapiens	26	43	[0.0, 500.0]
BIOMD0000000435	deBack2012	Taxonomy-Murinae	4	18	[0.0, 22.8]
BIOMD0000000436	Gupta2009	Taxonomy-Mammalia	12	77	[0.0, 300.0]
BIOMD0000000439	Smith2009	Taxonomy-Schizosaccharomyces pombe	20	19	[0.0, 50.0]
BIOMD0000000440	Sarma2012	Taxonomy-cellular organisms	11	25	[0.0, 2200.0]
BIOMD0000000441	Sarma2012	Taxonomy-cellular organisms	11	24	[0.0, 800.0]
BIOMD0000000442	Sarma2012B	Taxonomy-cellular organisms	13	30	[0.0, 850.0]

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Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000443	Sarma2012	Taxonomy-cellular organisms	19	47	[0.0, 2800.0]
BIOMD0000000444	Sarma2012	Taxonomy-cellular organisms	18	47	[0.0, 5000.0]
BIOMD0000000445	Pokhilko2013	Taxonomy-Arabidopsis thaliana	32	143	[0.0, 62.0]
BIOMD0000000446	Erguler2013	Taxonomy-cellular organisms	27	94	[0.0, 548.8]
BIOMD0000000447	Venkatraman 2012	Taxonomy-Rattus rattus	13	23	[0.0, 250.0]
BIOMD0000000448	nnmark2013	Taxonomy-Homo sapiens	27	68	[0.0, 100.0]
BIOMD0000000449	nnmark2013	Taxonomy-Homo sapiens	27	68	[0.0, 38.0]
BIOMD0000000452	Bidkhori2012	Taxonomy-cellular organisms	109	187	[0.0, 800.0]
BIOMD0000000453	Bidkhori2012	Taxonomy-cellular organisms	109	188	[0.0, 26.4]
BIOMD0000000458	Smallbone2013	Taxonomy-Escherichia coli	2	10	[0.0, 5.8]
BIOMD0000000459	Liebal2012	Taxonomy-Bacillus subtilis	3	7	[0.0, 1996.4]
BIOMD0000000460	Liebal2012	Taxonomy-Bacillus subtilis	3	7	[0.0, 300000.0]
BIOMD0000000461	Liebal2012	Taxonomy-Bacillus subtilis	3	6	[0.0, 157.6]
BIOMD0000000462	Proctor2012	Taxonomy-Homo sapiens	4	9	[0.0, 700.0]
BIOMD0000000463	Heldt2012	Taxonomy-Influenza A virus|Taxonomy-Mammalia	35	55	[0.0, 75.0]
BIOMD0000000464	Koo2013	Taxonomy-Homo sapiens|Gene Ontology-calcium ion import|Reactome-eNOS synthesizes NO	14	28	[0.0, 1985.0]
BIOMD0000000465	Koo2013	Taxonomy-Homo sapiens	16	23	[0.0, 1744.2]
BIOMD0000000466	Koo2013	Taxonomy-Homo sapiens	34	60	[0.0, 5000.0]
BIOMD0000000467	Koo2013	Taxonomy-Homo sapiens	19	18	[0.0, 2000.0]
BIOMD0000000468	Koo2013	Taxonomy-Homo sapiens	79	135	[0.0, 1956.8]
BIOMD0000000475	Amara2013	Taxonomy-Saccharomyces cerevisiae	23	26	[0.0, 1985.6]
BIOMD0000000476	Adams2012	Taxonomy-Arabidopsis thaliana	16	83	[0.0, 55.0]
BIOMD0000000477	Mol2013	Taxonomy-Homo sapiens|Reactome-Signaling by EGFR	43	82	[0.0, 30.0]
BIOMD0000000478	Besozzi2012	Taxonomy-Saccharomyces cerevisiae|Gene Ontology-protein kinase A signaling|Gene Ontology-cAMP-mediated signaling|Pathway Ontology-protein kinase A (PKA) signaling pathway	30	39	[0.0, 155.0]
BIOMD0000000479	Croft2013	Taxonomy-Saccharomyces cerevisiae	28	42	[0.0, 100.0]
BIOMD0000000481	tzel2012	Taxonomy-Bos taurus	16	69	[0.0, 45.0]
BIOMD0000000482	Noguchi2013	Taxonomy-Rattus	23	56	[0.0, 100000.0]
BIOMD0000000484	Cao2013	Taxonomy-cellular organisms	2	2	[0.0, 276.8]
BIOMD0000000485	Cao2013	Taxonomy-cellular organisms	2	8	[0.0, 4.2]
BIOMD0000000488	Proctor2013	Taxonomy-Mammalia|Pathway Ontology-Alzheimer disease pathway	64	73	[0.0, 907.4]
BIOMD0000000489	Sharp2013	Taxonomy-Mus musculus	35	87	[0.0, 30000.0]
BIOMD0000000490	Demin2013	Taxonomy-Homo sapiens	33	263	[0.0, 150.0]
BIOMD0000000491	Pathak2013	Taxonomy-Viridiplantae	57	172	[0.0, 56.0]
BIOMD0000000492	Pathak2013	Taxonomy-Viridiplantae	52	176	[0.0, 36.0]
BIOMD0000000493	Schittler2010	Taxonomy-cellular organisms	3	23	[0.0, 850.0]
BIOMD0000000502	Messiha2013	Taxonomy-Saccharomyces cerevisiae	8	52	[0.0, 800.0]
BIOMD0000000503	Messiha2013	Taxonomy-Saccharomyces cerevisiae|Gene Ontology-glycolytic process|Gene Ontology-pentose-phosphate shunt	28	192	[0.0, 256.2]
BIOMD0000000504	Proctor2013	Taxonomy-Homo sapiens	73	131	[0.0, 5000.0]
BIOMD0000000505	vanEunen2013	Taxonomy-Rattus	45	224	[0.0, 11.2]
BIOMD0000000506	vanEunen2013	Taxonomy-Rattus	45	224	[0.0, 70.0]
BIOMD0000000507	Gardner2000	Taxonomy-Escherichia coli	3	11	[0.0, 8.0]
BIOMD0000000508	Barrack2014	Taxonomy-Mammalia	16	48	[0.0, 52.5]
BIOMD0000000509	Barrack2014	Taxonomy-Mammalia	16	48	[0.0, 60.0]

Continued on next page

Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000510	Kerkhoven2013	Taxonomy-Trypanosoma brucei|Gene Ontology-glycolytic process|Gene Ontology-pentose-phosphate shunt	38	156	[0.0, 13.0]
BIOMD0000000511	Kerkhoven2013	Taxonomy-Trypanosoma brucei|Gene Ontology-glycolytic process|Gene Ontology-pentose-phosphate shunt	37	154	[0.0, 39.2]
BIOMD0000000512	Benson2014	Taxonomy-Homo sapiens	39	155	[0.0, 1490.4]
BIOMD0000000513	Kerkhoven2013	Taxonomy-Trypanosoma brucei	24	90	[0.0, 14.8]
BIOMD0000000514	Kerkhoven2013	Taxonomy-Trypanosoma brucei|Gene Ontology-glycolytic process|Gene Ontology-pentose-phosphate shunt	37	150	[0.0, 40.0]
BIOMD0000000515	Kerkhoven2013	Taxonomy-Trypanosoma brucei|Gene Ontology-glycolytic process|Gene Ontology-pentose-phosphate shunt	40	181	[0.0, 34.8]
BIOMD0000000516	Kerkhoven2013	Taxonomy-Trypanosoma brucei|Gene Ontology-glycolytic process|Gene Ontology-pentose-phosphate shunt	39	179	[0.0, 392.4]
BIOMD0000000517	Smallbone2013	Taxonomy-Murinae	4	20	[0.0, 187.8]
BIOMD0000000521	Ribba2012	Taxonomy-Homo sapiens	4	10	[0.0, 500.0]
BIOMD0000000522	Muraro2014	Taxonomy-Arabidopsis	16	69	[0.0, 6.0]
BIOMD0000000523	Kallenberger 2014	Taxonomy-Homo sapiens	18	12	[0.0, 672.6]
BIOMD0000000524	Kallenberger 2014	Taxonomy-Homo sapiens	18	12	[0.0, 900.0]
BIOMD0000000525	Kallenberger 2014	Taxonomy-Homo sapiens	18	15	[0.0, 758.8]
BIOMD0000000526	Kallenberger 2014	Taxonomy-Homo sapiens	18	15	[0.0, 1200.0]
BIOMD0000000528	Fribourg2014	Taxonomy-H1N1 swine influenza virus|Taxonomy-Homo sapiens	12	49	[0.0, 300.0]
BIOMD0000000529	Fribourg2014	Taxonomy-Influenza A virus (A/New Caledonia/20/1999(H1N1))|Taxonomy-Homo sapiens	12	49	[0.0, 300.0]
BIOMD0000000530	Schmitz2014	Taxonomy-Homo sapiens	10	17	[0.0, 5.4]
BIOMD0000000534	Dwivedi2014	Taxonomy-Homo sapiens|Gene Ontology-JAK-STAT cascade	39	51	[0.0, 400.0]
BIOMD0000000535	Dwivedi2014	Taxonomy-Homo sapiens|Gene Ontology-JAK-STAT cascade	39	51	[0.0, 0.4]
BIOMD0000000537	Dwivedi2014	Taxonomy-Homo sapiens|Gene Ontology-JAK-STAT cascade	40	53	[0.0, 200.0]
BIOMD0000000539	ois2005	Taxonomy-cellular organisms	6	11	[0.0, 400.0]
BIOMD0000000540	Yugi2014	Taxonomy-cellular organisms	21	22	[0.0, 100.0]
BIOMD0000000541	Yugi2014	Taxonomy-cellular organisms	30	44	[0.0, 3000.0]
BIOMD0000000543	Qi2013	Taxonomy-cellular organisms|Gene Ontology-interferon-gamma-mediated signaling pathway|Gene Ontology-interleukin-6-mediated signaling pathway	105	204	[0.0, 8000.0]
BIOMD0000000544	Qi2013	Taxonomy-cellular organisms|Gene Ontology-interferon-gamma-mediated signaling pathway|Gene Ontology-interleukin-6-mediated signaling pathway	102	192	[0.0, 8000.0]
BIOMD0000000545	Ouyang2014	Taxonomy-Arabidopsis thaliana	14	25	[0.0, 20.4]
BIOMD0000000546	Miao2010	Taxonomy-Influenza A virus (strain A/X-31 H3N2)|Taxonomy-Murinae	3	5	[0.0, 12.0]
BIOMD0000000547	Talemi2014	Taxonomy-Saccharomyces cerevisiae	11	52	[0.0, 200.0]
BIOMD0000000548	Sneppen2009	Taxonomy-Homo sapiens	3	4	[0.0, 85.0]
BIOMD0000000552	Ehrenstein2000	Taxonomy-Homo sapiens	2	4	[0.0, 451.2]

Continued on next page

Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000553	Ehrenstein1997	Taxonomy-Homo sapiens	2	4	[0.0, 451.2]
BIOMD0000000557	Reiterer2013	Taxonomy-Homo sapiens	25	50	[0.0, 300.0]
BIOMD0000000558	Cloutier2012	Taxonomy-Homo sapiens	2	8	[0.0, 198.6]
BIOMD0000000559	Ouzounoglou 2014	Taxonomy-Homo sapiens	89	23	[0.0, 5000.0]
BIOMD0000000560	Hui2016	Taxonomy-Mus musculus	62	113	[0.0, 1231.0]
BIOMD0000000563	Pritchard2014	Taxonomy-Pseudomonas syringae| Taxonomy-Arabidopsis	10	19	[0.0, 102.4]
BIOMD0000000564	Gould2013	Taxonomy-Arabidopsis thaliana	19	115	[0.0, 55.0]
BIOMD0000000565	Machado2014	Taxonomy-Escherichia coli	26	150	[0.0, 1844.6]
BIOMD0000000568	Mueller2015	Taxonomy-Mus musculus	24	79	[0.0, 427.8]
BIOMD0000000572	Costa2014	Taxonomy-Lactococcus lactis	26	112	[0.0, 1430.6]
BIOMD0000000573	Aguilera2014	Taxonomy-Homo sapiens	2	8	[0.0, 1750.0]
BIOMD0000000575	Sass2009	Taxonomy-Homo sapiens	49	245	[0.0, 1000.0]
BIOMD0000000576	Kolodkin2013	Taxonomy-Homo sapiens	34	61	[0.0, 1185.2]
BIOMD0000000579	Sengupta2015	Taxonomy-Homo sapiens	240	272	[0.0, 500.0]
BIOMD0000000580	Sonntag2012	Taxonomy-Mus musculus|Taxonomy-Homo sapiens	26	37	[0.0, 1636.2]
BIOMD0000000581	DallePezze2012	Taxonomy-Mus musculus|Taxonomy-Homo sapiens	25	32	[0.0, 631.6]
BIOMD0000000582	DallePezze2014	Taxonomy-Homo sapiens	23	55	[0.0, 64.0]
BIOMD0000000584	Mandlik2015	Taxonomy-Leishmania	21	34	[0.0, 500.0]
BIOMD0000000586	Karapetyan2016	Taxonomy-Saccharomyces cerevisiae	10	25	[0.0, 800.0]
BIOMD0000000587	Karapetyan2016	Taxonomy-Saccharomyces cerevisiae	10	25	[0.0, 400.0]
BIOMD0000000590	Hermansen2015	-	9	30	[0.0, 12.0]
BIOMD0000000591	Boehm2014	Taxonomy-Mus musculus	8	7	[0.0, 700.0]
BIOMD0000000596	Philipson2015	Helicobacter pylori, Mus musculus | biological process involved in symbiotic interaction; innate immune response; response to bacterium	15	56	[0.0, 3000.0]
BIOMD0000000597	Flis2015	Taxonomy-Arabidopsis thaliana	28	121	[0.0, 60.0]
BIOMD0000000598	Flis2015	Taxonomy-Arabidopsis thaliana	28	121	[0.0, 62.0]
BIOMD0000000599	Coggins2014	Taxonomy-Homo sapiens	30	40	[0.0, 1984.0]
BIOMD0000000600	re2011	transforming growth factor beta receptor signaling pathway	17	20	[0.0, 60000.0]
BIOMD0000000604	Elzbieta_Petelenz_Kurdziel2013	Taxonomy-Saccharomyces cerevisiae	29	127	[0.0, 5500.0]
BIOMD0000000605	PetelenzKuehn_osmoadaptation_HOG1att	Taxonomy-Saccharomyces cerevisiae	29	127	[0.0, 4000.0]
BIOMD0000000607	PetelenzKuehn_osmoadaptation_fps1D1	Taxonomy-Saccharomyces cerevisiae	29	127	[0.0, 80000.0]
BIOMD0000000611	Nayak2015	Taxonomy-Homo sapiens	66	106	[0.0, 3000.0]
BIOMD0000000612	Proctor2016	Taxonomy-Homo sapiens	35	72	[0.0, 200000.0]
BIOMD0000000613	Peterson2010	Taxonomy-Homo sapiens	31	268	[0.0, 2000.0]
BIOMD0000000614	Kamihira2000	Taxonomy-Homo sapiens	1	3	[0.0, 8000.0]
BIOMD0000000615	Kuznetsov2016	Taxonomy-Homo sapiens	4	12	[0.0, 8000.0]
BIOMD0000000616	Dunster2014	Taxonomy-Homo sapiens	4	8	[0.0, 300.0]
BIOMD0000000617	Walsh2014	Taxonomy-Homo sapiens	1	26	[0.0, 500.0]
BIOMD0000000620	Palmer2014	Taxonomy-Homo sapiens	34	52	[0.0, 50000.0]
BIOMD0000000621	Palmer2014	Taxonomy-Homo sapiens	34	52	[0.0, 50000.0]
BIOMD0000000622	NguyenLK2011	Taxonomy-cellular organisms	10	24	[0.0, 1910.0]
BIOMD0000000623	Orton2009	Taxonomy-Rattus norvegicus	25	54	[0.0, 197.6]

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Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000624	Sluka2016	Taxonomy-Homo sapiens	6	9	[0.0, 3000.0]
BIOMD0000000625	Leber2016	Taxonomy-Mus musculus|Taxonomy-Helicobacter pylori	17	50	[0.0, 550.0]
BIOMD0000000626	Ray2013	Taxonomy-Saccharomyces cerevisiae	6	20	[0.0, 55.2]
BIOMD0000000630	Venkatraman 2011	Taxonomy-cellular organisms	4	12	[0.0, 450.4]
BIOMD0000000631	DeCaluwe2016	Taxonomy-Arabidopsis thaliana	12	47	[0.0, 55.0]
BIOMD0000000640	DallePezze2016	Taxonomy-Mammalia	31	60	[0.0, 54.8]
BIOMD0000000642	Mufudza2012	Taxonomy-Homo sapiens	3	18	[0.0, 50.0]
BIOMD0000000646	Barr2016	Taxonomy-Homo sapiens	11	26	[0.0, 5000.0]
BIOMD0000000647	Kwang2003	KEGG Pathway-MAPK signaling pathway	11	11	[0.0, 68.8]
BIOMD0000000648	Padala2017	Taxonomy-Homo sapiens|Gene Ontology-MAPK cascade|KEGG Pathway-MAPK signaling pathway|KEGG Pathway-PI3K-Akt signaling pathway|KEGG Pathway-Wnt signaling pathway	53	103	[0.0, 882.4]
BIOMD0000000650	Owen1998	regulation of immune response to tumor cell	3	13	[0.0, 546.8]
BIOMD0000000651	Nguyen2016	Taxonomy-Homo sapiens|Pathway Ontology-the extracellular signal-regulated Raf/Mek/Erk signaling pathway|Pathway Ontology-mTOR signaling pathway	29	34	[0.0, 1600.0]
BIOMD0000000652	Padala2017	Taxonomy-Homo sapiens|Gene Ontology-MAPK cascade|KEGG Pathway-MAPK signaling pathway|KEGG Pathway-PI3K-Akt signaling pathway|KEGG Pathway-Wnt signaling pathway	53	102	[0.0, 1600.0]
BIOMD0000000653	Padala2017	Taxonomy-Homo sapiens|Gene Ontology-MAPK cascade|KEGG Pathway-MAPK signaling pathway|KEGG Pathway-PI3K-Akt signaling pathway|KEGG Pathway-Wnt signaling pathway	53	99	[0.0, 1196.0]
BIOMD0000000654	Padala2017	Taxonomy-Homo sapiens|Gene Ontology-MAPK cascade|KEGG Pathway-MAPK signaling pathway|KEGG Pathway-PI3K-Akt signaling pathway|KEGG Pathway-Wnt signaling pathway	53	101	[0.0, 50000.0]
BIOMD0000000655	Padala2017	Taxonomy-Homo sapiens|Gene Ontology-MAPK cascade|KEGG Pathway-MAPK signaling pathway|KEGG Pathway-PI3K-Akt signaling pathway|KEGG Pathway-Wnt signaling pathway	53	101	[0.0, 600.0]
BIOMD0000000656	Padala2017	Taxonomy-Homo sapiens|Gene Ontology-MAPK cascade|KEGG Pathway-MAPK signaling pathway|KEGG Pathway-PI3K-Akt signaling pathway|KEGG Pathway-Wnt signaling pathway	53	101	[0.0, 1868.4]
BIOMD0000000657	Araujo2016	Taxonomy-Homo sapiens|Pathway Ontology-M phase pathway	3	21	[0.0, 3000.0]
BIOMD0000000660	Barr2017	Taxonomy-Homo sapiens	15	31	[0.0, 1904.4]
BIOMD0000000662	Moore2004	Taxonomy-Homo sapiens	3	12	[0.0, 60.0]
BIOMD0000000663	Wodarz2007	Taxonomy-Human immunodeficiency virus 1|Taxonomy-Homo sapiens	3	7	[0.0, 19.2]
BIOMD0000000664	Muller2008	Taxonomy-Homo sapiens	6	20	[0.0, 80.0]

Continued on next page

Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000666	Pappalardo2016	Experimental Factor Ontology-response to dabrafenib|Reactome-PI3K/AKT Signaling in Cancer|Reactome-MAPK family signaling cascades|KEGG Pathway-MAPK signaling pathway|KEGG Pathway-PI3K-Akt signaling pathway	35	87	[0.0, 45.0]
BIOMD0000000667	Hornberg2005	regulation of MAPK cascade	103	83	[0.0, 15000.0]
BIOMD0000000668	Zhu2015	KEGG Drug-Gemcitabine (USAN/INN)|KEGG Drug-Birinapant (USAN/INN)|Brenda Tissue Ontology-pancreas|NCIt-Combination Chemotherapy|KEGG Drug-Gemcitabine (USAN/INN)|KEGG Drug-Birinapant (USAN/INN)|BioModels Database-MODEL1604270000	10	26	[0.0, 400.0]
BIOMD0000000670	Owen1998	regulation of immune response to tumor cell	3	8	[0.0, 122.4]
BIOMD0000000676	Chen2006	nitric oxide biosynthetic process	13	21	[0.0, 2.0]
BIOMD0000000677	Holmes2006	muscle contraction	1	10	[0.0, 4.0]
BIOMD0000000678	Tomida2003	Taxonomy-Homo sapiens	2	11	[0.0, 60.0]
BIOMD0000000679	Waugh2006	Taxonomy-Homo sapiens	3	11	[0.0, 24.0]
BIOMD0000000682	Wierschem2004	Taxonomy-Homo sapiens	5	34	[0.0, 600000.0]
BIOMD0000000683	Wodarz1999	Taxonomy-Homo sapiens	4	12	[0.0, 45.0]
BIOMD0000000684	Wodarz2003	Taxonomy-Homo sapiens	10	28	[0.0, 231.8]
BIOMD0000000685	Wodarz2003	Taxonomy-Homo sapiens|Brenda Tissue Ontology-cytotoxic T-lymphocyte|Cell Type Ontology-professional antigen presenting cell	4	17	[0.0, 55.0]
BIOMD0000000686	Wodarz2007	Taxonomy-Mus	5	16	[0.0, 76.0]
BIOMD0000000687	Wodarz2007	Taxonomy-Mus|Experimental Factor Ontology-cytomegalovirus infection|Brenda Tissue Ontology-cytotoxic T-lymphocyte	15	16	[0.0, 60.0]
BIOMD0000000688	Wodarz2007	Taxonomy-Mus|Experimental Factor Ontology-cytomegalovirus infection|Brenda Tissue Ontology-cytotoxic T-lymphocyte|Brenda Tissue Ontology-natural killer cell	16	20	[0.0, 60.0]
BIOMD0000000691	Wolf2000	Taxonomy-Saccharomyces	13	14	[0.0, 4.0]
BIOMD0000000692	Phillips2003	Taxonomy-Escherichia coli|KEGG Orthology-neurofibromin 1	8	12	[0.0, 0.8]
BIOMD0000000695	FelixGarza2017	Experimental Factor Ontology-psoriasis|Brenda Tissue Ontology-keratinocyte|Gene Ontology-response to blue light	12	55	[0.0, 2000.0]
BIOMD0000000696	Boada2016	-	11	41	[0.0, 700.0]
BIOMD0000000697	Ciliberto2003	NCIt-Xenopus laevis|Gene Ontology-regulation of cell cycle	13	25	[0.0, 1000.0]
BIOMD0000000698	Reed2004	Taxonomy-Homo sapiens	5	23	[0.0, 10.0]
BIOMD0000000700	Heldt2018	Taxonomy-Homo sapiens|Gene Ontology-DNA damage response, detection of DNA damage|Gene Ontology-regulation of cell cycle	23	56	[0.0, 1976.8]
BIOMD0000000703	Diedrichs2018	Reactome-unfolded protein [endoplasmic reticulum lumen]	11	81	[0.0, 3000.0]
BIOMD0000000704	Aguda1999	Taxonomy-Vertebrata|Gene Ontology-G2 DNA damage checkpoint	16	40	[0.0, 70.0]

Continued on next page

Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000705	Smith2010	Taxonomy-Eukaryota|NCIt-Apoptosis|Gene Ontology-metabolic process|Gene Ontology-post-translational protein modification|Gene Ontology-regulation of cell cycle	32	19	[0.0, 1982.8]
BIOMD0000000706	Smith2010	Taxonomy-Eukaryota|Gene Ontology-post-translational protein modification	45	128	[0.0, 120.0]
BIOMD0000000707	Revilla2003	Taxonomy-Human immunodeficiency virus 1|Taxonomy-Homo sapiens	5	10	[0.0, 200.0]
BIOMD0000000708	Liu2017	Taxonomy-Aves|Taxonomy-Homo sapiens	5	12	[0.0, 30000.0]
BIOMD0000000709	Liu2017	Taxonomy-Aves|Taxonomy-Homo sapiens	5	12	[0.0, 80000.0]
BIOMD0000000710	Vargas2012	Taxonomy-Influenza A virus|Taxonomy-Homo sapiens	7	17	[0.0, 90.8]
BIOMD0000000713	Aston2018	Taxonomy-Hepacivirus C|Taxonomy-Homo sapiens	3	8	[0.0, 500.0]
BIOMD0000000714	Reynolds2006	Taxonomy-Homo sapiens	4	25	[0.0, 765.4]
BIOMD0000000715	Huo2017	Taxonomy-Homo sapiens	4	11	[0.0, 15.0]
BIOMD0000000716	Lee2018	Taxonomy-H5N6 subtype|Taxonomy-Aves|Taxonomy-Homo sapiens	4	20	[0.0, 1500.0]
BIOMD0000000717	Lee2018	Taxonomy-H5N6 subtype|Taxonomy-Aves|Taxonomy-Homo sapiens	4	20	[0.0, 3000.0]
BIOMD0000000718	Li2008	Taxonomy-Caulobacter vibrioides	16	50	[0.0, 600.0]
BIOMD0000000719	Tsai2014	Taxonomy-Xenopus laevis|Gene Ontology-regulation of cell cycle	5	19	[0.0, 150.0]
BIOMD0000000720	Yan2012	Taxonomy-Mammalia|Gene Ontology-cell cycle	8	29	[0.0, 120.0]
BIOMD0000000721	Graham2013	Taxonomy-Homo sapiens	5	25	[0.0, 55.0]
BIOMD0000000722	Bianchi2015	Taxonomy-Mus musculus|Taxonomy-Rattus norvegicus	5	33	[0.0, 243.6]
BIOMD0000000724	Theinmozhi2018	Taxonomy-Homo sapiens	28	51	[0.0, 10000.0]
BIOMD0000000725	Sora2016	-	38	93	[0.0, 1200.0]
BIOMD0000000726	Ruan2017	Taxonomy-Rabies lyssavirus|Taxonomy-Homo sapiens|Taxonomy-Canis lupus familiaris	8	16	[0.0, 971.2]
BIOMD0000000727	Li2009	Taxonomy-Caulobacter vibrioides	30	96	[0.0, 300.0]
BIOMD0000000728	Norel1990	Taxonomy-Homo sapiens	2	4	[0.0, 7.0]
BIOMD0000000729	Goldbeter1996	Taxonomy-Eukaryota	3	15	[0.0, 45.0]
BIOMD0000000730	Gerard2009	Taxonomy-Mammalia	45	187	[0.0, 75.0]
BIOMD0000000731	Robertson-Tessi M 2012	Taxonomy-Homo sapiens|NCIt-Tumor-Infiltrating Immune Cell	16	48	[0.0, 50000.0]
BIOMD0000000732	Kirschner1998	Taxonomy-Homo sapiens|NCIt-Tumor-Infiltrating Immune Cell	3	14	[0.0, 3500.0]
BIOMD0000000733	Moore2004	Taxonomy-Homo sapiens	3	12	[0.0, 666.4]
BIOMD0000000736	Mouse_Iron_Distribution_Adequate_iron_diet_No_Tracer_	Taxonomy-Mus musculus	10	32	[0.0, 973.6]
BIOMD0000000737	Mouse_Iron_Distribution_Deficient_iro_diet_No_Tracer_	Taxonomy-Mus musculus	11	32	[0.0, 1412.0]
BIOMD0000000738	Mouse_Iron_Distribution_Rich_iron_diet_No_Tracer_	Taxonomy-Mus musculus	11	32	[0.0, 51.6]
BIOMD0000000742	Victor_Garcia 2018	Taxonomy-Homo sapiens	2	6	[0.0, 48.8]

Continued on next page

Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000743	Gallaher2018	Taxonomy-Homo sapiens	4	31	[0.0, 349.8]
BIOMD0000000744	Hu2019	Taxonomy-Homo sapiens	5	23	[0.0, 630.4]
BIOMD0000000746	Saad2017	Taxonomy-Homo sapiens	4	12	[0.0, 50.0]
BIOMD0000000747	Nagashima2002	Homo sapiens | blood coagulation	33	44	[0.0, 10000.0]
BIOMD0000000748	Phan2017	Taxonomy-Homo sapiens	4	8	[0.0, 200.0]
BIOMD0000000750	Lolas2016	Taxonomy-Homo sapiens	8	41	[0.0, 3.6]
BIOMD0000000752	Wilkie2013r	Taxonomy-Homo sapiens	3	9	[0.0, 500.0]
BIOMD0000000753	Figueredo2013	Taxonomy-Homo sapiens	2	8	[0.0, 9.8]
BIOMD0000000754	Figueredo2013	Taxonomy-Homo sapiens	3	13	[0.0, 1000.0]
BIOMD0000000756	Figueredo2013	Taxonomy-Homo sapiens	4	20	[0.0, 3000.0]
BIOMD0000000757	Abernathy2016	Taxonomy-Homo sapiens	7	32	[0.0, 300.0]
BIOMD0000000759	Breems2015	Taxonomy-Homo sapiens	6	24	[0.0, 64.4]
BIOMD0000000761	Cappuccio2006	Taxonomy-Mus musculus|Gene Ontology-T cell mediated cytotoxicity directed against tumor cell target|Gene Ontology-natural killer cell mediated cytotoxicity directed against tumor cell target	6	22	[0.0, 434.8]
BIOMD0000000762	Kuznetsov1994	T cell mediated immune response to tumor cell	2	8	[0.0, 1000.0]
BIOMD0000000763	Dritschel2018	T cell mediated immune response to tumor cell	3	8	[0.0, 50.0]
BIOMD0000000765	Mager2005	Taxonomy-Unknown	4	13	[0.0, 80.0]
BIOMD0000000766	Macnamara2015	Taxonomy-Homo sapiens	5	15	[0.0, 80.0]
BIOMD0000000767	Macnamara2015	Taxonomy-Homo sapiens	3	8	[0.0, 13.6]
BIOMD0000000768	Eftimie2010	Taxonomy-Homo sapiens	5	23	[0.0, 10.4]
BIOMD0000000769	Eftimie2017	Taxonomy-Homo sapiens	5	27	[0.0, 53.6]
BIOMD0000000770	Eftimie2017	Taxonomy-Homo sapiens	4	16	[0.0, 50.0]
BIOMD0000000771	Bajzer2008	-	3	8	[0.0, 38.0]
BIOMD0000000773	Wodarz2018	Taxonomy-Homo sapiens	3	21	[0.0, 4000.0]
BIOMD0000000774	Wodarz2018	Taxonomy-Homo sapiens	2	13	[0.0, 5000.0]
BIOMD0000000775	Iarosz2015	Taxonomy-Homo sapiens	4	15	[0.0, 621.6]
BIOMD0000000776	Monro2008	Taxonomy-Homo sapiens	2	7	[0.0, 1660.8]
BIOMD0000000777	Chakrabarty2010	-	3	9	[0.0, 244.0]
BIOMD0000000778	Wei2017	Taxonomy-Homo sapiens	3	14	[0.0, 35000.0]
BIOMD0000000779	dePillis2009	-	6	44	[0.0, 26.2]
BIOMD0000000780	Wang2016	Taxonomy-Homo sapiens	4	15	[0.0, 1658.4]
BIOMD0000000781	Wang2016	Taxonomy-Homo sapiens	3	10	[0.0, 400.0]
BIOMD0000000782	Wang2016	Taxonomy-Homo sapiens	2	6	[0.0, 187.2]
BIOMD0000000783	Dong2014	-	3	8	[0.0, 500.0]
BIOMD0000000784	Lopez2014	Taxonomy-Mus musculus	3	16	[0.0, 36.8]
BIOMD0000000785	Costa2003	T cell mediated immune response to tumor cell	2	3	[0.0, 100.0]
BIOMD0000000786	Lipniacki2004	Mus musculus | response to tumor necrosis factor	15	35	[0.0, 40000.0]
BIOMD0000000787	Frascoli2014	response to tumor cell; T cell mediated cytotoxicity; T cell mediated cytotoxicity directed against tumor cell target	2	6	[0.0, 285.6]
BIOMD0000000788	Schropp2019	Taxonomy-Homo sapiens	8	21	[0.0, 97.2]
BIOMD0000000789	Jenner2018	Taxonomy-Homo sapiens	3	8	[0.0, 150.0]
BIOMD0000000790	Alvarez2019	immune response to tumor cell	4	18	[0.0, 783.4]
BIOMD0000000791	Wilson2012	Taxonomy-Homo sapiens	5	13	[0.0, 347.6]
BIOMD0000000792	Hu2019	immune response to tumor cell	6	28	[0.0, 500.0]
BIOMD0000000793	Chen2011	Taxonomy-Homo sapiens	2	4	[0.0, 66.0]
BIOMD0000000794	Benary2019	Mus musculus | response to tumor necrosis factor	16	33	[0.0, 8500.0]

Continued on next page

Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000795	Chen2011	Taxonomy-Homo sapiens	2	6	[0.0, 1083.0]
BIOMD0000000796	Yang2012	Taxonomy-Homo sapiens	5	27	[0.0, 800.0]
BIOMD0000000797	Hu2018	immune response to tumor cell	4	19	[0.0, 50.0]
BIOMD0000000798	Sharp2019	Taxonomy-Homo sapiens	5	14	[0.0, 46.4]
BIOMD0000000799	Cucuianu2010	-	2	6	[0.0, 100.0]
BIOMD0000000800	Precup2012	-	3	10	[0.0, 1000.0]
BIOMD0000000801	Sturrock2015	Taxonomy-Homo sapiens	4	17	[0.0, 10000.0]
BIOMD0000000802	Hoffman2018	Taxonomy-Homo sapiens	4	11	[0.0, 5000.0]
BIOMD0000000803	Park2019	T cell homeostasis	9	10	[0.0, 10.0]
BIOMD0000000804	Koenders2015	Taxonomy-Homo sapiens	3	20	[0.0, 180.0]
BIOMD0000000805	Husari2013	Taxonomy-Homo sapiens	4	14	[0.0, 6.6]
BIOMD0000000806	Eftimie2019	Taxonomy-Mus musculus|Taxonomy-Vesicular stomatitis virus	6	32	[0.0, 476.6]
BIOMD0000000808	Kronik2008	-	6	40	[0.0, 3200.0]
BIOMD0000000809	Malinzi2018	regulation of immune response to tumor cell; dormancy process	5	13	[0.0, 19.6]
BIOMD0000000810	Ganguli2018	Human Disease Ontology-breast cancer	13	70	[0.0, 1000.0]
BIOMD0000000811	He2017	-	6	52	[0.0, 1267.8]
BIOMD0000000812	Galante2012	Homo sapiens | programmed cell death	4	16	[0.0, 300.0]
BIOMD0000000813	Anderson2015	Homo sapiens	3	12	[0.0, 400.0]
BIOMD0000000814	Víctor2019	-	3	14	[0.0, 2000.0]
BIOMD0000000815	Chrobak2011	Taxonomy-Mus musculus	2	6	[0.0, 280.4]
BIOMD0000000818	Lee2008	Taxonomy-Mammalia	10	18	[0.0, 12.0]
BIOMD0000000819	Nazari2018	-	10	34	[0.0, 500.0]
BIOMD0000000821	Yazdjer2019	angiogenesis	3	8	[0.0, 72.0]
BIOMD0000000823	Varusai2018	Taxonomy-Homo sapiens	16	38	[0.0, 800.0]
BIOMD0000000824	Lewkiewics2019	Taxonomy-Thymus	1	7	[0.0, 200.0]
BIOMD0000000825	Greene2019	-	2	12	[0.0, 80.0]
BIOMD0000000826	Sung_Young_Shin2018	Taxonomy-Homo sapiens	13	89	[0.0, 1956.8]
BIOMD0000000828	Jung2019	signaling	5	17	[0.0, 10.0]
BIOMD0000000829	Jung2019	signaling	11	51	[0.0, 200.8]
BIOMD0000000830	GiantsosAdams 2013	Homo sapiens	6	7	[0.0, 1089.4]
BIOMD0000000831	Smith1980	Taxonomy-Bos taurus||Taxonomy-Bos taurus|Gene Ontology-gonadotropin secretion|Gene Ontology-luteinizing hormone secretion|Mathematical Modelling Ontology-Ordinary differential equation model|BioModels Database-MODEL7898438988	3	9	[0.0, 7.2]
BIOMD0000000832	Shin2016	Taxonomy-Homo sapiens	20	61	[0.0, 100.0]
BIOMD0000000833	DiCamillo2016	|Taxonomy-Homo sapiens|Gene Ontology-insulin receptor signaling pathway|Bio Models Database-MODEL1604100005	61	222	[0.0, 10000.0]
BIOMD0000000835	Rao2014	Taxonomy-Rattus norvegicus	31	202	[0.0, 60.0]
BIOMD0000000837	Hanson2016	-	8	22	[0.0, 50000.0]
BIOMD0000000838	Tsur2019	-	3	8	[0.0, 155.2]
BIOMD0000000839	Almeida2019	circadian rhythm	8	29	[0.0, 45.0]
BIOMD0000000840	Caldwell2019	-	5	10	[0.0, 268.0]

Continued on next page

Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000842	Heitzler2012	Taxonomy-Homo sapiens||Taxonomy-Homo sapiens|Brenda Tissue Ontology-kidney|Gene Ontology-transmembrane receptor protein serine/threonine kinase signaling pathway|Gene Ontology-angiotensin-activated signaling pathway|BioModels Database-MODEL1012080000	20	35	[0.0, 339.6]
BIOMD0000000843	Dudziuk2019	-	10	39	[0.0, 409.6]
BIOMD0000000844	Viertel2019	-	9	82	[0.0, 1400.0]
BIOMD0000000845	Gulbudak2019	-	3	12	[0.0, 250.0]
BIOMD0000000846	Gulbudak2019	-	3	12	[0.0, 400.0]
BIOMD0000000847	Adams2019	-	4	17	[0.0, 100.0]
BIOMD0000000848	FatehiChenar 2018	immune response	9	28	[0.0, 13.2]
BIOMD0000000849	M_Berg2017	Bos taurus	13	72	[0.0, 60.0]
BIOMD0000000850	Jenner2019	-	3	4	[0.0, 659.2]
BIOMD0000000851	Ho2019	-	5	17	[0.0, 35.6]
BIOMD0000000852	Andersen2017	-	6	35	[0.0, 20000.0]
BIOMD0000000853	Smolen2018	-	9	31	[0.0, 4000.0]
BIOMD0000000854	Gray2016	Taxonomy-Mus musculus	4	9	[0.0, 14.4]
BIOMD0000000855	Cooper2015	Taxonomy-Homo sapiens	4	39	[0.0, 20.0]
BIOMD0000000857	Larbat2016	sucrose metabolic process; phenol-containing compound metabolic process; starch metabolic process	9	34	[0.0, 170.0]
BIOMD0000000858	Larbat2016	phenol-containing compound metabolic process; sucrose metabolic process; starch metabolic process	8	35	[0.0, 50.0]
BIOMD0000000859	Larbat2016	starch metabolic process; phenol-containing compound metabolic process; sucrose metabolic process	12	46	[0.0, 55.0]
BIOMD0000000861	Bachmann2011	Taxonomy-Mus musculus	26	32	[0.0, 300.0]
BIOMD0000000862	Proctor2017B	Homo sapiens	8	11	[0.0, 40000.0]
BIOMD0000000863	Kosinsky2018	Taxonomy-Mus musculus	6	27	[0.0, 40.2]
BIOMD0000000864	Proctor2017	Homo sapiens	6	9	[0.0, 5000.0]
BIOMD0000000865	Nikolaev2019	Taxonomy-Mus musculus	4	58	[0.0, 20.0]
BIOMD0000000866	Simon2019	Homo sapiens	3	6	[0.0, 3000.0]
BIOMD0000000867	Coulibaly2019	Taxonomy-Homo sapiens	10	57	[0.0, 70.0]
BIOMD0000000868	Simon2019	Homo sapiens	4	7	[0.0, 1990.0]
BIOMD0000000869	Simon2019	Homo sapiens	4	12	[0.0, 1448.0]
BIOMD0000000870	Simon2019	Homo sapiens	6	14	[0.0, 1943.0]
BIOMD0000000871	NIK_dependent_ p100_processing_ into_p52	Homo sapiens	8	20	[0.0, 500.0]
BIOMD0000000872	Verma2016	Homo sapiens	7	24	[0.0, 1000.0]
BIOMD0000000873	Soni2018	Leishmania	29	57	[0.0, 125.2]
BIOMD0000000874	Perelson1993	Taxonomy-Homo sapiens	4	10	[0.0, 4500.0]
BIOMD0000000875	Nelson2000	Taxonomy-Homo sapiens	4	8	[0.0, 30.0]
BIOMD0000000876	Aavani2019	Taxonomy-Homo sapiens	4	11	[0.0, 70.0]
BIOMD0000000877	Ontah2019	Taxonomy-Homo sapiens	4	12	[0.0, 70.0]
BIOMD0000000878	Lenbury2001	Taxonomy-Homo sapiens	3	13	[0.0, 420.0]
BIOMD0000000879	Rodrigues2019	Taxonomy-Homo sapiens	3	18	[0.0, 1.5]
BIOMD0000000880	Trisilowati2018	Taxonomy-Homo sapiens	4	14	[0.0, 183.0]
BIOMD0000000881	Kogan2013	-	4	27	[0.0, 3000.0]
BIOMD0000000882	Munz2009	Taxonomy-Homo sapiens	3	5	[0.0, 6.0]
BIOMD0000000883	Giani2019	Taxonomy-Homo sapiens	63	134	[0.0, 77.6]

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Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000884	Cortes2019	Taxonomy-Homo sapiens	3	9	[0.0, 50.0]
BIOMD0000000885	Sumana2018	Taxonomy-Homo sapiens	4	11	[0.0, 600.0]
BIOMD0000000887	Lim2014	Taxonomy-Homo sapiens	4	10	[0.0, 549.2]
BIOMD0000000888	Unni2019	Homo sapiens	4	21	[0.0, 80.0]
BIOMD0000000889	Fribourg2014	Taxonomy-Homo sapiens	12	50	[0.0, 150.0]
BIOMD0000000890	Bhattacharya 2014	Taxonomy-Homo sapiens	3	11	[0.0, 5000.0]
BIOMD0000000891	Khajanchi2019	Taxonomy-Homo sapiens	3	12	[0.0, 430.8]
BIOMD0000000892	Sandip2013	Taxonomy-Homo sapiens	4	11	[0.0, 27.4]
BIOMD0000000893	Gonzalez Miranda2013	-	3	6	[0.0, 9.0]
BIOMD0000000894	Bose2011	Taxonomy-Homo sapiens	3	8	[0.0, 143.6]
BIOMD0000000895	Schokker2013	Taxonomy-Gallus gallus	9	49	[0.0, 120.4]
BIOMD0000000896	Szymanska2009	Taxonomy-Homo sapiens	9	18	[0.0, 1118.0]
BIOMD0000000897	Khajanchi2015	Taxonomy-Homo sapiens	2	10	[0.0, 800.0]
BIOMD0000000898	Jiao2018	Taxonomy-Homo sapiens	5	22	[0.0, 150.0]
BIOMD0000000899	Ota2015	Homo sapiens | GDP-dissociation inhibitor binding	12	15	[0.0, 1000.0]
BIOMD0000000903	perez2019	Taxonomy-Homo sapiens	5	28	[0.0, 53.6]
BIOMD0000000905	Dubey2007	Taxonomy-Homo sapiens	5	15	[0.0, 72.0]
BIOMD0000000906	Dubey2007	Taxonomy-Homo sapiens	3	9	[0.0, 29.2]
BIOMD0000000907	HeberleRazquin Navas2019	Taxonomy-Homo sapiens	25	68	[0.0, 214.4]
BIOMD0000000908	dePillis2013	Taxonomy-Homo sapiens	7	43	[0.0, 1500.0]
BIOMD0000000909	dePillis2003	Taxonomy-Homo sapiens	4	17	[0.0, 100.0]
BIOMD0000000910	Isaeva2008	Taxonomy-Homo sapiens	3	10	[0.0, 250.0]
BIOMD0000000911	Merola2008	Taxonomy-Homo sapiens	3	9	[0.0, 137.2]
BIOMD0000000912	Caravagna2010	Taxonomy-Homo sapiens	3	14	[0.0, 500.0]
BIOMD0000000913	dePillis2008	Taxonomy-Homo sapiens	6	37	[0.0, 350.0]
BIOMD0000000914	Guillen2013	Taxonomy-Mus musculus	5	10	[0.0, 85.0]
BIOMD0000000915	Sun2018	Taxonomy-Homo sapiens	9	30	[0.0, 18000.0]
BIOMD0000000916	Kraan199	Taxonomy-Homo sapiens	5	5	[0.0, 4.4]
BIOMD0000000917	Phillips2007	Taxonomy-Homo sapiens	3	21	[0.0, 48.0]
BIOMD0000000919	Ledzewicz2013	Taxonomy-Homo sapiens||Taxonomy-Homo sapiens|Mathematical Modelling Ontology-Ordinary differential equation model	2	9	[0.0, 15.2]
BIOMD0000000920	Jarrett2015	Taxonomy-Mus musculus	4	16	[0.0, 594.0]
BIOMD0000000921	Khajanchi2017	Taxonomy-Homo sapiens	5	24	[0.0, 225.2]
BIOMD0000000922	Turner2015	Taxonomy-Lutzia|Experimental Factor Ontology-malaria	3	9	[0.0, 22.8]
BIOMD0000000924	Smith2011	Taxonomy-Homo sapiens	6	28	[0.0, 79.8]
BIOMD0000000925	Dunster2016	Homo sapiens | blood coagulation	14	22	[0.0, 6000.0]
BIOMD0000000926	Rhodes2019	Taxonomy-Homo sapiens	8	78	[0.0, 600.0]
BIOMD0000000927	Grigolon2018	Taxonomy-unidentified plant	3	9	[0.0, 444.4]
BIOMD0000000928	Baker2017	-	4	14	[0.0, 18.0]
BIOMD0000000929	Li2016	Taxonomy-Homo sapiens	5	39	[0.0, 500.0]
BIOMD0000000930	Liu2017	Taxonomy-Homo sapiens	4	17	[0.0, 800.0]
BIOMD0000000931	Voliotis2019	Taxonomy-Mus	3	22	[0.0, 35.0]
BIOMD0000000933	Kosiuk2015	Taxonomy-unclassified eukaryotes	3	14	[0.0, 27.0]
BIOMD0000000934	Linke2017	NCIt-Saccharomyces cerevisiae|Gene Ontology-regulation of cell cycle	13	23	[0.0, 80.0]
BIOMD0000000935	Ferrel2011	Taxonomy-Xenopus laevis|Gene Ontology-regulation of cell cycle	2	8	[0.0, 10.6]

Continued on next page

Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000000936	ferrel2011	Taxonomy-Xenopus laevis|Gene Ontology-regulation of cell cycle	1	4	[0.0, 7.6]
BIOMD0000000937	Ferrel2011	Taxonomy-Xenopus laevis|Gene Ontology-regulation of cell cycle	3	12	[0.0, 12.0]
BIOMD0000000938	Gerard2013	Taxonomy-Mus musculus|Gene Ontology-regulation of cell cycle	3	12	[0.0, 60.0]
BIOMD0000000939	Iwamoto2010	Taxonomy-Homo sapiens|Gene Ontology-DNA damage response, detection of DNA damage	54	138	[0.0, 5500.0]
BIOMD0000000940	Tang2019	Taxonomy-Homo sapiens	20	42	[0.0, 1368.4]
BIOMD0000000941	Gerard2010	Taxonomy-Mammalia|Gene Ontology-regulation of cell cycle	8	22	[0.0, 50.0]
BIOMD0000000942	Sible2007	Taxonomy-Xenopus laevis|Gene Ontology-regulation of cell cycle	7	27	[0.0, 200.0]
BIOMD0000000943	Hat2016	Taxonomy-Homo sapiens|Gene Ontology-cell cycle	33	101	[0.0, 150000.0]
BIOMD0000000944	Goldbeter2013	OMIT-Amphibians	3	15	[0.0, 40.0]
BIOMD0000000947	Lee2017	Taxonomy-Homo sapiens|KEGG Drug-Acetaminophen (JP18/USP)	9	12	[0.0, 188.8]
BIOMD0000000948	Landberg2009	KEGG Compound-Resorcinol|ChEBI-5-alkylresorcinol	4	6	[0.0, 80.0]
BIOMD0000000949	Chitnis2008	Taxonomy-Homo sapiens|Experimental Factor Ontology-malaria	7	24	[0.0, 12000.0]
BIOMD0000000952	Rodenfels2019	Taxonomy-Danio rerio|Gene Ontology-regulation of cell cycle	11	26	[0.0, 70.0]
BIOMD0000000953	Queralt2006	Taxonomy-Saccharomyces cerevisiae (strain ATCC 204508 / S288c)	13	46	[0.0, 500.0]
BIOMD0000000954	Pandey2018	Taxonomy-Homo sapiens	18	53	[0.0, 1536.4]
BIOMD0000000959	Kok2020	Homo sapiens | regulation of type I interferon-mediated signaling pathway	41	87	[0.0, 50.0]
BIOMD0000000965	LeBeau1999	Homo sapiens | calcium-mediated signaling	5	17	[0.0, 250.0]
BIOMD0000000966	Cui2008	Taxonomy-Escherichia coli	7	7	[0.0, 5.0]
BIOMD0000000967	McLean1991	Taxonomy-Human immunodeficiency virus|Taxonomy-Homo sapiens	4	10	[0.0, 100.0]
BIOMD0000000968	Palmer2008	Taxonomy-Homo sapiens	10	11	[0.0, 100.0]
BIOMD0000000972	Tang2020	Taxonomy-Severe acute respiratory syndrome coronavirus 2|Taxonomy-Homo sapiens	8	22	[0.0, 150.0]
BIOMD0000000973	Dasgupta2020	Taxonomy-Homo sapiens	2	8	[0.0, 100.0]
BIOMD0000000979	Malkov2020	Taxonomy-Severe acute respiratory syndrome coronavirus 2|Taxonomy-Homo sapiens	5	6	[0.0, 350.0]
BIOMD0000000985	Fabry1984	receptor-mediated endocytosis	7	13	[0.0, 80.0]
BIOMD0000000986	Aubry1995	Taxonomy-Dictyostelium discoideum AX2	6	12	[0.0, 293.6]
BIOMD0000000987	Aubry1995	Taxonomy-Dictyostelium discoideum AX2	9	11	[0.0, 152.8]
BIOMD0000001004	Intosalmi2015	Taxonomy-Mus sp.	10	18	[0.0, 100.0]
BIOMD0000001005	Bae2017	Taxonomy-Mus musculus|Taxonomy-Rattus norvegicus	41	133	[0.0, 100.0]
BIOMD0000001006	Ciliberto2005	Taxonomy-Homo sapiens	7	27	[0.0, 1250.0]
BIOMD0000001007	Zhang2007	Taxonomy-Homo sapiens	3	32	[0.0, 150.0]
BIOMD0000001009	Zhang2007	Taxonomy-Homo sapiens	4	33	[0.0, 28.8]
BIOMD0000001010	Zhang2007	Taxonomy-Homo sapiens	2	26	[0.0, 126.0]
BIOMD0000001011	Triana2020	Taxonomy-Homo sapiens	3	5	[0.0, 550.0]
BIOMD0000001012	Triana2020	Taxonomy-Homo sapiens	4	10	[0.0, 1000.0]
BIOMD0000001015	Jarrah2014	Taxonomy-Mus musculus	6	20	[0.0, 200.0]

Continued on next page

Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
BIOMD0000001016	Bakshi2020	Taxonomy-Homo sapiens	8	14	[0.0, 100.0]
BIOMD0000001017	Bakshi2020	Taxonomy-Homo sapiens	13	22	[0.0, 20000.0]
BIOMD0000001018	Bakshi2020	Taxonomy-Homo sapiens	18	31	[0.0, 20000.0]
BIOMD0000001021	Lavigne2021	Taxonomy-Homo sapiens	6	10	[0.0, 8.8]
BIOMD0000001022	Creemers2021	Taxonomy-Homo sapiens	4	7	[0.0, 80.0]
BIOMD0000001023	Alharbi2020	Taxonomy-Homo sapiens	3	16	[0.0, 38.8]
BIOMD0000001024	Chaudhury2020	Taxonomy-Homo sapiens	2	4	[0.0, 240.0]
BIOMD0000001025	Chaudhury2020	Taxonomy-Homo sapiens|Human Disease Ontology-leukemia	3	14	[0.0, 1000.0]
BIOMD0000001026	Kurlovics2021	Taxonomy-Homo sapiens||Brenda Tissue Ontology-erythrocyte|ChEBI-metformin|Gene Ontology-drug transport|Mathematical Modelling Ontology-Ordinary differential equation model|Bio Models Database-MODEL2103170002	1	12	[0.0, 200.0]
BIOMD0000001027	Zake2021	Taxonomy-Mus sp.	20	97	[0.0, 15.0]
BIOMD0000001028	Zake2021	Taxonomy-Homo sapiens	21	104	[0.0, 150.0]
BIOMD0000001029	Zake2021	Taxonomy-Homo sapiens	21	104	[0.0, 80.0]
BIOMD0000001030	Sontag2017	Taxonomy-Vertebrata	2	6	[0.0, 200.0]
BIOMD0000001031	Tuwairqi2020	Taxonomy-Homo sapiens	3	4	[0.0, 30.0]
BIOMD0000001032	Tuwairqi2020	Taxonomy-Homo sapiens	4	7	[0.0, 49.6]
BIOMD0000001033	Almuallem2020	Taxonomy-Homo sapiens	6	31	[0.0, 113.8]
BIOMD0000001034	Mendrazitsky 2007	Taxonomy-Mycobacterium tuberculosis variant bovis BCG|Taxonomy-Homo sapiens	4	12	[0.0, 50.0]
BIOMD0000001035	Tuwairqi2020	Taxonomy-Homo sapiens	4	11	[0.0, 259.6]
BIOMD0000001036	Cappuccio2007	Taxonomy-Homo sapiens	3	13	[0.0, 1500.0]
BIOMD0000001037	Alharbi2019	Taxonomy-Homo sapiens	2	6	[0.0, 10.0]
BIOMD0000001038	Alharbi2019	Taxonomy-Homo sapiens	3	10	[0.0, 100.0]
BIOMD0000001039	Zake2021	Taxonomy-Mus sp.	20	97	[0.0, 6.0]
BIOMD0000001040	Kurlovics2021	Taxonomy-Homo sapiens||Brenda Tissue Ontology-erythrocyte|ChEBI-metformin|Gene Ontology-drug transport|Mathematical Modelling Ontology-Ordinary differential equation model|Bio Models Database-MODEL2103170001	1	11	[0.0, 120.0]
BIOMD0000001043	Wodarz2001	Taxonomy-Homo sapiens	3	9	[0.0, 1500.0]
BIOMD0000001045	Moore2004	Taxonomy-Influenza A virus (strain A/Hong Kong/1/1968 H3N2)|Taxonomy-Homo sapiens	3	2	[0.0, 130.0]
BIOMD0000001047	Collier1996	Taxonomy-Drosophila	4	14	[0.0, 24.4]
BIOMD0000001048	Siddhartha2002	Taxonomy-Homo sapiens	3	11	[0.0, 1994.4]
BIOMD0000001052	Alharbi2020	Taxonomy-Homo sapiens	3	16	[0.0, 30.0]
BIOMD0000001053	Garde2020	Taxonomy-Bacillus subtilis	6	6	[0.0, 5.5]
BIOMD0000001054	Pearce2021	Taxonomy-Homo sapiens	7	11	[0.0, 200.0]
BIOMD0000001056	Chulian2021	Taxonomy-Homo sapiens	3	9	[0.0, 4000.0]
BIOMD0000001057	Nikolov2020	Taxonomy-Homo sapiens	2	7	[0.0, 1298.0]
BIOMD0000001058	Novak2022	Taxonomy-Homo sapiens	10	48	[0.0, 212.0]
BIOMD0000001059	Stucki2005	Taxonomy-Mus	10	23	[0.0, 1062.6]
BIOMD0000001060	Frank2021	Taxonomy-Homo sapiens	2	20	[0.0, 24.4]
BIOMD0000001072	Phillips2013	Taxonomy-Mammalia	6	53	[0.0, 270000.0]
BIOMD0000001077	Adlung2021	Taxonomy-Homo sapiens	15	29	[0.0, 130.0]
BIOMD0000001102	Burbano2023	-	23	28	[0.0, 62.0]
MODEL0910896131	Guyton1972	Taxonomy-Homo sapiens	2	8	[0.0, 60000.0]
MODEL0975191032	Chang2008	Taxonomy-Homo sapiens	115	219	[0.0, 1000.0]

Continued on next page

Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
MODEL1002160000	Cabrero2011	Taxonomy-Homo sapiens	8	64	[0.0, 100.0]
MODEL1004010000	Kuwahara2010	Taxonomy-Escherichia coli	31	55	[0.0, 2.0]
MODEL1004010001	Kuwahara2010	Taxonomy-Escherichia coli	31	55	[0.0, 3.0]
MODEL1004010002	Kuwahara2010	Taxonomy-Escherichia coli	31	55	[0.0, 0.15]
MODEL1004070000	Haut1974	Rattus | pentose-phosphate shunt	31	49	[0.0, 750.0]
MODEL1004070001	Vaseghi1999	Saccharomyces cerevisiae | pentose-phosphate shunt	6	18	[0.0, 1986.0]
MODEL1005050000	Salazar2009	Taxonomy-Arabidopsis thaliana	15	76	[0.0, 60.0]
MODEL1005200000	Twycross2010	Taxonomy-Embryophyta	43	16	[0.0, 1000.0]
MODEL1007200000	Nijhout2006	Taxonomy-Homo sapiens	23	108	[0.0, 5.0]
MODEL1008060000	Munz2009	Taxonomy-Homo sapiens	3	9	[0.0, 107.6]
MODEL1008060002	Munz2009	Taxonomy-Homo sapiens	5	10	[0.0, 400.0]
MODEL1009220000	Martins2004	Saccharomyces cerevisiae	19	110	[0.0, 290.0]
MODEL1009230000	Munz2009	Taxonomy-Homo sapiens	3	6	[0.0, 1985.2]
MODEL1011010000	Bruck2008	Saccharomyces cerevisiae | glycolytic process	17	93	[0.0, 0.4]
MODEL1101100000	Bakker1997	Taxonomy-Trypanosoma brucei	38	50	[0.0, 55.0]
MODEL1101170000	Nakano2010	Taxonomy-Homo sapiens	182	286	[0.0, 60.0]
MODEL1102210000	Wei2011	Taxonomy-Homo sapiens	190	316	[0.0, 84.8]
MODEL1102210001	Telesco2011	Taxonomy-Homo sapiens	119	58	[0.0, 1712.0]
MODEL1103210001	Jamshidi01_RBC_Metabolic Network	Homo sapiens	39	410	[0.0, 150.0]
MODEL1108260010	Jesty1993	Taxonomy-Homo sapiens	4	8	[0.0, 12.0]
MODEL1108260015	Qiao2004	Taxonomy-Homo sapiens	6	20	[0.0, 503.2]
MODEL1109160000	Kogan2001	-	30	32	[0.0, 33.2]
MODEL1109160001	Kogan2001	-	40	37	[0.0, 11.8]
MODEL1112100000	nsson2005	Taxonomy-Arabidopsis thaliana	1012	522	[0.0, 2000.0]
MODEL1112110004	Silber2007	Taxonomy-Homo sapiens	7	22	[0.0, 1980.0]
MODEL1112150000	Tiemann2011	Taxonomy-Mus musculus	8	23	[0.0, 6.0]
MODEL1112260002	Smith2010	Taxonomy-Homo sapiens	54	135	[0.0, 300.0]
MODEL1202030000	Dupeux2011	Taxonomy-Arabidopsis thaliana	14	24	[0.0, 10.0]
MODEL1202030001	Dupeux2011	Taxonomy-Arabidopsis thaliana	8	12	[0.0, 10.0]
MODEL1202170000	Nazaret2008	Taxonomy-Eukaryota	12	49	[0.0, 94.8]
MODEL1203220000	Mol2013	Taxonomy-Leishmania	64	131	[0.0, 200.0]
MODEL1204040000	Houser2012	Taxonomy-Saccharomyces cerevisiae	15	46	[0.0, 8000.0]
MODEL1204060000	Kubota2012	Taxonomy-Rattus	23	26	[0.0, 350.0]
MODEL1204240000	Sorokina2011	Taxonomy-Ostreococcus tauri	38	73	[0.0, 236.2]
MODEL1204280001	Sarma2012	Taxonomy-Xenopus laevis	12	23	[0.0, 500.0]
MODEL1204280002	Sarma2012	Taxonomy-Mus musculus	11	22	[0.0, 350.0]
MODEL1204280003	Sarma2012	Taxonomy-Mus musculus	11	22	[0.0, 500.0]
MODEL1204280004	Sarma2012	Taxonomy-Mus musculus	11	26	[0.0, 600.0]
MODEL1204280005	Sarma2012	Taxonomy-Xenopus laevis	12	23	[0.0, 266.8]
MODEL1204280006	Sarma2012	Taxonomy-Mus musculus	11	22	[0.0, 400.0]
MODEL1204280007	Sarma2012	Taxonomy-Mus musculus	11	22	[0.0, 300.0]
MODEL1204280008	Sarma2012	Taxonomy-Mus musculus	11	26	[0.0, 294.8]
MODEL1204280009	Sarma2012	Taxonomy-Xenopus laevis	12	23	[0.0, 500.0]
MODEL1204280010	Sarma2012	Taxonomy-Mus musculus	11	22	[0.0, 350.0]
MODEL1204280011	Sarma2012	Taxonomy-Mus musculus	11	22	[0.0, 500.0]
MODEL1204280012	Sarma2012	Taxonomy-Mus musculus	11	26	[0.0, 600.0]
MODEL1204280013	Sarma2012	Taxonomy-Xenopus laevis	12	23	[0.0, 400.0]
MODEL1204280014	Sarma2012	Taxonomy-Mus musculus	11	22	[0.0, 350.0]
MODEL1204280015	Sarma2012	Taxonomy-Mus musculus	11	22	[0.0, 280.0]
MODEL1204280016	Sarma2012	Taxonomy-Mus musculus	11	26	[0.0, 350.0]

Continued on next page

Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
MODEL1204280017	Sarma2012	Taxonomy-Xenopus laevis	25	37	[0.0, 1000.0]
MODEL1204280018	Sarma2012	Taxonomy-Mus musculus	24	37	[0.0, 1000.0]
MODEL1204280019	Sarma2012	Taxonomy-Mus musculus	24	37	[0.0, 1503.0]
MODEL1204280021	Sarma2012	Taxonomy-Xenopus laevis	25	37	[0.0, 1541.2]
MODEL1204280022	Sarma2012	Taxonomy-Mus musculus	24	37	[0.0, 1000.0]
MODEL1204280023	Sarma2012	Taxonomy-Mus musculus	24	37	[0.0, 1572.4]
MODEL1204280025	Sarma2012	Taxonomy-Xenopus laevis	25	37	[0.0, 1200.0]
MODEL1204280029	Sarma2012	Taxonomy-Xenopus laevis	25	37	[0.0, 1200.0]
MODEL1204280030	Sarma2012	Taxonomy-Mus musculus	24	37	[0.0, 5000.0]
MODEL1204280031	Sarma2012	Taxonomy-Mus musculus	24	37	[0.0, 5000.0]
MODEL1204280032	Sarma2012	Taxonomy-Mus musculus	27	45	[0.0, 5000.0]
MODEL1204280033	Sarma2012	Taxonomy-Xenopus laevis	12	23	[0.0, 700.0]
MODEL1204280034	Sarma2012	Taxonomy-Mus musculus	11	22	[0.0, 421.6]
MODEL1204280035	Sarma2012	Taxonomy-Mus musculus	11	22	[0.0, 559.6]
MODEL1204280037	Sarma2012	Taxonomy-Xenopus laevis	12	23	[0.0, 201.2]
MODEL1204280038	Sarma2012	Taxonomy-Mus musculus	11	22	[0.0, 500.0]
MODEL1204280039	Sarma2012	Taxonomy-Mus musculus	11	22	[0.0, 346.4]
MODEL1208030000	Mandlik2013	Taxonomy-Leishmania	6	27	[0.0, 30.0]
MODEL1303010000	Lopez2013	Taxonomy-Homo sapiens	74	106	[0.0, 10000.0]
MODEL1303140000	Mazemondet 2012	Taxonomy-Homo sapiens	5	12	[0.0, 302.0]
MODEL1303260000	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	15	93	[0.0, 154.4]
MODEL1303260001	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	15	95	[0.0, 154.0]
MODEL1303260002	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	15	105	[0.0, 182.0]
MODEL1303260003	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	15	106	[0.0, 100.0]
MODEL1303260004	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	16	111	[0.0, 97.2]
MODEL1303260005	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	16	111	[0.0, 103.8]
MODEL1303260006	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	20	128	[0.0, 156.8]
MODEL1303260007	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	20	129	[0.0, 109.0]
MODEL1303260008	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	20	132	[0.0, 472.0]
MODEL1303260009	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	20	142	[0.0, 518.4]
MODEL1303260010	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	20	142	[0.0, 465.6]
MODEL1303260011	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	20	142	[0.0, 300.0]
MODEL1303260012	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	20	145	[0.0, 200.0]
MODEL1303260013	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	20	146	[0.0, 607.6]
MODEL1303260014	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	20	155	[0.0, 573.6]
MODEL1303260015	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	20	155	[0.0, 575.4]
MODEL1303260016	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	20	166	[0.0, 526.6]

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Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
MODEL1303260017	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	20	168	[0.0, 546.2]
MODEL1303260018	Smallbone2013	Saccharomyces cerevisiae | glycolytic process	20	168	[0.0, 305.0]
MODEL1304300000	Hofmeyr1996	regulation of catalytic activity	3	8	[0.0, 2.0]
MODEL1308080003	Cao2010	Taxonomy-Escherichia coli	14	30	[0.0, 100000.0]
MODEL1403250001	vanEunen2012	Taxonomy-Saccharomyces cerevisiae	13	88	[0.0, 2.0]
MODEL1403250002	Rao2014	Taxonomy-Saccharomyces cerevisiae	8	50	[0.0, 3.6]
MODEL1409050001	Rutkis2013	-	20	87	[0.0, 6500.0]
MODEL1409240002	Qosa2014	Taxonomy-Mus musculus|Taxonomy-Homo sapiens	6	14	[0.0, 46.2]
MODEL1411130000	Miao2014	-	8	12	[0.0, 10.0]
MODEL1412200000	Talemi2015	-	5	19	[0.0, 284.4]
MODEL1502270000	Weisse2015	-	22	32	[0.0, 0.15]
MODEL1503180002	Smallbone2015	-	4	7	[0.0, 10000.0]
MODEL1503180004	Smallbone2015	-	3	11	[0.0, 15.0]
MODEL1503180005	Smallbone2015	-	3	9	[0.0, 100.0]
MODEL1503180006	Smallbone2015	-	3	12	[0.0, 66.0]
MODEL1504010000	Shestov2014	-	21	144	[0.0, 4.4]
MODEL1504080004	Costa2015	-	18	145	[0.0, 200.0]
MODEL1504080005	Costa2015	-	18	140	[0.0, 200.0]
MODEL1504080006	Costa2015	-	18	149	[0.0, 200.0]
MODEL1504160000	Wu2011	-	6	11	[0.0, 30.0]
MODEL1505110000	Millard2016	-	64	440	[0.0, 87.8]
MODEL1506070001	Nguyen2014	-	21	60	[0.0, 1250.0]
MODEL1509020000	Ashall2009	-	14	38	[0.0, 50000.0]
MODEL1509050002	Perrett2014	-	11	39	[0.0, 275.2]
MODEL1510230001	Rantasalo2016	-	13	33	[0.0, 3.2]
MODEL1510230002	Rantasalo2016	-	13	33	[0.0, 0.7]
MODEL1510230003	Rantasalo2016	-	13	33	[0.0, 2.8]
MODEL1510230004	Rantasalo2016	-	18	37	[0.0, 0.3]
MODEL1510230005	Rantasalo2016	-	18	37	[0.0, 0.06]
MODEL1601250000	Tortolina2015	-	456	862	[0.0, 264.6]
MODEL1603150000	Saa2016	-	5	40	[0.0, 1000.0]
MODEL1603270000	D1 LTP time window	-	87	143	[0.0, 20.0]
MODEL1604100000	Mukhopadhyay 2013	-	53	22	[0.0, 0.4]
MODEL1608100000	Aguilera2017	-	2	4	[0.0, 267.2]
MODEL1608100001	Aguilera2017	-	7	18	[0.0, 1500.0]
MODEL1609100001	Adams2013	-	6	7	[0.0, 1000.0]
MODEL1610100003	Proctor2017	-	7	9	[0.0, 10.0]
MODEL1611030000	Guisoni2016	-	5	5	[0.0, 27.2]
MODEL1611160001	Leber2016	-	42	117	[0.0, 43.4]
MODEL1611160002	Leber2016	-	37	76	[0.0, 860.8]
MODEL1701170000	Yapo2017	-	52	115	[0.0, 800.0]
MODEL1701170001	Yapo2017	-	64	152	[0.0, 500.0]
MODEL1704110000	Proctor2017	-	9	17	[0.0, 500000.0]
MODEL1704110001	Proctor2017	-	16	31	[0.0, 1879.2]
MODEL1704110002	Proctor2017	-	11	17	[0.0, 500000.0]
MODEL1704110003	Proctor2017	-	14	21	[0.0, 2000000.0]
MODEL1704110004	Proctor2017	-	46	84	[0.0, 1917.8]

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Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
MODEL1704190000	Phosphatase_activities_on_PI_3_4_5_P3_and_PI_3_4_P2	-	12	31	[0.0, 1187.6]
MODEL1705030000	DallePezze2016	-	29	50	[0.0, 30.0]
MODEL1705030001	DallePezze2016	-	29	53	[0.0, 48.8]
MODEL1705170000	Proctor2017	-	10	15	[0.0, 1989.4]
MODEL1705170001	Proctor2017	-	5	8	[0.0, 5000.0]
MODEL1705170002	Proctor2017	-	7	11	[0.0, 10000.0]
MODEL1705170003	Proctor2017	-	12	18	[0.0, 1988.6]
MODEL1705170004	Proctor2017	-	17	25	[0.0, 10000.0]
MODEL1705170005	Proctor2017	-	50	83	[0.0, 1981.8]
MODEL1708210000	PanRTK_model_for_single_cell_line	Taxonomy-Homo sapiens	59	124	[0.0, 600.0]
MODEL1808280013	Ducrot2009	-	10	35	[0.0, 1990.6]
MODEL1812050001	Rohan_D_Gidvani2012	-	13	24	[0.0, 1987.2]
MODEL1907260001	dePillis2005	Taxonomy-Mus musculus	3	23	[0.0, 52.8]
MODEL1907310002	Arciero2004	Taxonomy-Homo sapiens	4	20	[0.0, 255.2]
MODEL1908270001	Nikolopoulou 2018	-	3	31	[0.0, 210.8]
MODEL1908270002	ODea2007	-	24	70	[0.0, 1531.8]
MODEL1908290001	Hetmanski2019	-	58	97	[0.0, 1978.8]
MODEL1909050002	Ribba2018	-	2	13	[0.0, 300.0]
MODEL1909090002	Wei2019	-	5	27	[0.0, 17.6]
MODEL1909090003	Shariatpanahi 2018	-	4	29	[0.0, 195.4]
MODEL1909100002	Louzoun2014	-	4	24	[0.0, 494.6]
MODEL1909160002	Reyes2017	Taxonomy-Mus musculus	8	29	[0.0, 3000.0]
MODEL1910240001	Petrov2018	-	5	13	[0.0, 8000.0]
MODEL1911120004	Gupta2019	-	4	26	[0.0, 500.0]
MODEL1911130007	Hutchinson2016	Taxonomy-Homo sapiens	16	49	[0.0, 828.4]
MODEL1911140002	Tiwari2019	Taxonomy-Mus musculus	17	38	[0.0, 6500.0]
MODEL1911150001	Tsur2019	-	3	10	[0.0, 1896.4]
MODEL1911190001	Day2015	immune response	6	39	[0.0, 488.8]
MODEL1911270001	Sigal2019	Taxonomy-Mus musculus	7	35	[0.0, 500.0]
MODEL1911270002	Yu2019	Taxonomy-Mus musculus	4	17	[0.0, 46.2]
MODEL1912090002	Kawka2014	Wnt signaling pathway	9	19	[0.0, 20000.0]
MODEL1912100004	Nguyen2013	Taxonomy-Homo sapiens	22	49	[0.0, 10000.0]
MODEL1912100005	Nguyen2013	Taxonomy-Homo sapiens	7	19	[0.0, 50.4]
MODEL1912120002	Sandip2013	Taxonomy-Homo sapiens	4	11	[0.0, 37.2]
MODEL1912120005	Mkango2019	Taxonomy-Homo sapiens	6	29	[0.0, 35.0]
MODEL2001080001	Guimera2017	cell redox homeostasis; mitochondrion	15	9	[0.0, 803.2]
MODEL2001090001	Leeuwen2007	Taxonomy-Homo sapiens	11	29	[0.0, 5.0]
MODEL2001090002	Nanda2013	Homo sapiens	5	28	[0.0, 1000.0]
MODEL2001130003	Kronik2010	Taxonomy-Homo sapiens	7	15	[0.0, 600.0]
MODEL2001160001	dePillis2007	Taxonomy-Homo sapiens	4	15	[0.0, 288.4]
MODEL2003030002	Back2018	-	5	12	[0.0, 100.0]
MODEL2003060002	Grigolon2018	Taxonomy-unidentified plant	3	9	[0.0, 50.0]
MODEL2003170002	Vibert2017	-	6	14	[0.0, 1.4]
MODEL2003180001	Gidvani2012	Taxonomy-Saccharomyces cerevisiae	13	24	[0.0, 1.0]
MODEL2003190008	Garira2019	-	6	65	[0.0, 1901.0]
MODEL2003200001	Jafarnejad2019	Taxonomy-Homo sapiens	49	71	[0.0, 0.6]
MODEL2004300002	Konrath2020	-	7	41	[0.0, 10.0]

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Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
MODEL2005070001	Hatzimanikatis 1999	cell cycle	3	19	[0.0, 0.002]
MODEL2005130001	Ma2005	Homo sapiens	3	36	[0.0, 8.0]
MODEL2007090001	Nwabugwu2013	Taxonomy-Homo sapiens|NCIt-Tumor-Infiltrating Immune Cell	16	48	[0.0, 14000.0]
MODEL2009230001	Hardiansyah2019	Taxonomy-Homo sapiens	8	29	[0.0, 5.0]
MODEL2011030003	Bouchnita2020	-	6	11	[0.0, 250.0]
MODEL2012040002	Khandibharad 2022	-	63	83	[0.0, 384.0]
MODEL2103010001	Hetmanski2021	-	58	99	[0.0, 1856.0]
MODEL2103290001	Sharma2022	-	8	88	[0.0, 5000.0]
MODEL2109110006	Mullinax2016	Taxonomy-Homo sapiens	4	17	[0.0, 101.4]
MODEL2112290001	Berzins2022	BioModels Database-BIOMD0000000222	30	229	[0.0, 13.0]
MODEL2201210001	Sivery2016	-	7	21	[0.0, 12.8]
MODEL2201250001	Potassium_ balance_2021	-	15	72	[0.0, 60.0]
MODEL2202020001	Slaviero2021	-	27	196	[0.0, 40.0]
MODEL2206230001	Mosbacher2022	-	98	280	[0.0, 10.0]
MODEL2209260001	Grignard2022	-	6	12	[0.0, 37.6]
MODEL2210070001	Maier2022	-	44	63	[0.0, 2500.0]
MODEL2301180001	Li2021	-	4	11	[0.0, 16.0]
MODEL2306170002	Harish2009	-	38	71	[0.0, 35.0]
MODEL2306300001	Davies2023	Taxonomy-Homo sapiens	29	108	[0.0, 433.6]
MODEL2307050001	Sobotta2017	-	35	74	[0.0, 1995.2]
MODEL2307110001	Mothes2020	-	7	17	[0.0, 502.4]
MODEL2307130001	Konrath2023	-	20	29	[0.0, 1990.0]
MODEL2311140001	DeboraSoncini 2024	-	18	157	[0.0, 1971.2]
MODEL2401050001	Lang2024	-	124	325	[0.0, 2000.0]
MODEL2426780967	MODEL2426	Taxonomy-Saccharomyces cerevisiae	17	88	[0.0, 5.2]
MODEL2427021978	MODEL2427	Taxonomy-Saccharomyces cerevisiae	17	88	[0.0, 5.2]
MODEL2502170001	Hayashi2013	Taxonomy-Mus musculus	51	67	[0.0, 1263.6]
MODEL2502170002	Hayashi2013	Taxonomy-Mus musculus	26	42	[0.0, 1643.2]
MODEL2503310002	Kurpad2023	Taxonomy-Homo sapiens	3	13	[0.0, 20000.0]
MODEL2937159804	Shimoni2009	Taxonomy-Escherichia coli	6	14	[0.0, 640.0]
MODEL3631586579	MODEL3631	Taxonomy-Embryophyta	436	1661	[0.0, 0.4]
MODEL4780784080	Hayashi1999	Taxonomy-Rattus	10	16	[0.0, 22.0]
MODEL4816599063	mahaney2000	Taxonomy-Spodoptera frugiperda	10	17	[0.0, 25.0]
MODEL4821294342	Kierzek2001	Taxonomy-Escherichia coli	8	18	[0.0, 269.6]
MODEL4992089662	Banaji2005	Homo sapiens | blood circulation	36	106	[0.0, 220.2]
MODEL5954483266	Koster1988	Taxonomy-Xenopus laevis	2	12	[0.0, 600.0]
MODEL6185511733	Qiao2007	Taxonomy-Eukaryota	9	12	[0.0, 0.6]
MODEL6185746832	Qiao2007	Taxonomy-Eukaryota	14	18	[0.0, 500.0]
MODEL6623617994	Lambeth2002	Taxonomy-Homo sapiens	20	85	[0.0, 11.6]
MODEL6624091635	MODEL6624	Taxonomy-Lactococcus lactis	23	151	[0.0, 0.5]
MODEL6624199343	Martins2001	Taxonomy-Saccharomyces cerevisiae	3	13	[0.0, 50000.0]
MODEL6655501972	Nijhout2004	-	6	43	[0.0, 10000.0]
MODEL7743212613	Radulescu2008	Taxonomy-Mammalia	5	41	[0.0, 15000.0]
MODEL7743315447	Radulescu2008	Taxonomy-Mammalia	6	40	[0.0, 30000.0]
MODEL7743358405	Radulescu2008	Taxonomy-Mammalia	8	37	[0.0, 32000.0]
MODEL7743444866	Radulescu2008	Taxonomy-Mammalia	18	100	[0.0, 1016.0]
MODEL7743528808	Radulescu2008	-	19	100	[0.0, 1972.8]
MODEL7743576806	Radulescu2008	-	21	100	[0.0, 60000.0]
MODEL7743608569	Radulescu2008	Taxonomy-Mammalia	30	95	[0.0, 60000.0]

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Model ID	Reference	Domain	Sys. Dim.	Par. Dim.	Time Span
MODEL7743631122	Radulescu2008	Taxonomy-Mammalia	39	90	[0.0, 60000.0]
MODEL8236441887	Yang2008	Taxonomy-Homo sapiens	19	73	[0.0, 78.0]
MODEL8236480549	Yang2008	Taxonomy-Homo sapiens	20	67	[0.0, 20000.0]
MODEL8236520494	Yang2008	Taxonomy-Homo sapiens	7	24	[0.0, 10.0]
MODEL8262229752	MODEL8262	Taxonomy-Escherichia coli	21	25	[0.0, 700.4]
MODEL8459127548	You2010	Taxonomy-Saccharomyces cerevisiae	30	37	[0.0, 1946.0]
MODEL8478881246	Basak_Cell_2007	Taxonomy-Homo sapiens	32	101	[0.0, 1720.8]
MODEL8938094216	dAlcantara2003	Taxonomy-Homo sapiens	14	16	[0.0, 10.0]
MODEL9070467164	Bhalla2002	Taxonomy-Mus musculus	86	169	[0.0, 1610.2]
MODEL9071122126	Bhalla1999	Taxonomy-Homo sapiens	59	108	[0.0, 513.2]
MODEL9071773985	Bhalla2001	Taxonomy-Homo sapiens	66	139	[0.0, 295.8]
MODEL9077438479	Bhalla2002	-	26	43	[0.0, 136.8]
MODEL9079179924	Bhalla2002	Taxonomy-Mus musculus	75	136	[0.0, 1597.2]
MODEL9079740062	Bhalla2004	Taxonomy-Mammalia	26	43	[0.0, 20.0]
MODEL9080388197	Bhalla2004	Taxonomy-Mammalia	12	20	[0.0, 87.2]
MODEL9080747936	Bhalla2004	Taxonomy-Mammalia	44	84	[0.0, 777.8]
MODEL9081220742	Bhalla2004	Taxonomy-Mammalia	174	337	[0.0, 1399.2]
MODEL9085850385	Bhalla2004	Taxonomy-Mammalia	54	94	[0.0, 537.8]
MODEL9086207764	Hayer2005	Taxonomy-Mammalia	273	570	[0.0, 0.4]
MODEL9086518048	Hayer2005	Taxonomy-Mammalia	273	580	[0.0, 140.0]
MODEL9086628127	Hayer2005	Taxonomy-Mammalia	14	28	[0.0, 12500.0]
MODEL9086926384	Hayer2005	Taxonomy-Mammalia	77	148	[0.0, 148.4]
MODEL9086953089	Hayer2005	Taxonomy-Mammalia	104	196	[0.0, 1566.4]
MODEL9087255381	Hayer2005	Taxonomy-Mammalia	275	586	[0.0, 882.0]
MODEL9087474843	Hayer2005	Taxonomy-Mammalia	276	586	[0.0, 482.0]
MODEL9087766308	Soud1999	Taxonomy-Homo sapiens	5	4	[0.0, 0.1]
MODEL9087988095	Soud1999	Taxonomy-Homo sapiens	5	4	[0.0, 0.4]
MODEL9088169066	Soud1999	Taxonomy-Homo sapiens	5	4	[0.0, 0.3]
MODEL9088294310	Soud1999	Taxonomy-Homo sapiens	5	4	[0.0, 0.15]
MODEL9089491423	Ajay_Bhalla_2004	Taxonomy-Rattus	181	349	[0.0, 205.6]
MODEL9089538076	Ajay_Bhalla_2004_PKM_MKP3_Tuning	Taxonomy-Rattus	182	353	[0.0, 8000.0]
MODEL9089914876	Ajay_Bhalla_2004_Feedback_Tuning	Taxonomy-Rattus	179	347	[0.0, 5000.0]
MODEL9147091146	Ajay_Bhalla_2007_Bistable	Taxonomy-Rattus	71	132	[0.0, 1436.8]
MODEL9147232940	Ajay_Bhalla_2007_PKM	Taxonomy-Rattus	59	104	[0.0, 1.6]
MODEL9147975215	Asthagiri2001	Taxonomy-Cricetulus griseus	36	47	[0.0, 8000.0]